



- Multicentricity is common

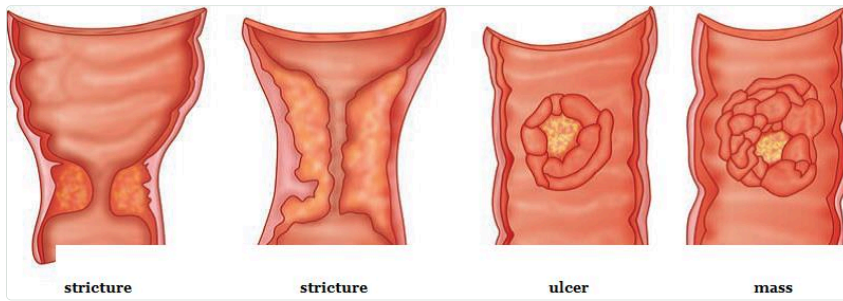


Figure 15.1 — Macroscopic appearances of colorectal cancer

Incidence of colorectal cancer in different parts:

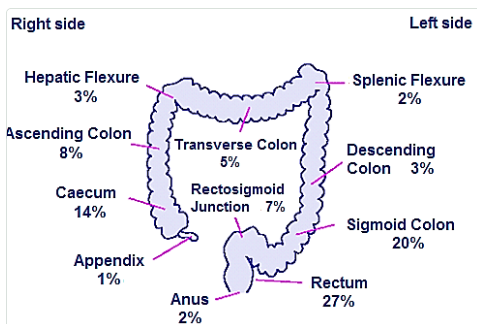


Figure 15.2 — Distribution of colorectal cancer

The most common site incidence is rectosigmoid part of the colon

Spread:

A- Direct: - mucosa → submucosa → muscle layer → serosa → nearby structures or Organs. It may lead to internal fistula as colo-vesical or recto-vaginal fistulae.

- **Right colon** → right ureter, abdominal wall, gonadal vessels, duodenum, small intestine, peritoneum, liver.
- **Transverse colon** → duodenum, stomach, liver, intestine, peritoneum.
- **Left colon** → spleen, abdominal wall, intestine, left ureter, gonadal vessels, peritoneum.

Rectum:

to urinary bladder, vagina → causing fistula → to sacrum, prostate, seminal vesicles, and peritoneum.

B- Lymphatic spread:

- 1- Epicolic lymph nodes
- 2- Para-colic lymph nodes



3- Mesocolic lymph nodes

4- Mesenteric lymph nodes: along the mesenteric vessels

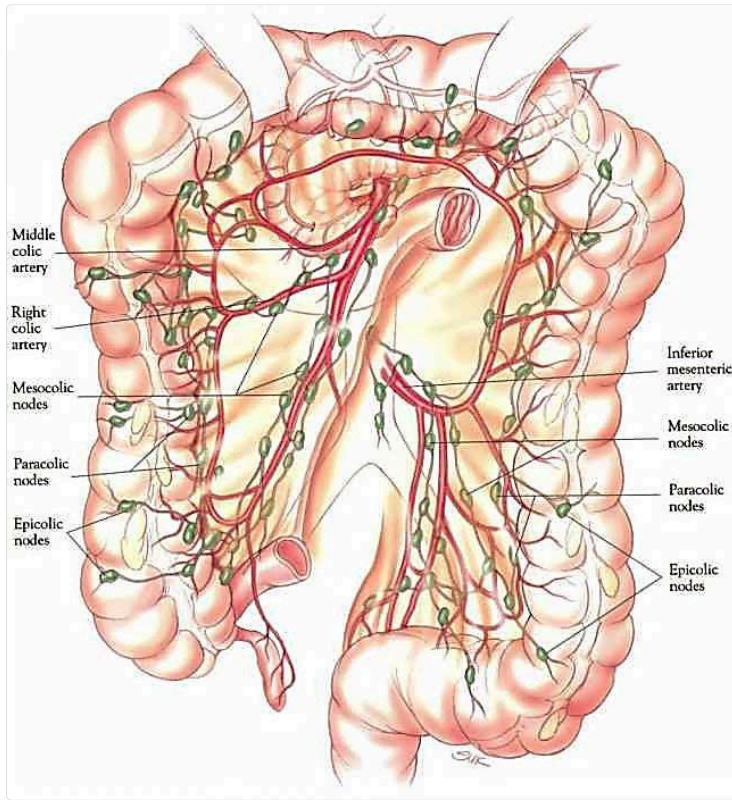


Figure 15.3 — Lymphatics of the colon

C- Blood spread:

- Liver → most common (Portal circulation).
 - Lung, bone, brain.

Staging:

Duke's staging for carcinoma of colon

Stage A:

Tumor in the bowel but not beyond muscularis propria.

Stage B:

Trans-mural invasion of bowel wall.

Stage C:

Any degree of bowel wall invasion with lymph node metastasis.

- C1—Epicolic and paracolic nodes.



- C2—Intermediate and proximal lymph nodes (mesenteric lymph nodes).

■ In modified Duke's :

Stage D : Presence of distant metastasis; is added

TNM classification of cancer colon:

■ T= Tumour

- **T1**: The tumor has grown through the muscularis mucosa and extends into the submucosa.
- **T2**: The tumor has grown through the submucosa and extends into the muscularis propria .
- **T3**: The tumor has grown through the muscularis propria and into the outer muscle layers of the colon or rectum but not through them. It has not reached any nearby organs
- **T4a**: The cancer has grown through the serosa the outermost lining of the intestines.
- **T4b**: The cancer has grown through the wall of the colon or rectum and is attached or invades into nearby tissues or organs.

■ N= nodes

N0: No cancer in nearby lymph nodes.

N1: Cancer cells are found in or near 1 to 3 nearby lymph nodes

N2: Cancer cells are found in 4 or more nearby lymph nodes

■ M= metastasis

To The **liver**, lungs, bones , brain.

M0: No distant spread is detected.

M1a:

The cancer has spread to ONE distant organ

M1b:

The cancer has spread to MORE than ONE distant organ or it has spread to distant parts of the peritoneum.

Differences between cancer right and left colon:

■ Right-side colorectal cancer characteristics:

Colorectal cancer that starts on the right side (Cecum, Ascending Colon, Right Half of Transverse Colon, Hepatic Flexure)

is seen more often in women and people over 65

- is diagnosed more frequently in African Americans
- often begins with flat, sessile polyps that go unnoticed until they are fairly large • is usually diagnosed at a later stage



- often presents with symptoms such as anemia and bleeding
- has a higher occurrence of mucinous tumors, BRAF-mt or KRAS-mt mutations, MSI-H • responds well to immunotherapy

■ Left-side colorectal cancer characteristics:

Colorectal cancer that begins on the left side (Splenic Flexure, Descending and Sigmoid Colon, Left Half of Transverse Colon, Rectosigmoid and Rectum) occurs more often than right-sided colorectal cancers

- is seen more often in men and in people under 65
- has tumors that grow along the wall of the colon
- often presents with constipation, narrow stools, and other bowel habit changes • causes bowel obstructions more often than right-sided colorectal cancer
- is usually diagnosed in its early stages
- is associated with FAP (familial adenomatous polyposis)
- has a higher occurrence of HER2 amplifications
- overall better prognosis than right-sided tumors
- responds better to standard chemotherapy and targeted therapies

Clinical picture of colorectal cancers:

■ General symptoms for all:

- 1- old aged male
- 2- history of FAP of colon (mainly left colon)
- 3- history of inflammatory bowel disease
- 4- loss of weight, anaemia, cachexia

I- Cancer of the cecum

The right colon is larger in diameter, contain fluid stool, low bacterial load and low pressure. The cancer comes more in the form of mass or ulcer

- Weight loss
- Unexplained anaemia
- Pain in the right iliac fossa (D.D.)
- Bleeding per-rectum (occult or obovious)
- Irregular bowel habits
- Late → intestinal obstruction: Colicky abdominal pain – Distension - progressive constipation - Repeated vomiting
- Firm to hard fixed right iliac fossa mass of variable size.



II. Cancer left colon & sigmoid:

- The left colon is narrower than right, contain more solid stool and the pressure is higher • The cancer lesion is commonly stricture (growth along the wall).
- 1. Pain in left side of the abdomen:
 - Left hypochondrium (splenic flexure)
 - Left iliac fossa (sigmoid colon)
 - Ana pain (rectum)
- 2. Altered bowel habits (intermittent diarrhea and progressive constipation (common) 3. Bleeding per-rectum more frequent
- 4. Attacks of colonic obstruction
 - Recurrent colicky pains
 - Abdominal distension
 - Rarely vomiting
 - Constipation
- 5. Intestinal obstruction full picture
- 6. Mass is felt → hard & fixed

III- Cancer rectum

- 1- **Tenesmus** (sense of incomplete evacuation of rectum+ mucus and blood with Little stool)
- 2- Bleeding per-rectum
- 3- Anal pain
- 4- Mass can be felt on P/R examination in low rectal cancers →hard fixed mass with bleeding surface.
- 5- Altered bowel habits
- 6- Colonic obstruction

Investigations:

I- Lab.

- 1- CBC for the presence of anaemia and increased ESR
- 2- Stool analysis for occult blood in stool
- 3- **CEA (Carcino-Embryonic Antigen)** for follow up: A rising CEA level is an indication for colonoscopic examination or CT scan abdomen (For local recurrence or metastasis)
- 4- Other routine lab. Investigations



II-Endoscopy:

for visualization of the lesions size and site and **multi-centricity** and take biopsies for histo-pathology (most valuable investigation).

a)Sigmoidoscope:

to examine the rectum and sigmoid colon b) **Colonoscopy** to examine the entire colon till the cecum

III-Radiological:

1- Barium enema → detects the lesion as:

a) mass as filling defect b) stricture as apple-core, shouldering, irregularities and proximal dilatation c) ulcer as filling defect

2- C.T.

scan with contrast:

for staging (detect extra-luminal spread and metastases in liver).

3- Endo- rectal ultra-sound for rectal carcinoma spread locally



Figure 15.4 — Colonoscopy: source comparison view



Figure 15.5 — Colonoscopy: source mass image

Normal colonoscope mass



Figure 15.6 — Barium enema: source stricture image

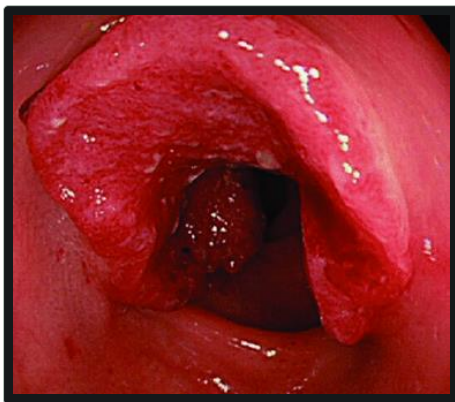


Figure 15.7 — Endoscopy: source ulcer image

Malignant ulcer barium enema; stricture

Management

- Management of cancer colon depends on some factors:
 - 1- Site of the cancer (right, left, transverse, sigmoid, rectum, anal canal).
 - 2- Multi-centricity of the tumour.
 - 3- Stage of the tumour and the presence of Metastases
 - 4- Operable or non-operable tumour.
 - 5- Presence of complications as intestinal Obstruction or Perforation.
 - 6- General condition of the patient.

I- Cancer cecum or ascending colon

- Operable + non- obstructed Or obstructed carcinoma
- Steps:
 - 1- Colonic **preparation 3 days before surgery**. Done in elective cases
- Oral fluids intake



- Laxatives
- Neomycin & metronidazole oral
- Repeated enemas
- Colonic preparation
- **one day before surgery:**
total bowel irrigation with normal saline through naso-gastric tube
- On table irrigation during surgery

Right hemi-colectomy and ileo-transverse anastomosis:

- a. Ligation of the ileo-colic and right colic vessels that supply the Right colon
- b. Resection of the whole right colon.
- c. Removal of the regional lymph nodes.
- d. Ileo-transverse anastomosis (anastomose the terminal ileum to the right transverse colon).

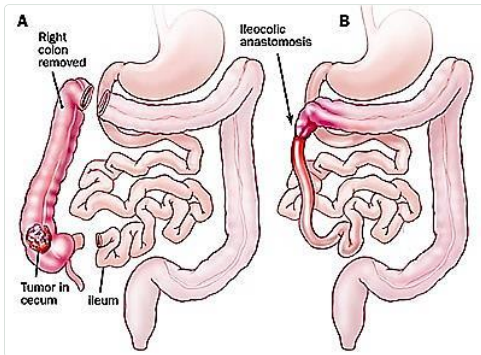


Figure 15.8 — Right hemi-colectomy

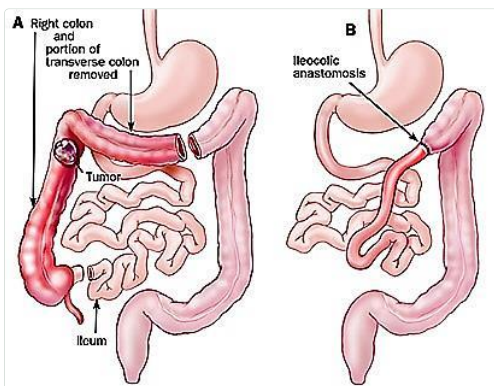


Figure 15.9 — Cancer ascending colon- extended right hemi-colectomy

II - Cancer transverse colon:

A - Operable non - obstructed case Or obstructed cases:

Surgery :

- 1- Transverse colectomy + colo-colic anastomosis



2-transverse colectomy and ileo-colic anastomosis (In tumour in the right transverse colon)

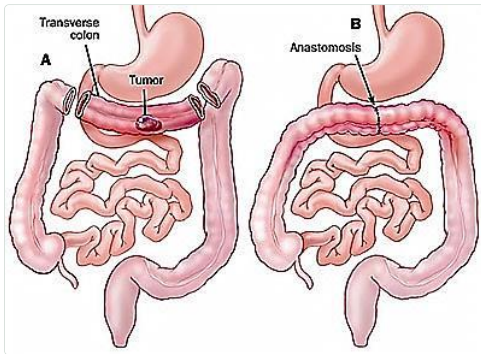


Figure 15.10 — Cancer transverse colon

III - Cancer left colon

A - Non - obstructed Carcinoma

Steps:

good colonic preparation due to the presence of high bacterial load in the left side.

- **Colonic preparation:**
 - Oral fluids + low residue diet for 3 days + purgatives
 - Repeated daily enemas
 - Oral Flagyl and neomycin (intestinal antiseptic)
 - Total bowel irrigations by normal saline 24 hours before surgery through naso-gastric tube.(rapid preparation)
 - Not done in obstructed patients

Values of colon preparation:

- The idea of gut preparation is to get an empty colon with reduced bacterial load.
- This reduces contamination during surgery and thereby incidence of wound infection and leakage following surgery are minimized
- **Surgery: using surgical staplers (EEA)**
 - 1- **Descending colectomy + colo-colic anastomosis**
 - 2- **In cancer sigmoid colon → sigmoid colectomy + Colo-rectal anastomosis.**

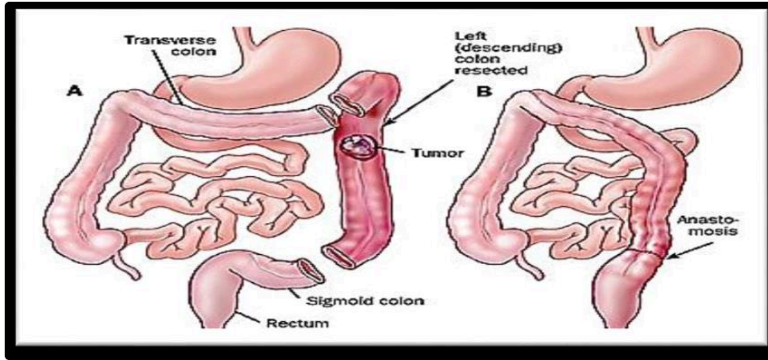


Figure 15.11 — Descending colon cancer

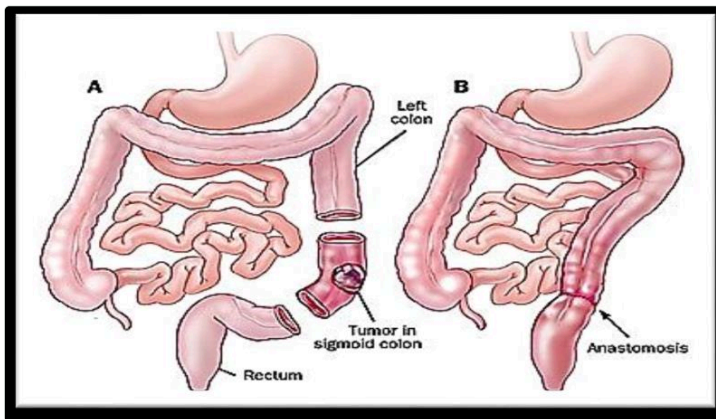


Figure 15.12 — Sigmoid colon cancer

B - Obstructed carcinoma of left colon:

■ Steps:

- 1- No colonic preparation
- 2- Naso- gastric tube suction
- 3- Intravenous fluids and correction of electrolyte disturbances. 4- Antibiotic are started pre-operatively to control infections (3rd generation cephalosporins and Metronidazole I.V.).

5 - Surgery:

■ First stage: (Hartmann's procedure)

a) Resection:

- 1- Descending colectomy in cancer descending colon
- 2 -Sigmoid colectomy in cancer sigmoid colon.

b) Proximal colostomy or ileostomy → to divert stool away from site of surgery (**temporal colostomy**).

Second stage: 6 to 8 weeks later: Using surgical staplers

- 1- Restoration of bowel continuity by anastomosis of the two ends.



2- Closure of the colostomy.

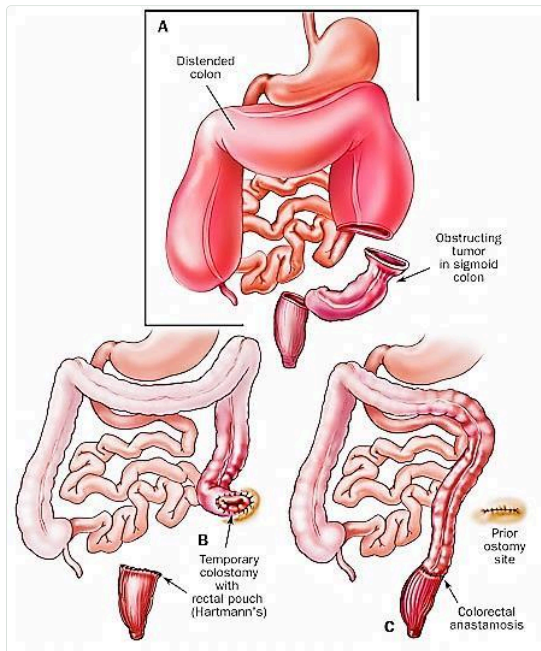


Figure 15.13 — Obstructing cancer sigmoid

IV - Cancer rectum

A - Non - obstructed case:

1- As above in cancer left colon

2- Surgery:

■ In upper and middle rectal cancers → Low anterior resection. Using

■ surgical staplers:

Resection of lower sigmoid + rectum till 2 cm below the lesion + colo-rectal or colo-anal anastomosis.

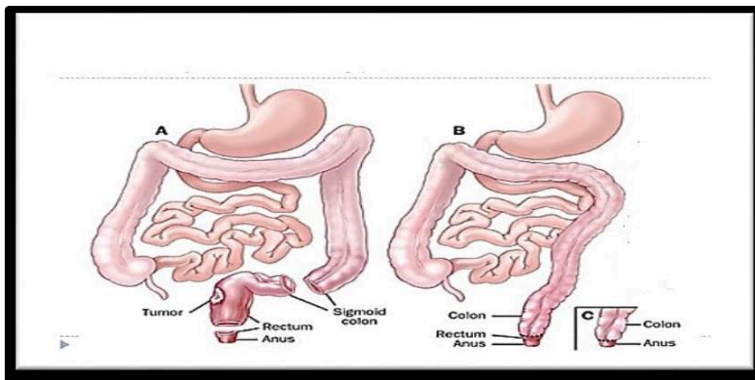


Figure 15.14 — Low anterior resection



■ Low rectal cancer:

■ Abdomino - Perineal Resection (APR)

a) Mobilization of the lower sigmoid and rectum (abdominal) b) Mobilization the anal canal (trans-perineal) c) Removal of the whole specimen through the abdomen.

d) Terminal end-colostomy (permenant colostomy).

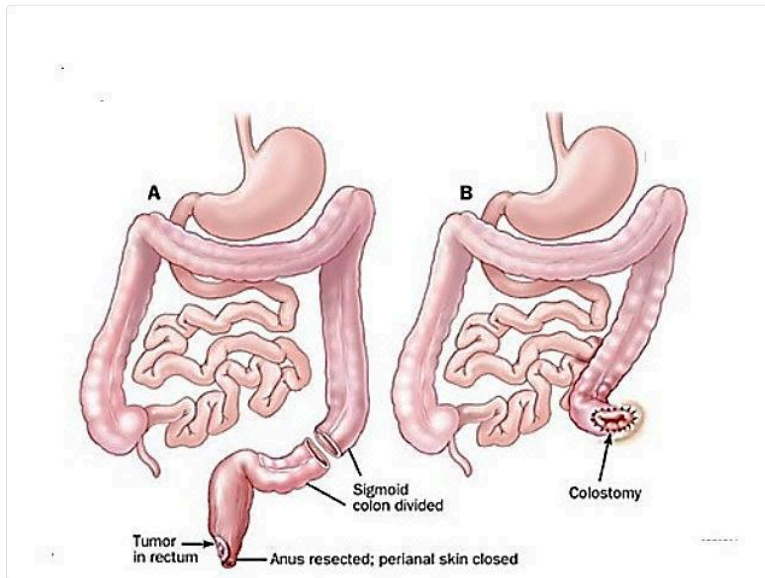


Figure 15.15 — Abdomino-Perineal Resection (APR)

B - Obstructed carcinoma: upper rectum : using staplers:

First stage:

Hartmann's procedure as before.

Second stage:

Reversal of Hartmann's (re-anastomosis and closure of colostomy)

N.B. indications of Total colectomy in cancer colon:

- 1- Done in multi- centric cancers
- 2- Done in Cancer complicating familial adenomatous polyposis

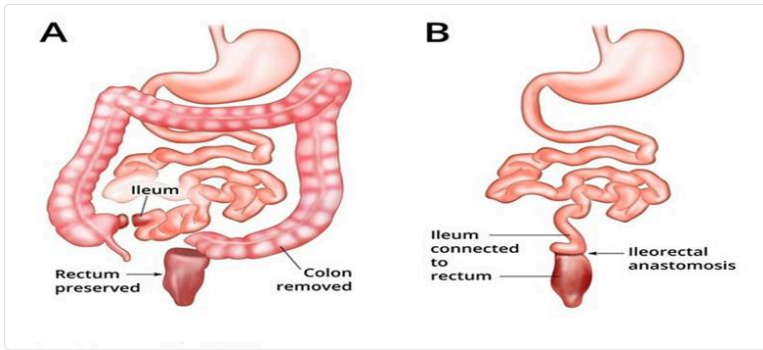


Figure 15.16 — Total colectomy and ileo-rectal anastomosis

Anorectal diseases



16



Anatomical points:

Venous drainage of rectum and anal canal:

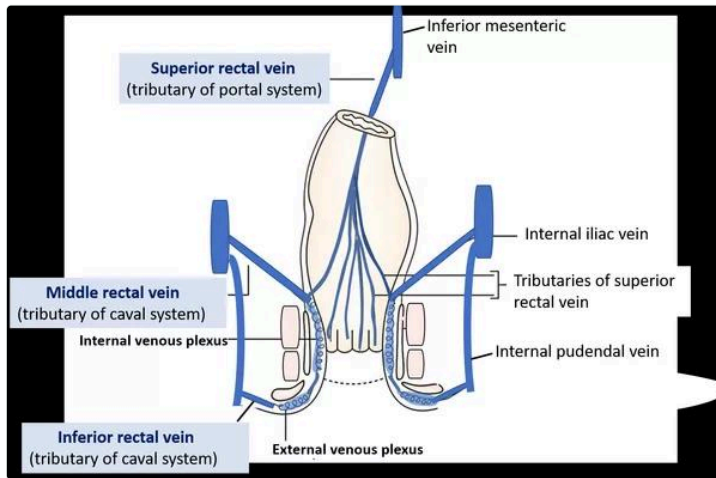


Figure 16.1 — Venous drainage of the rectum and anal canal

Hemorrhoids = piles

Haemorrhoides → **Haima** = blood **Roos** = flowing

Piles → **Pila** = ball

Definition:

- These are the dilated veins within the anal canal in the sub-epithelial region formed by radicals of Superior, Middle and Inferior rectal veins.
- Piles can be described as masses (“cushions”) of tissue within the anal canal that contain blood vessels and the surrounding connective tissue (hemorrhoidal cushions).

Classifications:

I- According to Location-

1. Internal Piles –It is covered with mucous membrane.

It arises from Internal Hemorrhoidal plexus & above dentate line.

It is insensitive (autonomic).

2. External piles – It is situated outside the anal orifice & is covered by skin. It arises from External Hemorrhoidal plexus & below dentate line – it is sensitive (spinal nerves)

3. Internal + External haemorrhoids.

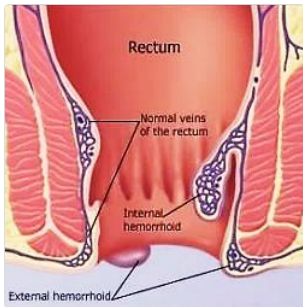


Figure 16.2 — Position of internal and external piles

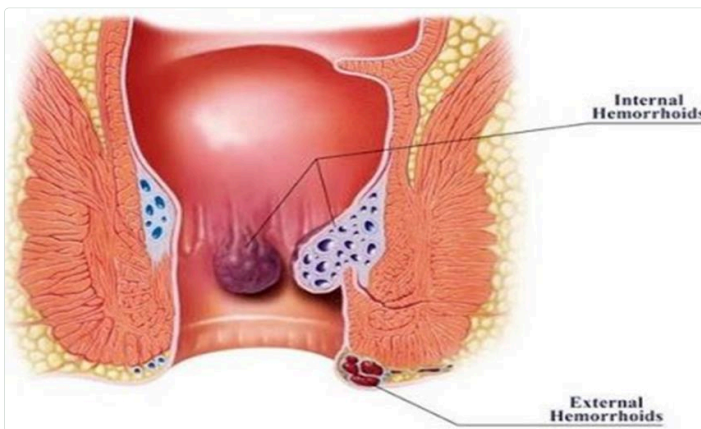


Figure 16.3 — Internal and external hemorrhoids

II- According to etiology-

1. **Primary** – Due to diets & habits
2. **Secondary** – Due to some other underlying disease

Primary piles:

predisposing factors:

- The haemorrhoidal veins are situated in anal sub-mucosa in longitudinal direction & does not have support of any other surrounding tissue
- Long term standing or sitting (gravity)
- Straining at defecation or micturition
- Chronic constipation
- Diarrhea → mucosal irritation and prolapsing down
- Low fiber diet prone to constipation

Secondary piles:

- Pregnancy: pressure of the gravid uterus on pelvic veins → venous stasis and congestion • labor → high anal pressure → prolapsed piles and anal fissure
 - Portal hypertension: dilated porto-systemic collaterals in lower rectum



Degrees of internal piles:

1- 1st-degree

Piles Project into anal lumen internally

2- 2nd-degree

Protrusion outside anal canal at defecation with spontaneous Reduction

3- 3rd-degree

Protrusion outside anal canal at defecation or straining needs digital repositioning

4-4th-degree:

Permanently prolapsed outside the anus = irreducible piles – prone for strangulation.

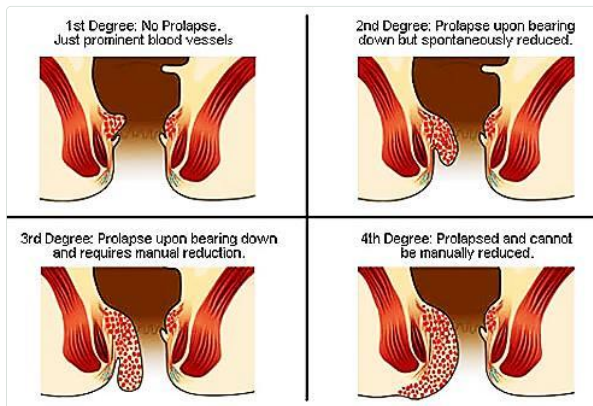


Figure 16.4 — Degrees of piles

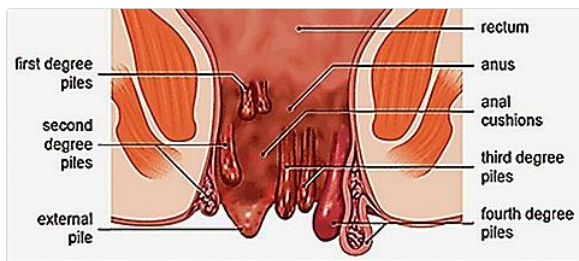


Figure 16.5 — Degrees of piles

Position of piles intra-anally (mother piles):

- Right anterior (11- o'clock)
- Right posterior (7- o'clock)
- Left lateral (3- o'clock)



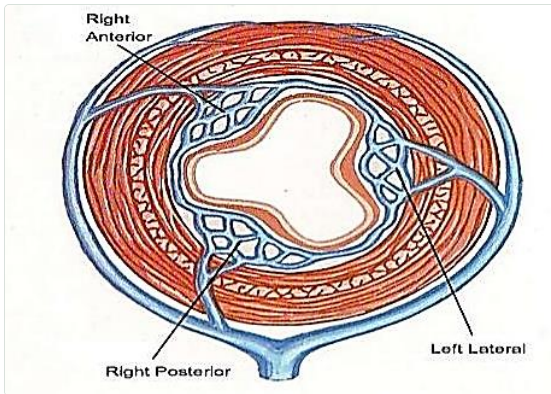


Figure 16.6 — Positions of internal piles

Clinical picture of internal piles:

- Bleeding (bright red) caused by hard stool and straining.
- Swelling / Prolapse of soft tissue (according to the degree)
- Straining / Pain / Discomfort
- Constipation
- Itching, Irritation and discharge
- Incomplete evacuation
- Anemia in recurrent prolonged cases

Anal Examination:

- 1- Position of the patient: Sim's = left lateral - Lithotomy
- 2 - Separate the buttocks apart gently - Focus the light source
Observe the peri-anal region anal verge
- 3- 3rd-degree = Inner red/purple mucosa & outer skin covered pile
- 4th-degree = Evident irreducible prolapsed pile
- External piles → appear like peri-anal hematoma at anal verge
- 4- P/R examination: detect piles, detect associated pathology as rectal mass
- 5- Proctoscope: to detect site and degrees of pile and associated pathology



Figure 16.7 — Bleeding third -degree piles



Figure 16.8 — External piles: source clinical image

External piles: (peri - anal hematoma – Thrombosed external pile):

- It is situated outside the anal orifice & is covered by skin.
- It arise from External Hemorrhoidal plexus & below dentate line
- it is sensitive
- Thrombosed external piles sometimes referred to as 5 days' disease
- It usually follow severe constipation and straining at defecation with rupture of small venules causing hematoma.
- Clinically:
 - Patient presented with painful peri-anal swelling
 - P/R: very tender, tense and bluish swelling at the anal verge about 1 or 2 cm



Figure 16.9 — Thrombosed external pile

Complications of internal piles:

- 1- Anemia: due to chronic blood loss
- 2- Strangulation in 4th degree piles

Treatment of piles

■ Grade I, II internal piles :

A - Medical treatment

- 1- Treatment of constipation and the Predisposing Factors
- 2- Venotonics to stop bleeding
- 3- Hemostatic agents
- 4- Decongestant local ointment

B - Out - patient procedures:

1- Sclero- therapy (induction of fibrosis of the pedicle) using phenol in almond oil

2- Rubber band ligation of the pile pedicle → ideal for grade II

The pile sloughs after 7 - 10 days

■ Grade III, IV:

■ Surgical management:

1- open hemorrhoidectomy:

Electro-cautery:

Dissection of the pile till its pedicle intra-anally with electro-cautery device with double ligation and division.

- 2- Stapling of hemorrhoids using stapler
- 3- LASER hemorrhoidectomy



- Treatment of external thrombosed piles:

1.Conservative:

local decongestant – venotonics – analgesics – laxatives

2.Surgical:

incision and drainage of the hematoma under local or regional anesthesia

Peri - anal abcess

Types:

- 1- subcutaneous peri-anal abcess
- 2- submucous abcess
- 3- Inter-sphincteric abcess
- 4- supra-levator abcess
- 5- ischio-rectal abcess → present in the ischio-rectal fossa

Causes and risk factors:

- 1- Blocked anal glands is the most important starting triggering factor for abcess formation
- 2- Chron's disease
- 3- Diabetics and other immune-compromized situations
- 4- Infected anal fissure
- 5- Diarrhea - constipation

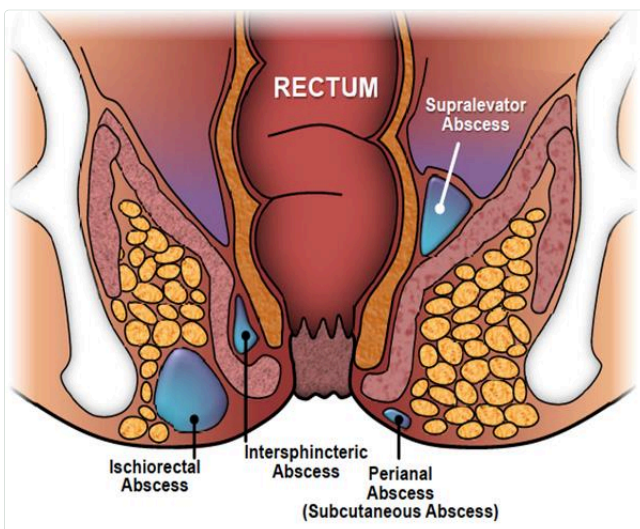


Figure 16.10 — Types of peri-anal abscesses



Clinical picture :

- fever, severe anal pain
- external tender peri-anal swelling is evident in 1 & 2
- tender swelling intra-anal in 3 & 4
- ischiorectal swelling tender red edematous in 5
- pus may come out from the anus in opened abcess
- sometimes the patient may get reflex difficult urination
- sometimes the patient needs examination under anaesthesia to diagnose higher abcesses



Figure 16.11 — Perianal abscess: source clinical view



Figure 16.12 — Perianal abscess

Treatment:

- Incision and Drainage under G.A. with proper irrigation with saline and anti-septic solutions.

Complications → peri-anal fistula – recurrent abcess



Peri - anal fistula

Anatomy :

1- The external sphincter muscle

- Is a striated muscle under voluntary control
- Formed of three components: subcutaneous, superficial, and deep muscle. • Its deep part is continuous with the puborectalis and forms the anorectal ring, which is palpable upon digital examination.

2- The internal sphincter muscle is a smooth muscle under autonomic control and is an extension of the circular muscle of the rectum.

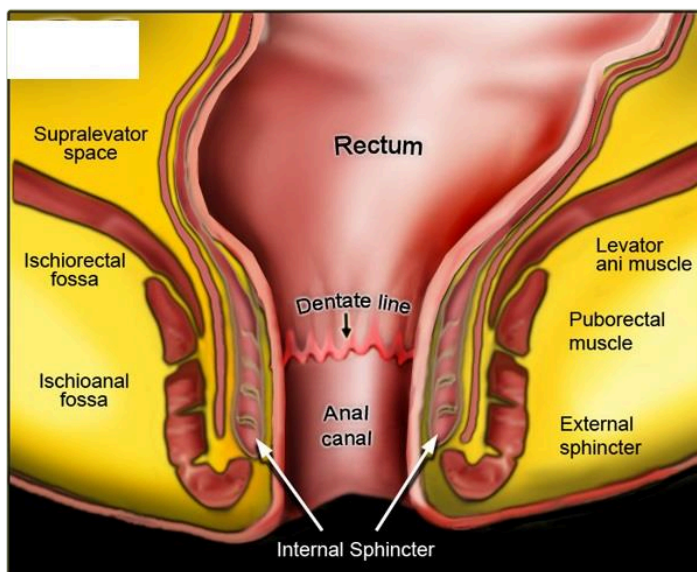


Figure 16.13 — Anal sphincter anatomy

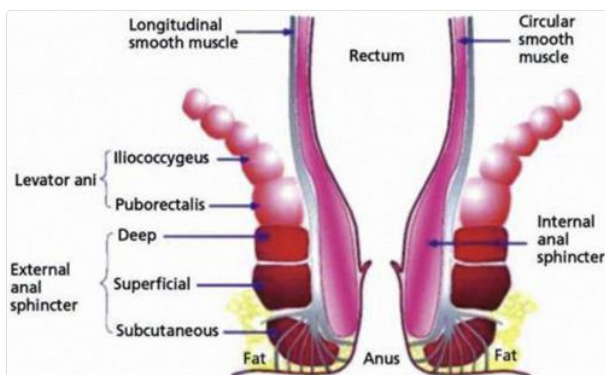


Figure 16.14 — Anatomy of anal sphincters



Etiology of anal fistula :

1 - previous anorectal abscess.

- There are 8 to 10 anal crypt glands at the level of the dentate line in the anal canal, arranged circumferentially.
- These glands penetrate the internal sphincter and end in the Inter-sphincteric plane.
- An infection begins in the anal canal glands and extends into the anal sphincters to cause an anorectal abscess.
- After surgical or spontaneous drainage in the perianal skin, A granulation tissulined tract is occasionally left behind, causing fistula symptoms.

2 - Crohn disease:

multiple and recurrent fistulae

3- Chronic Anal fissures

4- Actinomycoses, tuberculosis infections.

Classification of anal fistula

Parks Classification

The most commonly used classification for anal fistula. There are four types of fistulae that result from crypto-glandular infections:

1. Inter-sphincteric

2. Trans-sphincteric

3. Supra-sphincteric

4. Extra-sphincteric

1- An intersphincteric fistula:

a) It is the result of a perianal abscess b) **Common course** - It begins at the dentate line, then pass via the internal sphincter to the inter-sphincteric space between the internal and external anal sphincters, and finally terminates in the perianal skin c) Incidence – 70 % of all anal fistulas

2- A trans-sphincteric fistula:

a) **Common course** - It passes from the internal opening at the dentate line through the internal and external anal sphincters into the ischiorectal fossa and then terminates in the perianal skin b) Incidence – 25 % of all anal fistulas





3- A supra-sphincteric fistula:

a) Common course - It passes from the internal opening at the dentate line to the intersphincteric space, pass superiorly to above the puborectalis, and then curves downward lateral to the external anal sphincter into the ischiorectal fossa and finally to the peri-anal skin
b) Incidence - 5 % percent of all anal fistulas

4- An extra-sphincteric fistula:

a) **Common course** - It runs from the perianal skin via the ischiorectal fossa, tracking upward and through the levator ani muscles to the rectal wall, completely outside the sphincters, with or without connection to the dentate line
b) Incidence - 1% of all anal fistulas

Old classifications:

1- Low peri-anal fistula → internal opening below the anorectal ring

2- High peri-anal fistula → internal opening above the anorectal ring

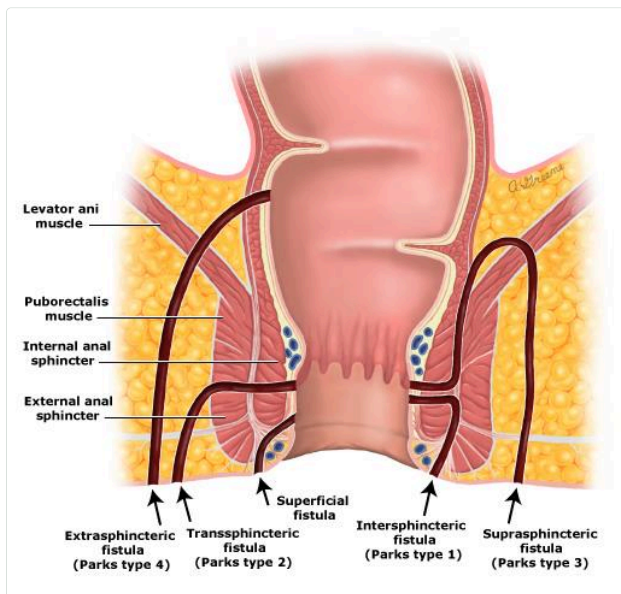


Figure 16.15 — Different types and anatomy of anal fistulae

Clinical presentation:

- History of previous pain, swelling, and spontaneous or surgical drainage of an anorectal abscess.
- History of Inflammatory bowel disease (Chron's)

Signs and symptoms of fistula:

- Perianal discharge



- Pain
- Swelling
- Bleeding
- Skin excoriation
- External opening on peri-anal skin

Examination:

1. Looking for an external opening that appears as an open sinus or elevation of granulation tissue.
2. Spontaneous discharge of pus or blood via the external opening may be apparent or expressible on digital rectal examination.



Figure 16.16 — Peri-anal fistula: external opening

3. Goodsall's law:

- An external opening opens in posteriorly has curved track that opens in the Midline
- An anterior opening has straight course and the internal opening to one side of the midline

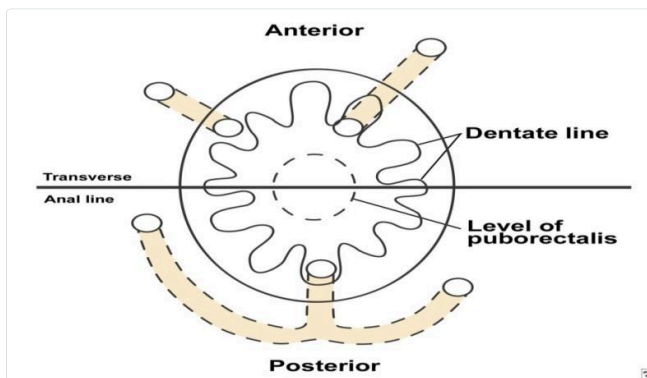


Figure 16.17 — Goodsall's law



4. Digital rectal examination (DRE):

- a) may reveal a fibrous tract or cord beneath the skin
 - b) determine the relationship between the anorectal ring and the position of the tract
 - c) The sphincter tone should be assessed
5. **Anoscopy** is usually required to identify the internal opening.
6. **Proctoscopy** is also indicated in the presence of rectal disease (eg, Crohn disease)

Investigations:

1- Fistulography:

A dye is injected through the external opening to outline the course of the fistula tract (of low values nowadays).

2 - Endoanal or endorectal ultrasonography

Help to define the sphincter anatomy and thereby help to differentiate inter-sphincteric from trans-sphincteric lesions.

3 - Magnetic resonance imaging Fistulogram : MRI

- The most accurate in detecting the fistulous tracts.
- Detects the relation of the track to the sphincters
- Detects the presence of peri-anal or peri-rectal abscesses

Management of peri - anal fistula:

Intraoperative considerations include the following:

Examine the patient under anesthesia to confirm the extent of the fistula

Identifying the internal opening to prevent recurrence is imperative by probing and

- hydrogen peroxide injection through the external opening

Fistulotomy:

- 1- The laying-open technique (fistulotomy) is useful for 85-95% of primary fistulas (ie, submucosal, inter-sphincteric, and low trans-sphincteric)
- 2- A probe is passed into the tract through the external and internal openings.
- 3- The overlying skin, subcutaneous tissue, and internal sphincter muscle are divided with electrocautery, and the entire fibrous tract is opened and curetted.

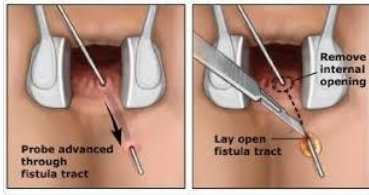


Figure 16.18 — Fistulotomy

Seton Placement:

This technique is useful in patients with the following conditions:

1. **Complex fistulas** (i.e, high trans-sphincteric, supras-phincteric, extra-sphincteric) or multiple fistulas
2. **Recurrent fistulas** after previous fistulotomy
3. **Poor sphincter tone:**
 - The purposes of setons are to drain, to promote fibrosis, and to cut through the fistula. • Setons can be made from large silk suture.
 - The seton is tightened on subsequent visits until it is pulled through over 6-8 weeks.

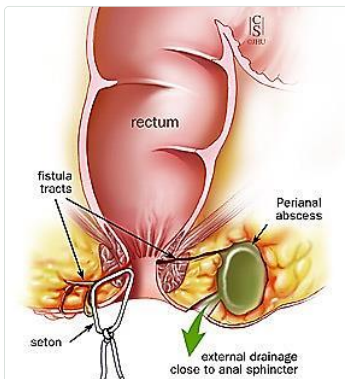


Figure 16.19 — Seton placement: anatomy

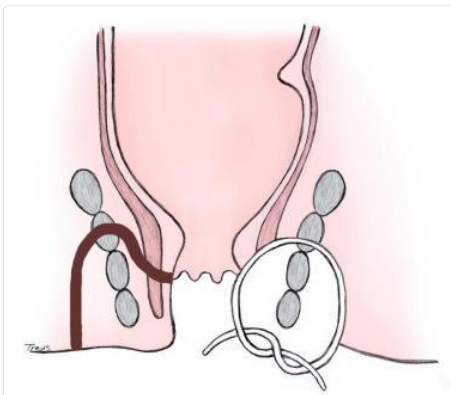


Figure 16.20 — Seton placement



Other lines :

1. Proximal colostomy temporary in high and complex anal fistula
2. Local fibrin glue injection inside the fistulous tract.
3. Anal fistula plug placed inside the tract to induce fibrosis

Anal fissure

Anal fissures are tears in the mucosa of anal canal.

Types:

They may be **acute or chronic**

Causes:

1- trauma to the anal mucosa → fissure in midline posterior

- Hard stool
- Straining in chronic constipation
- Diarrhea
- Instruments (thermometer, enema tip, endoscope, or ultrasound probe)
- During labour

2- Other causes:

fissure is placed in abnormal sites

- Chron's disease
- Syphilis
- Anal cancer

Site of anal fissure:

In the lower anal canal in the **midline posterior** at the **crypta magna** of the mucosa. **why ?**

a) because of the configuration of the muscle that surrounds the anus are oval shaped with better support at the sides and weakest posteriorly b) posterior mucosa most commonly exposed to trauma c) posterior part of the anal canal has less blood flowing to it than the other parts of the anal canal preventing fissures from healing





Patho-physiology of anal fissure:

- Trauma to the anal mucosa → tear → infection → fissure → lead to spasm of the internal anal sphincter → failure to relax which may lead to pulling of the fissure edges apart → **gapping and failure of healing**.

Clinical picture of anal fissure:

A- symptoms:

1- Anal pain:

- a) it is more severe in acute than chronic fissure
- b) increase with defecation and relieved after defecation
- c) may lead to constipation due to spasm of internal sphincters
- d) may lead to dysuria

2- Constipation due to spasm or stenosis

3- Anal pruritis and discharge

4- Bleeding per-rectum

B- Anal examination:

■ **An acute anal fissure** looks like a linear tear with hyperemic edematous edges + tight anal spasm.

- **A chronic anal fissure** frequently is associated with a **triad of findings** that includes :

1- A tag of skin at the edge of the anus (**sentinel pile**)

2- Thickened edges of the fissure with muscle fibers of the internal sphincter visible at the base of the fissure, (button hole)

3- Enlarged **anal papilla** at the upper end of the fissure inside the anal canal.

c- Anoscope or proctoscopy for associated lesions as piles or masses



Figure 16.21 — Anal fissure: source clinical view





Figure 16.22 — Anal fissure

General treatment for anal fissures

- In acute fissures, medical (non-operative) therapy is successful in the majority of patients.
 - Of acute fissures, 80% to 90% will heal with conservative measures as compared with chronic (recurrent) fissures, which show only a 40% rate of healing.
- 1- Excess oral fluids
 - 2- Treatment of constipation
 - 3- High fiber diet
 - 4- Bulk forming stools
 - 5- Avoid spicy foods
 - 6- Avoid sharp foods
 - 7- Local anaesthetic ointment
 - 8- Local decongestant (ointment and suppository)
 - 9- Nitroglycerine preparation locally
 - 10- Botulinum toxin locally to relief the spasm of the sphincter and allows the healing of the fissure
- **Surgery:**
 - In failure of medical treatment
 - **It includes partial lateral internal sphincterotomy** to relief the internal sphincter spasm allowing healing of the fissure

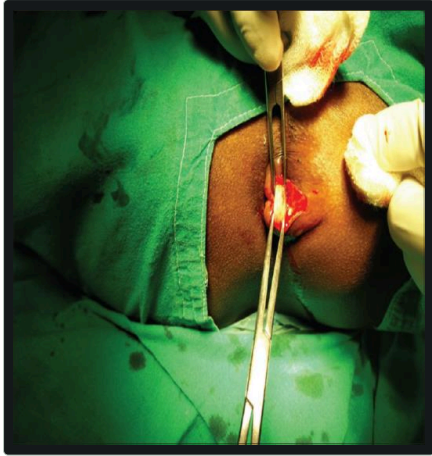


Figure 16.23 — Lateral sphincterotomy: operative view

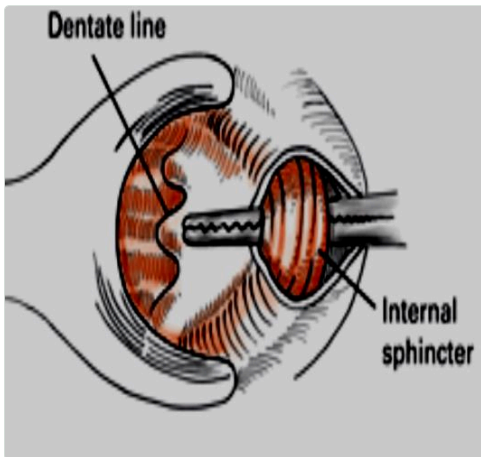


Figure 16.24 — Partial internal lateral sphincterotomy

Pilonidal sinus

Pilonidal means a 'nest of hairs'

A pilonidal sinus (PNS) is a track in the natal cleft between the buttocks over the sacrococcygeal area

Types:

1- congenital 2- Acquired

Risk factors include:

- Male gender
- **Extensive hairy individuals** (presence of hair within the natal cleft).
- **Sitting occupations**



- Presence of a deep natal cleft, • **Family history** is seen in 38% of patients with pilonidal disease.
- Obesity is a risk factor for recurrent disease (deep natal cleft)

Patho - physiology

(1) The invader , hair

(2) The force , causing hair penetration; occupation

(3) The nature of the skin, extensive hairy

a- Upon prolonged sitting, hair follicles can break and open a pit in the skin b- Debris may collect in this pit, followed by development of a sinus with a short tract c- Hair penetrates into the subcutaneous tissues through dilated hair follicles → initiates inflammatory process in the subcutaneous tissue d- Pilonidal abscess will form e- Improper treatment leads to PNS f- A PNS usually contains hair, dirt, and debris

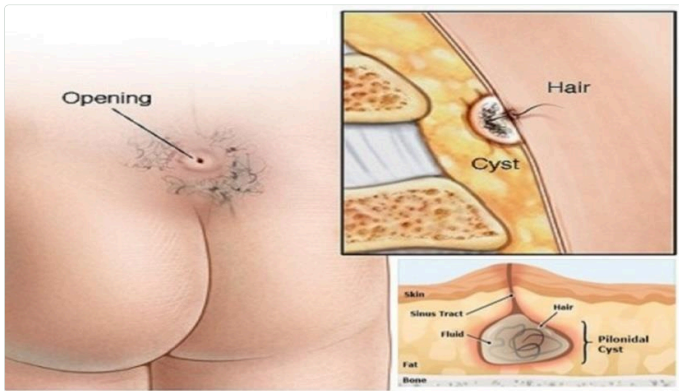


Figure 16.25 — Pilonidal sinus

Clinical picture:

A - Acute pilonidal abscess

- Typical findings of an abscess, including severe pain redness, warmth, local tenderness, and fluctuation.

b- Chronic cases

Patients may present with intermittent swelling and drainage, including purulent, mucoid, or bloody fluid from this area.

- Chronic pilonidal disease often manifests as recurrent or persistent discharge from a sinus at sacrococcygeal region.
- Loose hair may be seen projecting from the site of sinus





Treatment:

A- Pilonidal abscess:

incision and drainage under G.A

B- Pilonidal sinus

1- Non-surgical treatment

- 80% phenol is injected into the sinus, left there for 1 minute, and then expressed out of the cavity. The sinus is then curetted.
- Fibrin glue placed inside the sinus track allows healing in some cases

2- Surgical treatment:

Surgical options for management of a chronic pilonidal sinus include the following: a) **Excision and lying open of the sinus tract** → mainly used in markedly infected sinuses followed by regular dressings with healing by 2nd intension.

The average time for wound healing to occur is about 6 weeks

b) Excision with primary closure

Used in low infected sinuses

Wide and deep excision to the sacrum

Then closure of the wound leaving drain inside

Healing occurs by primary intension

c) Using flaps techniques:

used commonly in recurrent cases d) After management the hair at the lower back and sacro-coccygeal area should be permanently shaved to avoid recurrence

N.B. D.D. Of painful anal conditions:

1. Acute anal abscess
2. Peri-anal hematoma
3. Anal fissure
4. Thrombosed or strangulated prolapsed piles
5. Anal carcinoma
6. Proctalgia
7. parasitic infestations



Hepatology and Hepatic Surgery



17



Anatomy of the Liver

- **Surgically:** the liver divided into 8 segments (4 for each lobe)

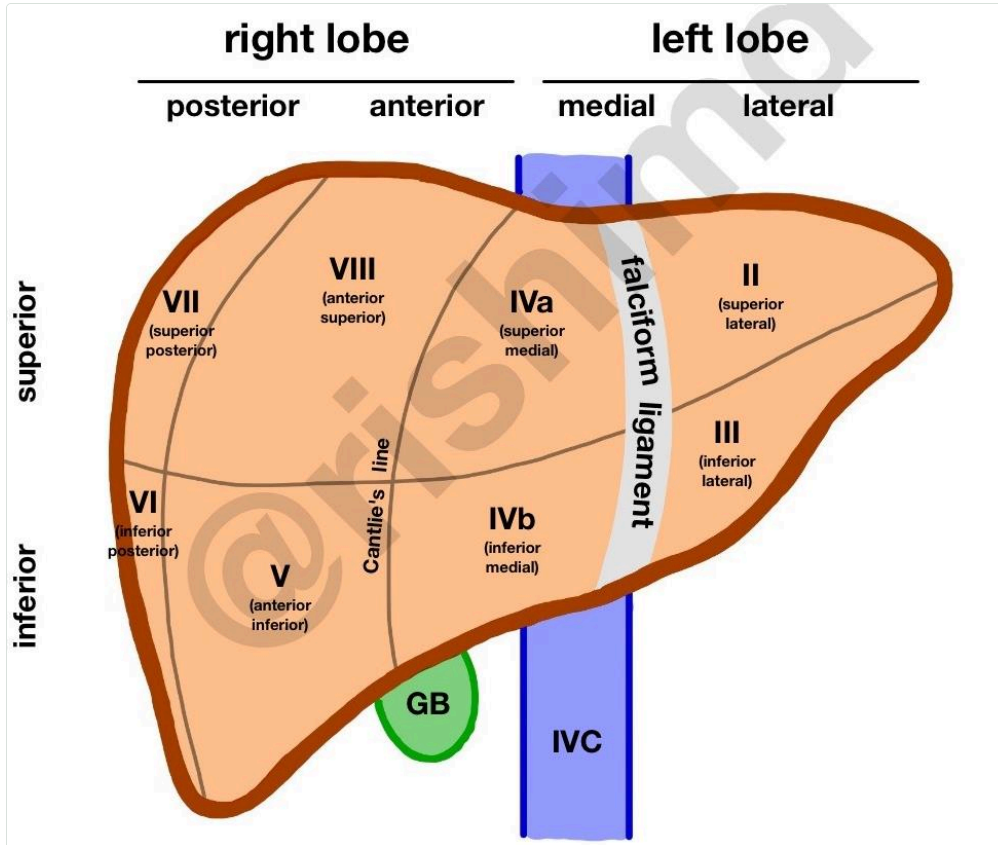


Figure 17.1 — Segments of the liver

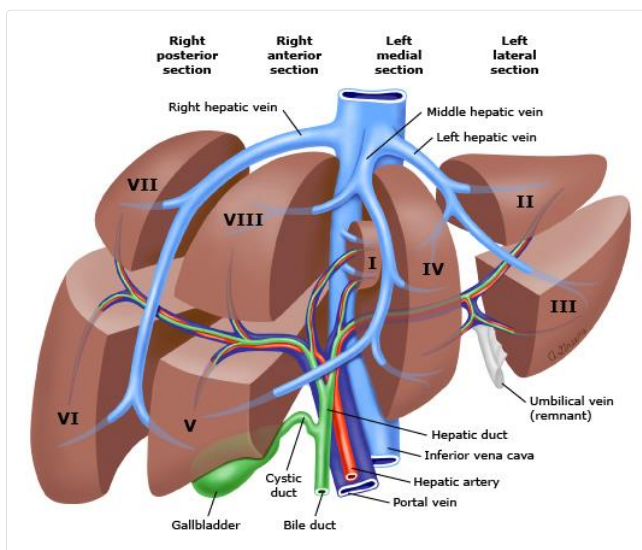


Figure 17.2 — Segments of the liver



Blood supply of the liver:

The liver has rich blood supply

1) Portal vein blood:

through the **portal vein** (portal circulation) = 80%

2) Systemic blood:

through the **common hepatic artery** = 20%

Branch of the celiac trunk (pure oxygenated blood)

3) The blood enters the liver passes to the portal vein and hepatic artery branches in the portal tracts → to the hepatic sinusoids between the hepatocytes then to the central veins then collected to emerges through the hepatic veins.

4) The blood leaves the liver through the 3 hepatic veins to the IVC (systemic)

5) The arrangement of the structures at the free border of lesser omentum:

- Common bile duct (ant. to right)
- Common hepatic artery (ant. to left)
- Portal vein (most posterior)

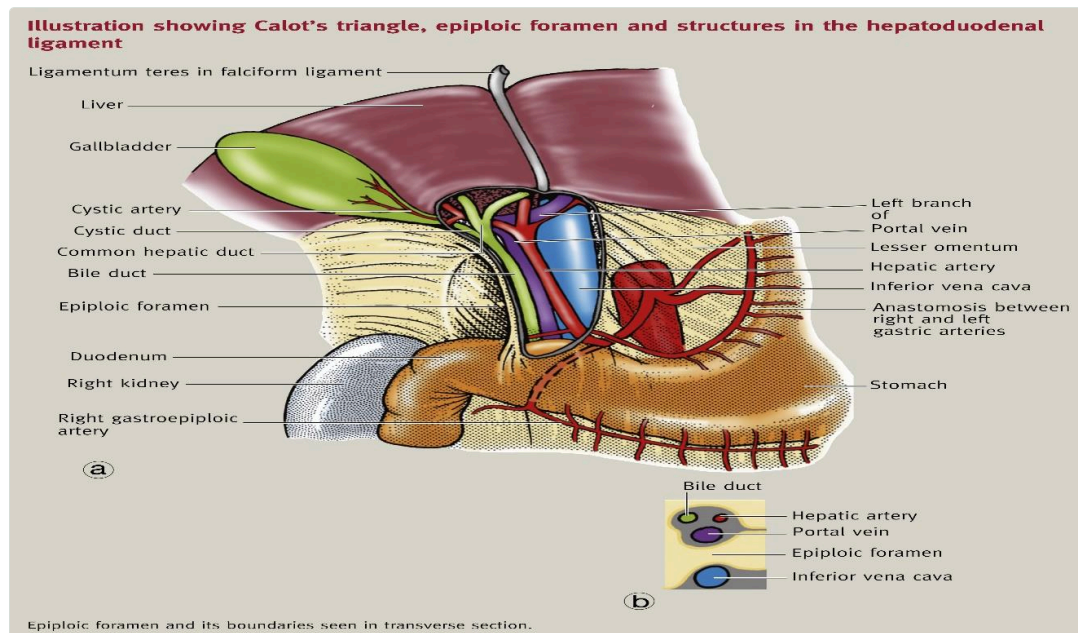


Figure 17.3 — Hepatoduodenal structures

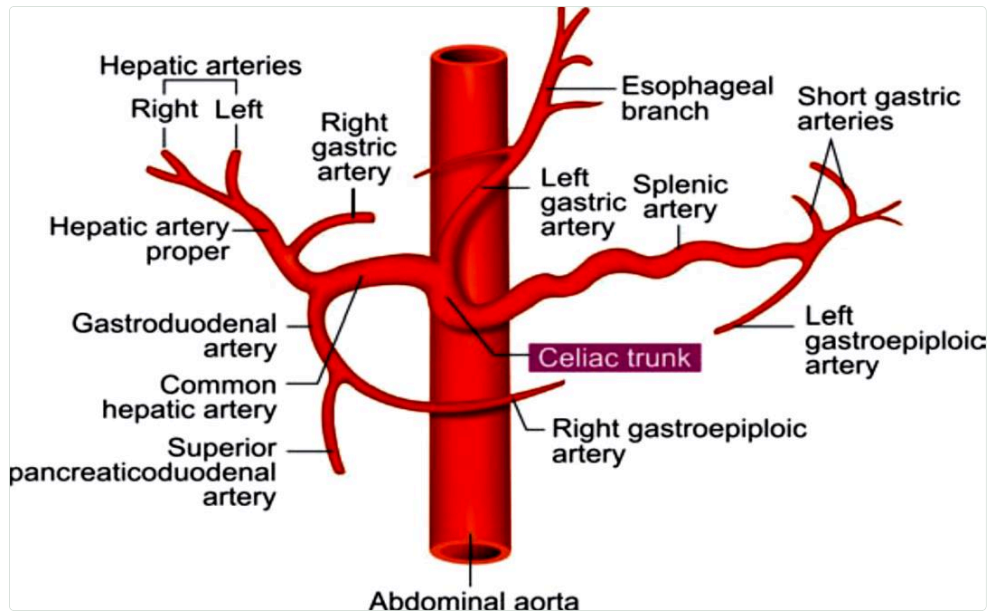


Figure 17.4 — Branches of the celiac trunk

Branches of the celiac trunk

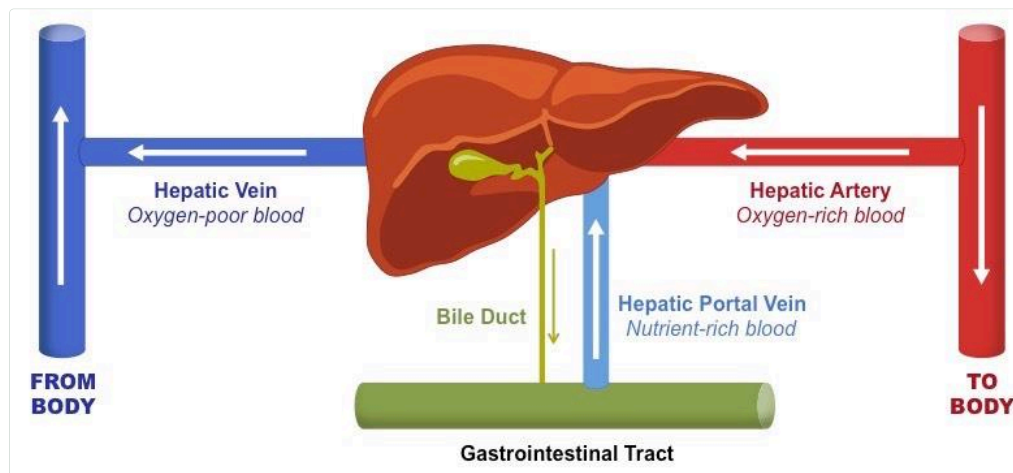


Figure 17.5 — Hepatic blood flow

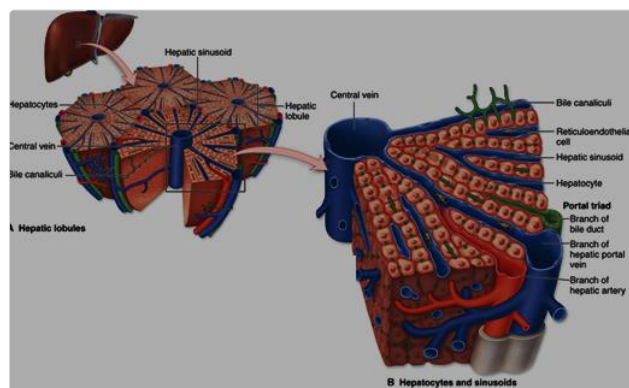


Figure 17.6 — Hepatic microcirculation



The hepatic lobule : the hepatic lobule is the basic unit of the liver. In the centre, there is a central tributary of the hepatic vein and at the periphery, the portal tracts containing branches of the portal vein, hepatic artery and the bile duct. Cords of liver cells and sinusoids extend between these two systems. The portal tracts are surrounded by a limiting plate of liver cells.

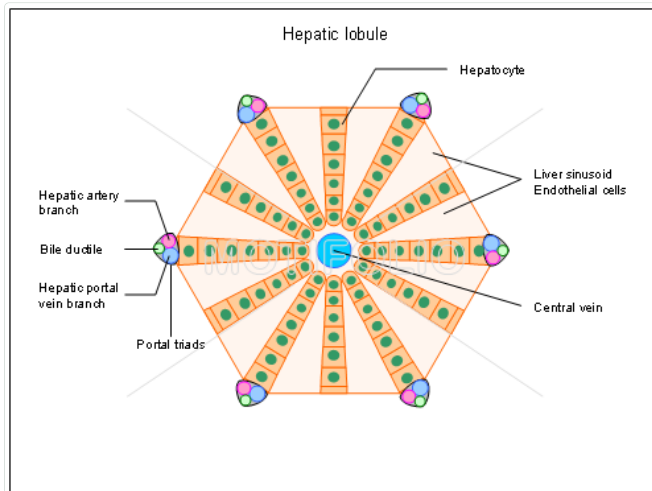


Figure 17.7 — Diagram of a hepatic lobule

The acinus is the functional hepatic unit. This consists of parenchyma supplied by the smallest portal tracts containing portal vein radicals, hepatic arterioles and bile ductules. The hepatocytes near this triad (zone 1) are well supplied with oxygenated blood and are more resistant to damage than the cells nearer the terminal hepatic (central) veins (zone 3).

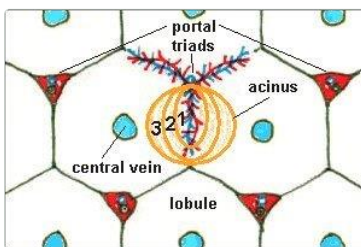


Figure 17.8 — Diagram of an acinus. Zones 1, 2 and 3 represent areas supplied by blood, with zone 1 being best oxygenated.

The sinusoids lack a basement membrane and are loosely surrounded by fenestrated endothelial cells and Kupffer cells (phagocytic cells). Sinusoids are separated by plates of liver cells (hepatocytes). The subendothelial space between the sinusoids and hepatocytes is the space of Disse, which contains a matrix of basement membrane and stellate cells. Activated stellate cells produce extracellular matrix components including collagen and participate in the fibrosis of the liver.

SYMPTOMATOLOGY AND INVESTIGATIONS OF LIVER DISEASES



18

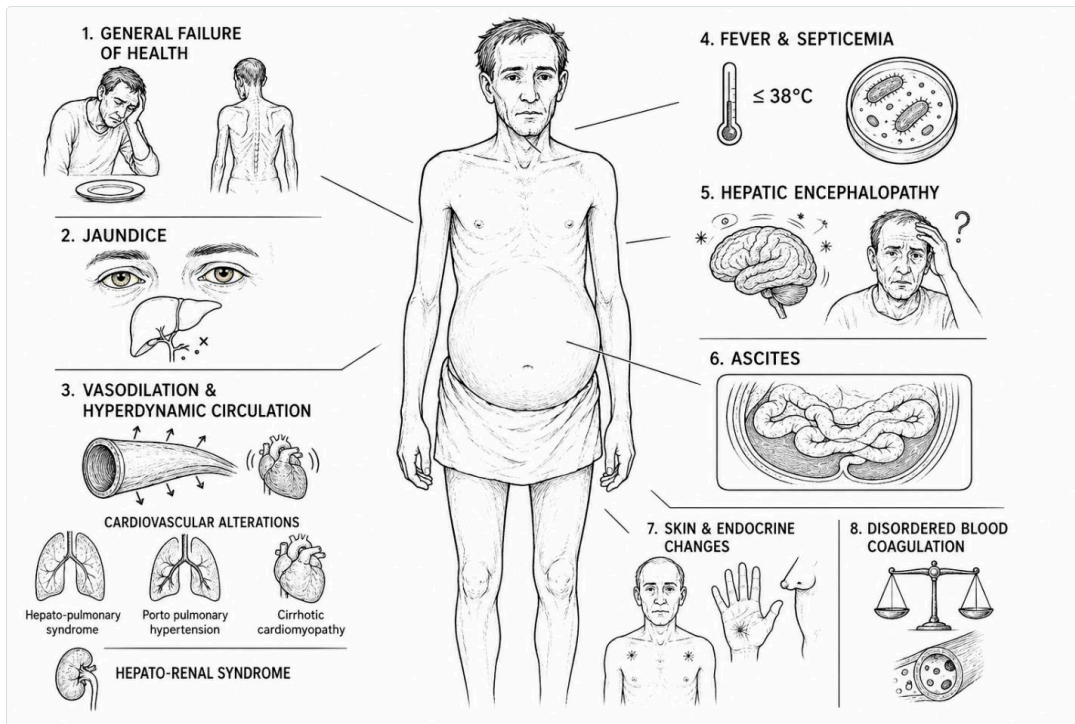


Figure 18.1 — Physical signs of liver disease

General failure of health

- The most conspicuous feature is easy fatigability.
- Muscle wasting, anorexia and malnutrition.

Jaundice

- due to failure of the liver cells to metabolize bilirubin.

Vasodilation and hyperdynamic circulation

- Cardiovascular Alterations:
 - Hepato-pulmonary syndrome.
 - Porto pulmonary hypertension.
 - Cirrhotic cardiomyopathy.
- Hepato-renal syndrome.

Fever and septicemia

- Patients with decompensated cirrhosis show a continuous low-grade fever which rarely exceeds 38°C.
- Septicemia is frequent in terminal hepato-cellular failure.



Hepatic Encephalopathy

- Neuropsychiatric abnormalities occurs in patients with acute or chronic liver disease.

Ascites

is free fluid within the peritoneal cavity.

Skin and endocrine changes.

Disordered blood coagulation.

INVESTIGATIONS OF LIVER DISEASES

Blood tests

Liver function tests :

- Serum albumin:
 - Is the most important serum protein synthesized by the liver.
 - It has a long half-life (21 days).
 - Not a good indicator of acute liver injury, decreased only in chronic liver diseases.
 - Total protein in chronic liver diseases is not of diagnostic value (may be $\uparrow$ or $\downarrow$ or normal).
 - Prothrombin time (PT): Prolongation of PT in liver disease may be due to impaired synthesis of clotting factors or impaired absorption of vitamin K.

Liver biochemistry :

- Bilirubin : increased serum bilirubin in liver disease.
 - Aminotransferases :
 - These enzymes are present in the hepatocytes and thus their elevations indicate liver cell damage.
 - Alanine aminotransferase (ALT) more specific to the liver so that a rise only occurs with liver disease.
 - Aspartate aminotransferase (AST) is also present in heart, muscle, kidney and brain. High levels are seen in hepatic necrosis, myocardial infarction, muscle injury.

Marked elevation of ALT and AST 10-100 times in acute viral hepatitis.

Mild to moderate elevation of ALT and AST (2-5 folds) in liver cirrhosis.



Mild to moderate elevation with ALT/AST ratio >1 in chronic active hepatitis.

- Alkaline phosphatase (ALP) and gamma glutamyl transferase (GGT):
- Raised in obstructive jaundice.
- Slight to moderate elevation occur in hepatitis and cirrhosis.
- GGT elevated in alcoholic even with normal liver.
- GGT is more specific for liver disease than ALP.

Additional blood investigations :

- Viral markers : including hepatitis A, B, C, D, and E.
 - Biochemical tests :
 - α -Fetoprotein.
 - Increased in hepatocellular carcinoma , slightly raised in patients with hepatitis and chronic liver disease.
 - α 1Anti-trypsin deficiency can produce cirrhosis.
 - Raised urinary copper, low serum copper and ceruloplasmin occur in Wilson's disease

Immunological tests :

- Serum immunoglobulins :
 - In primary biliary cirrhosis, raised IgM.
 - Autoimmune hepatitis , raised IgG.
 - In liver cirrhosis IgA is mainly increased.
 - In chronic hepatitis there is a main increase in IgG.
 - Serum autoantibodies:
 - Anti-mitochondrial antibodies (AMA's) found in 95% of patients with primary biliary cirrhosis (PBC), and are also found in autoimmune hepatitis.
 - Anti-nuclear, anti-smooth muscle, liver/kidney microsomal antibodies increased in autoimmune hepatitis.
 - Anti-nuclear cytoplasmic antibodies (ANCA)
 - found in primary sclerosing cholangitis.

Genetic analysis : e.g. haemochromatosis “HFE gene”.

Imaging techniques

Ultrasound examination (US)

- The first imaging modality used to investigate liver disease.
- non-invasive, safe cheap technique.
- The normal liver appears as a relatively homogeneous structure.



Figure 18.2 — Liver ultrasound

- **Color Doppler** will demonstrate vascularity within a lesion.



Figure 18.3 — Liver ultrasonography

Computed tomography (CT)

- **Triphasic CT** is the investigation of choice for the diagnosis of hepatocellular carcinoma.



Figure 18.4 — Computed tomography of the liver



Figure 18.5 — Abdominal CT

Magnetic resonance imaging (MRI)

- MRI is the most sensitive investigation of focal liver disease
- **Magnetic resonance cholangiopancreatography (MRCP)** is usually the next test if a biliary abnormality is present on US examination.

Plain X-ray:

rarely used but may show gallstones or pancreatic calcification

Fibroscan:

(Elastography)



- is a special ultrasound technology that measures liver stiffness (hardness) which reflect liver fibrosis , also it measures fatty changes in the liver (sensitivity and specificity 80-95% compared to liver biopsy).

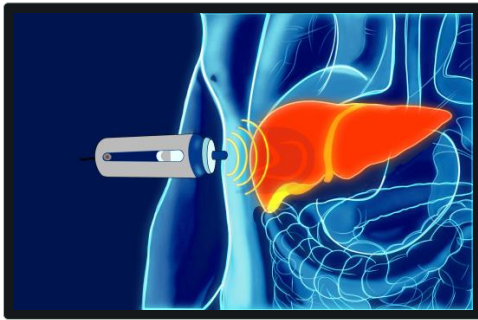


Figure 18.6 — FibroScan



Figure 18.7 — FibroScan

Endoscopy

Upper GI endoscopy

- Is used for diagnosis and treatment of varices, detection of portal hypertensive gastropathy, associated lesions such as peptic ulcers.

Endoscopic Retrograde Cholangiopancreatography (ERCP)

This technique outlines the biliary and pancreatic ducts. It involves the passage of an endoscope into the second part of the duodenum and cannulation of the ampulla.

Contrast is injected into both systems and the patient is screened radiologically.





Figure 18.8 — Endoscopic retrograde cholangiopancreatography

Therapeutic ERCP involves the following:

- Removal of common bile duct stones.
- Drainage of the biliary system by a stent through an obstruction.

Endoscopic ultrasound (EUS)

A less invasive method for bile duct imaging. EUS-guided fine-needle aspiration of tumors e.g. pancreatic tumors.

Percutaneous transhepatic cholangiography (PTC) : A fine needle is passed into the liver, contrast is injected until a biliary radicle is identified PTC is performed if

ERCP fails

Liver biopsy :

Histological examination of the liver is valuable in the differential diagnosis of diffuse or localized parenchymal disease.

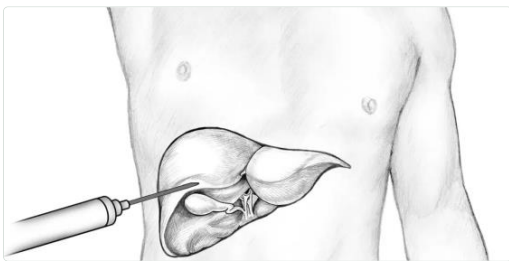


Figure 18.9 — Liver biopsy

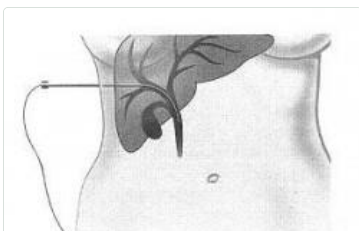


Figure 18.10 — Percutaneous transhepatic cholangiography





Laboratory diagnostic findings of liver diseases

The most important liver function tests include the following:

1-Albumin and other proteins:

Albumin Is the most important serum protein synthesized by the liver. It has a long half-life (21 days) and is therefore not a good indicator of acute liver injury. Furthermore, there is a high reserve of hepatic albumin synthesis and thus albumin is decreased only in chronic liver diseases with severe cells destruction.

As regards immunoglobulins, there is a polyclonal increase of gamma globulin (due to ↓clearance or ↑synthesis) in chronic liver disease with reversed albumin/globulin ratio. In liver cirrhosis IgA is mainly ↑and to a less extent other immunoglobulins. In chronic hepatitis there is a main ↑ in IgG. Total protein in chronic liver diseases is not of diagnostic value (may be ↑ or ↓ or normal).

2- Liver enzymes:

1- Aminotransferases:

They include alanine aminotransferase (ALT or SGPT) and aspartate aminotransferase (AST or SGOT). These enzymes are present in the hepatocytes and thus their elevations indicate liver cell damage. ALT is found primarily in the liver whereas AST is found in many other tissues including heart, muscles, and

RBCs. Thus, ALT measurement is more specific for liver diseases than AST. In hepatocytes ALT is found exclusively in cytoplasm while AST exists in both cytoplasm and mitochondria. In acute viral hepatitis, there is marked elevation of ALT and AST

(10-100 times). The rise of ALT is usually more than AST (ALT/AST ratio >1) because more damage of cytoplasm and slower clearance of ALT. Serum AST has a shorter half- life (17 hour) than ALT (47 hour). Hence, the serum ALT level takes a longer time to normalize than the AST after the injury has subsided. In liver cirrhosis, there is mild to moderate elevation of ALT and AST (2-5 folds). The rise of AST is more than ALT

(ALT/AST ratio is <1) because the damage affecting the mitochondria of liver cells with release of more AST. In chronic active hepatitis, there is also mild to moderate elevation of ALT and AST with ALT/AST ratio >1.



II-Alkaline phosphatase (ALP) and gamma glutamyl transferase (GGT): These enzymes are found in biliary tract epithelium. In cholestasis (obstructive jaundice), there is marked elevation of these enzymes (more than 3 folds). This marked rise is due to increase enzyme synthesis by cells lining biliary canaliculi. Slight to moderate elevation (1-2-fold) occur in patients with other liver diseases as hepatitis and cirrhosis. GGT is more specific for liver disease than ALP. The latter is also elevated in non-hepatic disorders as bone diseases. GGT also provides a more sensitive index of cholestasis than ALP. Elevated levels of GGT are also seen in alcoholic even with normal liver.

3-Bilirubin:

Bilirubin is the end product of heme degradation. Heme oxygenase oxidizes heme to biliverdin, and biliverdin reductase reduces biliverdin to bilirubin.

Hepatocellular processing of bilirubin involves: Carrier mediated uptake by hepatic cells, Conjugation to glucuronic acid and Excretion into bile. The gut bacterial deconjugate most bilirubin to colorless urobilinogen. The urobilinogen is largely excreted in feces as stercobilinogen and stercobilin. 10-20% of urobilinogen are reabsorbed in the ileum and colon and escape enterohepatic circulation and pass to the kidney to be excreted in urine.

Unconjugated (indirect) bilirubin is insoluble in water and is tightly complexed to serum albumin and not excreted in urine. Conjugated bilirubin (direct) is water soluble and only loosely bound to albumin and excreted in urine. Jaundice occurs when the equilibrium between production and clearance of bilirubin is disturbed

A-Unconjugated hyperbilirubinemia (unconjugated bilirubin is usually > 80%): it is due to:

I-Overproduction as in hemolytic anemia (hemolytic jaundice)

II-Decrease uptake as in Gilbert syndrome. This syndrome is relatively common and has a benign course. It results from defect of hepatic uptake and/or decrease of conjugation. Bilirubin level is increased but rarely exceeds 5 mg/dl. Liver enzymes are normal, with no evidence of hemolysis (normal reticulocyte count, no hemolytic anemia...).

III- Decrease conjugation as in Physiologic jaundice of newborn and Crigler-Najjar syndrome type. The severe form of this syndrome is characterized by complete absence of UGT (uridine diphosphate glucuronosyl transferase) which is the enzyme responsible for bilirubin conjugation. It might be fatal within 18 months of birth. Total Bilirubin is more than 20 mg/dl.

Laboratory findings of hemolytic jaundice:

- Increase bilirubin (mainly indirect)
- Increase urobilinogen in urine; Urine is normal in color (absent bilirubin)
- Stool is dark (increase stercobilin),



- Normal liver enzymes or only slight ↑ of AST (present in RBCs)
- Evidence of hemolysis: as ↑ reticulocytes

B-Conjugated hyperbilirubinemia (Conjugated bilirubin is usually > 50%):

It is due to:

A. **Impaired secretion** (Defect of bilirubin transfer into bile as in Dubin Johnson syndrome)

B.Cholestasis:

due to intrahepatic or extrahepatic obstruction.

Intrahepatic due to hepatic malignancy, primary biliary cirrhosis or other causes

Extrahepatic due to gallstones, tumors or other causes

Laboratory findings of obstructive jaundice(cholestasis):

- Increase bilirubin (mainly direct).
- Marked rise of GGT and ALP
- Normal ALT and AST except in late stages with secondary liver cell damage from bile
- Urine is dark (positive for bilirubin)
- Urobilinogen is decreased or absent
- Stool is clay color
- prothrombin time (PT) is prolonged
- Increase cholesterol.

C- Conjugated and unconjugated hyperbilirubinemia (hepatic jaundice): it is due to hepatic diseases as acute viral hepatitis. In hepatic jaundice both types of bilirubin are increased (direct and indirect). Liver enzymes are increased. Urine is dark due to the presence of direct bilirubin. Urobilinogen increases in early stage and then decrease

4- Prothrombin time (PT):

PT is dependent on normal hepatic synthesis of coagulation factors. Many of these factors are vitamin K dependent (factor II, VII, IX, X). Prolongation of PT in liver disease may be due to impaired synthesis of clotting factors or impaired absorption of vitamin K. In the latter PT is normalize after parenteral injection of vitamin K.

Acute viral hepatitis:

1 pre-icteric stage:

ALT is increased early and is more sensitive and specific than AST.

Bilirubin is still normal.



2- Icteric stage:

Marked elevation (10-100 times) of ALT and AST and ALT/AST ratio is usually >1. There is Increase both direct and indirect bilirubin (few cases are Anicteric, and bilirubin is normal). ALP and GGT showed mild elevations. Serum albumin is

normal. PT is prolonged in severe cases. Urine: is dark brown (Positive for bilirubin and bile salts).

Immunological (serological) Diagnosis of different types of viral hepatitis:

Hepatitis A:

is a single stranded RNA picornavirus. Incubation period is 2-6 week.

Usually benign, self-limited disease. Rarely causes fulminant hepatitis; and never causes chronic hepatitis. Mode of transmission is Fecal/oral. There are two types of Hepatitis A virus (HAV) antibodies:

First:

is IgM, which appears 2 to 4 weeks after infection and is present for 2 to 6 months.

Second:

is IgG, which appears a few weeks after IgM and is present for the rest of the person's life.

The presence of Hepatitis A virus IgM diagnoses recent infection. Testing HAV total antibodies (IgG plus IgM), measures both IgG and IgM antibodies, but does not differentiate between them. A positive HAV-Ab total test (without isolated testing for

IgM) indicates prior HAV exposure but does not differentiate current from previous

HAV infection. A negative HAV-Ab total test means there has been no prior exposure to the virus and the hepatitis A vaccination should be administered

Hepatitis B:

It is double stranded -DNA virus. Incubation period is 2-6 months.

Chronic hepatitis occurs in 5-10% of cases. It has increased risk of hepatocellular carcinoma. Mode of transmission is parenteral (as contaminated blood transfusion or contaminated syringes), sexual or perinatal from infected mother to their newborn.

Hepatitis B serological markers include:

Hepatitis B surface antigen (HBsAg): Is the earliest marker, appears in the late days of incubation period and denotes infectivity.



Hepatitis B surface antibody (HBsAb or anti-HBs): The presence of anti-HBs is generally interpreted as recovery (the acute infection is completely resolved) and immunity from hepatitis B virus infection. Anti-HBs also develops in a person who has been successfully vaccinated against hepatitis B. HBsAb does not develop in those with chronic hepatitis. Those who are immune due to natural infection will have HBsAb and HBeAb. Immunity due to vaccination will have only HBsAb.

Hepatitis B core antibody (HBcAb or anti-HBc):

Appears at the onset of symptoms in acute hepatitis B and persists for life. This is the only positive marker in window period (gap between disappearance of HBsAg and the appearance of HBsAb). Two types are measured in blood:

First the anti-HBc total antibodies test which detects both IgM and IgG antibodies, thus a positive result indicates acute, chronic, or past infection.

Second is IgM antibody to hepatitis B core antigen (HBcAb IgM or IgM anti-HBc): Its positivity indicates recent infection with hepatitis B virus. It is detected within 2 weeks after HBsAg and persists for 3-6 months. Hepatitis B core antigen is not detected in blood. The most important markers for diagnosis of recent infection are HBsAg and HBcAb IgM.

Hepatitis B envelop antigen (

HBeAg):

it is not found in every case. If positive this denotes rapid viral multiplication and high infectivity

Hepatitis B e antibody (HBeAb): is not found in every case. It appears after disappearance of HBeAg

HBV DNA by PCR:

hepatitis B virus DNA (HBV DNA) is now the most important and reliable marker for monitoring viral replication. The main advantages of PCR are its extreme sensitivity and quantitative PCR assays help to monitor viral replication and response to treatment. It can diagnose some cases of early acute HBV infection (before the appearance of HBsAg)

Hepatitis C:

is a single stranded -RNA virus. More than 50% of cases develop chronic hepatitis. Incubation period varies from 1-6 month. Mode of transmission is parenteral. There is increased risk of hepatocellular carcinoma. Diagnosis of hepatitis C is by detection of Hepatitis C Antibody (HCV-Ab) which is



usually positive within 2-5 months after infection. For early diagnosis, Hepatitis C RNA by PCR is done. It may be positive 1-2 weeks after infection. Quantitative PCR has a higher value for monitoring of therapy.

Hepatitis D (Delta virus):

It is RNA defective or incomplete virus. It cannot replicate without the presence of HBV. It is diagnosed by detection of HDV IgM antibody or by PCR. Transmission is parenteral or sexual. Incubation period varies from 4-8 weeks. Chronicity and hepatocellular carcinoma could occur.

Hepatitis E:

it is a single stranded RNA virus. Mode of transmission is fecal/oral. It never causes chronicity. Incubation period varies from 2-8 weeks. It has a high mortality rate among pregnant women reaching 20%. Diagnosis of recent infection is by detection of HEV-IgM antibody or by PCR.

Hepatitis G:

It is a single stranded RNA virus. Mode of transmission is parenteral.
Most cases recovered completely, and chronicity is very rare or absent

Laboratory findings of liver cirrhosis:

- Albumin is low and gamma globulin is high with reversed A/G ratio. Total protein is variable
- Mild to moderate elevation of AST and ALT (AST>ALT).
- ALP is normal or mildly elevated.
- GGT is mildly elevated (especially alcoholic).
- Bilirubin is usually normal or elevated in late stages.
- PT is prolonged and not corrected by Vit K
- Decreased platelets count (due to hypersplenism and or decreased thrombopoietin production from the liver). The latter is a glycoprotein hormone produced by the liver and kidney, which stimulates platelets production from megakaryocytes in the bone marrow.
- Serum protein electrophoresis (SPE): polyclonal increase of gamma globulin with beta gamma bridge (due to increase of IgA mainly). However, nowadays SPE is not commonly used for diagnosis of liver cirrhosis.
- Mild Increases in α -fetoprotein (AFP) concentrations are common in cirrhotic and chronic hepatitis patients, even in the absence of hepatocellular carcinoma (HCC). AFP is the most common tumor marker used for screening of HCC and is usually present at higher concentrations in HCC.

Chronic hepatitis:

Definition: chronic inflammation of liver cells for at least 6 months



It is commonly due to viral infection (HBV or HCV or HBV +HDV) or autoimmune or drug induced. Clinical and laboratory features of chronic hepatitis are also observed occasionally in patients with such hereditary/metabolic disorders as Wilson disease (copper overload).

Laboratory findings of chronic hepatitis

- Albumin is low and gamma globulin is high (IgG mainly).
- Mild to moderate elevation of ALT and AST (ALT>AST)
- Positive HBsAg, IgG anti-HBc and HBV DNA in chronic hepatitis caused by HBV. In some cases, HBeAg may be also positive.
- Positive HCV-Ab and HCV RNA in chronic hepatitis caused by HCV
- Auto antibodies may be found particularly in autoimmune type (ANA, ASMA, LKM). In chronic hepatitis due to viral infection, autoantibodies are uncommon or found at low titer
- Antinuclear antibody (ANA) and Anti smooth muscle antibody (ASMA) are commonly found in type 1 autoimmune hepatitis (often seen in adult particularly young women. It is the most common type)
- Anti-liver kidney microsome antibody type 1 (LKM 1) and anti-liver cytosol (anti-liver cytosol 1) are commonly found in type 2 autoimmune hepatitis (often affects pediatrics from 2-18 years. It is less common than type 1)

Diagnosis of Wilson disease:

- Slit-lamp examination of the eyes for Kayser-Fleischer rings
- Blood tests to measure levels of ceruloplasmin (a copper-carrying protein). Serum ceruloplasmin is usually low in Wilson disease.
- Measurement of copper excreted in the urine: In Wilson disease, 24-hour urinary copper excretion (normally, ≤ 30 mcg/day) is increased and usually > 100 mcg/day. If serum ceruloplasmin is low and urinary copper excretion is high, diagnosis is clear
- Liver biopsy: If the diagnosis is still unclear (indeterminate values for ceruloplasmin and urinary copper), the diagnosis is made by doing a liver biopsy to measure hepatic copper concentration. A diagnosis of Wilson disease is established if the hepatic copper concentration is ≥ 250 mcg/g dry weight.
- Serum copper levels are sometimes measured; however, they may be variable. In general, Total serum copper (which includes copper incorporated in ceruloplasmin) in Wilson disease is usually decreased in proportion to the decreased ceruloplasmin in the circulation but serum free non-ceruloplasmin bound copper concentration is elevated.

ACUTE HEPATITIS



19



Acute damage of the liver cells can be caused by several agents:

- Viral infections : Virus A, B , D , C , E , Epstein- Barr virus , Cytomegalovirus.
- Non viral infection : Toxoplasmosis – Leptospirosis – Q fever.
- Drugs : e. g .Paracetamol.
- Alcohol.
- Poisons : e.g. Amanita phalloides (mushrooms).
- Other : Pregnancy , circulatory insufficiency , Wilson's disease .

Acute Viral Hepatitis

Acute viral hepatitis is the most common cause of liver disease in the world. With its sequelae, Viral hepatitis is responsible for 1-2 million deaths annually.

Etiology:

- Enterically transmitted viruses: HAV, HEV. - Blood-borne viruses: HBV, HDV and HCV.

Hepatitis A virus (HAV):

Hepatitis A is the most common viral hepatitis occurring worldwide, often in epidemics. The disease is commonly seen in the autumn and affects children and young adults.

Spread of infection is mainly by the feco-oral route and arises from the ingestion of contaminated food.

HAV is a picornavirus, It replicates in the liver, and is excreted in bile and feces for about 2 weeks before the onset of clinical illness and for up to 7 days after. The disease is maximally infectious just before the onset of jaundice.

Investigations :

Liver biochemistry : in the prodromal stage: the serum bilirubin is usually normal, however, there is bilirubinuria and increased urinary urobilinogen. A raised serum AST or ALT, which can sometimes be very high, precedes the jaundice. In the icteric stage, the serum bilirubin reflects the level of jaundice.

Haematological tests :There is leucopenia with a relative lymphocytosis. The prothrombin time (PT) is prolonged in severe cases. The erythrocyte sedimentation rate (ESR) is raised.





. Viral markers:

see [Figure 19.1](#). An anti-HAV IgM means an acute infection, HAV IgG antibodies are common in the general population over the age of 50 years and has no role in diagnosis of acute infection, also in areas of high prevalence most children have antibodies following asymptomatic infection.

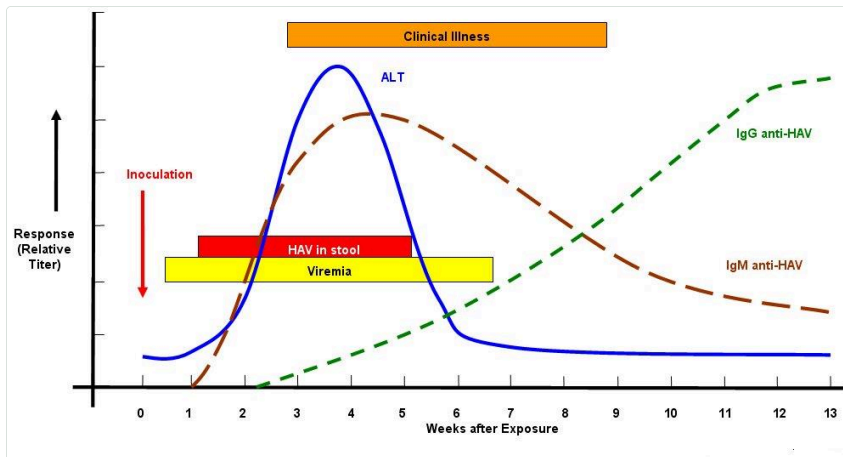


Figure 19.1 — Time course of the events and serological changes of acute Hepatitis A infection

Other tests : Further tests are not necessary in the presence of an IgM antibody, but liver biochemistry must be followed to establish a return to normal levels.

Differential diagnosis:

is from all other causes of jaundice, but in particular from other types of viral and drug-induced hepatitis.

Course and prognosis : The prognosis is excellent, with most patients making a complete recovery. The mortality in young adults is 0.1 % but it increases with age.

Death is due to fulminant hepatic failure. During convalescence, 5-15% of patients may have relapse of the hepatitis but this settles spontaneously. Occasionally, a more severe jaundice with cholestasis will run a prolonged course and is called 'cholestatic viral hepatitis'. Treatment is by reassurance. HAV hepatitis never progresses to chronic liver disease.

Treatment :

There is no specific treatment, and rest and dietary measures are unhelpful. Corticosteroids have no benefit. Admission

Prevention : Hygiene (e.g., hand washing) and sanitation (e.g., clean water sources) Boiling, chlorination of water



Vaccination -

Active immunization:

HAV vaccine (inactivated vaccine) , it is effective& safe, the protection starts after 15 days and lasts for more than 20 years. It is given in two doses followed by a booster at 6 or 12 months. It is recommended for travelers to endemic areas, patients with CLD and close contacts of a patient.

Post-exposure: Theoretically immunoglobulins can be given to contacts of patients with acute HAV infections

Hepatitis E virus (HEV) :

HEV is an RNA virus causing hepatitis clinically very similar to hepatitis A. It is enterically transmitted, usually by contaminated water. Epidemics have been seen in many developing countries and sporadically in developed countries, in patients who have had contacts with farm animals or travel abroad. It has mortality from fulminant hepatic failure of 1-2%, which rises to 20% in pregnant women especially in the second and third trimester. There is no carrier state and it does not progress to chronic liver disease except in some immunosuppressed patients in solid -organ transplant recipients. An ELISA for IgG and IgM anti-HEV is available for diagnosis. HEV RNA can

be detected in the serum or stools by PCR. Prevention and control depend on good sanitation and hygiene; a vaccine has been developed.

Hepatitis B Virus (HBV)

- It is a DNA, hepadna virus (see [Figure 19.2](#)) , there are 8 genotypes (A to H).
- It has an outer envelope: HBsAg.
- The HBV core contains: -HB core antigen (HBcAg) -HBe antigen (HBeAg) -Partially double-stranded DNA -DNA polymerase/reverse transcriptase enzyme.

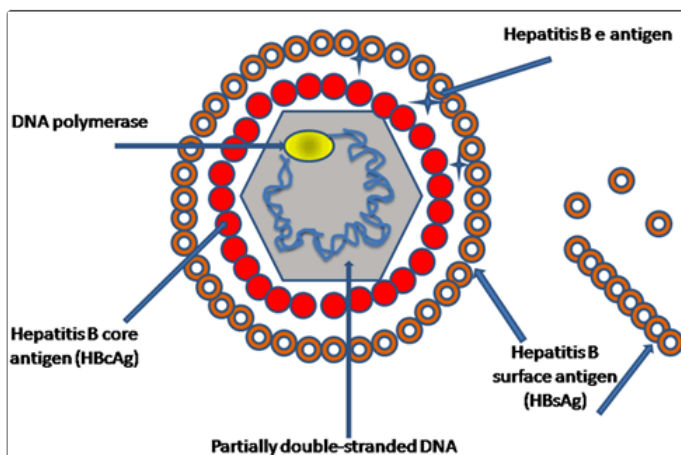


Figure 19.2 — Hepatitis B Virus

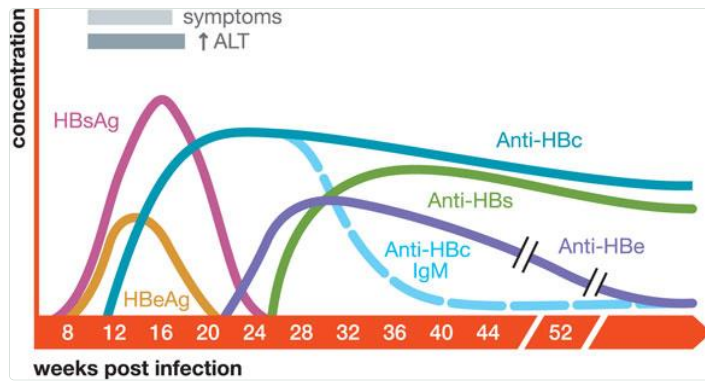


Figure 19.3 — Time course of the events and serological changes of Hepatitis B infection

- The transmission is by the parenteral, sexual and vertical route (at time of birth & close contact afterwards). Interfamily spread can also occurs by household articles .

- Incubation period: 2-6 months. The clinical attack may be mild or severe.

Serological diagnosis in the acute stage depends on IgM anti- HBV , HBVDNA and HBsAg (Figure 19.3) .After the attack, persistent infection may occur and can result in chronicity and cirrhosis and hepatocellular carcinoma.

Most patients require only supportive care, however antiviral therapy is recommended for severe acute hepatitis B, especially if there is

- INR >1.5
- prolonged jaundice
- acute liver failure
- severe coagulopathy

Preferred drugs are

- Tenofovir and Entecavir

Prophylaxis: hepatitis B immunoglobulin and vaccination.

Hepatitis D virus (HDV) : It is an incomplete RNA virus, it is unable to replicate on its own but is activated in the presence of HBV .

Mode of transmission: blood- born, sexual, vertical.

Incubation period: 4-7 weeks. HDV infection is declining due to HBV vaccination. It is endemic in Mediterranean basin.

Infection occurs as:

- 1- Co-infection with HBV at the same time.
- 2- Super infection of a chronic HBV carrier. Synthesis of HDV depresses the appearance of HBV viral markers. Prophylaxis: vaccination against hepatitis B.



Hepatitis C virus (HCV):

- HCV is an important cause of chronic liver disease worldwide, it was highly prevalent in Egypt .
- HCV is a single-stranded RNA of the flaviviridae family ,the RNA genome is encoding a polyprotein product consisting of structural (capsid and envelope) and non structural viral protein .(Figure 19.4)

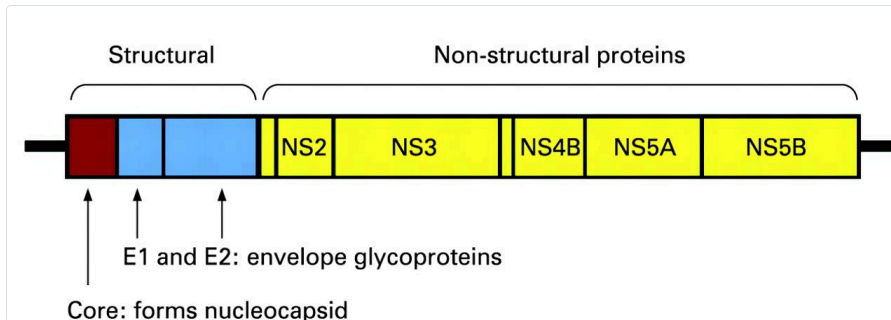


Figure 19.4 — HCV genome

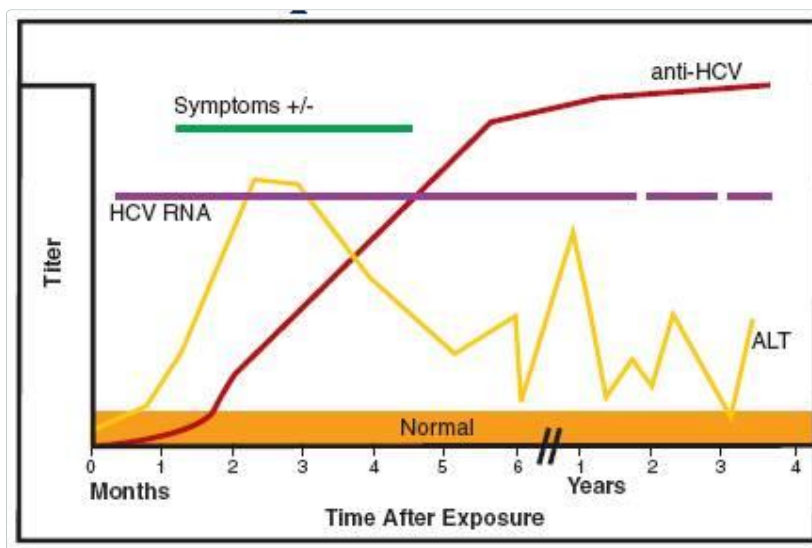


Figure 19.5 — Serology and Virology of Acute and Chronic HCV

- HCV has at least 8 major genotypes. Genotype 4 is the most prevalent in Egypt .
Incubation period: 1-6 months.

Transmission:

mainly parenteral. There is a low frequency of sexual and vertical transmission

Age:

usually above 16 years.



Clinical attack:

Usually mild and sub-clinical . The elevation of transaminases and the appearance of HCV RNA precede the appearance of HCV antibodies ([Figure 19.5](#)) .

Persistent infection is common (about 80%). Infection is linked with chronic hepatitis, cirrhosis and HCC.

- Prophylaxis: Prevention of nosocomial infection and supervision of blood donation

General clinical Picture of viral hepatitis

1- Anicteric Hepatitis : Mild attack with no jaundice , it may presents as gastroenteritis and is usually underdiagnosed

2- Icteric hepatitis:

Preicteric stage: It may lasts for few days. Patient feels generally unwell with marked malaise. Anorexia & nausea Pain may occurs in right hypochondrium associated with fever. This is followed by darkening of urine & light stool. In hepatitis B serum sickness-like syndrome occurs in less than 10% of cases with manifestations of fever, maculopapular rash and polyarthritits of the small joints.

Icteric stage: 2-4 weeks, with the appearance of jaundice, symptoms decrease: fever decreases, appetite improves and vomiting stops. Persistent vomiting and/or drowsiness or confusion indicates urgent hospital a as they may reflect liver failure. Liver is enlarged in 70% of patients, soft & tender, splenomegaly and lymphadenopathy may develop in some patients .

Convalescence stage: Recovery is rapid in children. Fatigue persists for some weeks. Clinical & biochemical recovery is usual within 6 months

Sequela of an acute attack of viral hepatitis:

1-Complete recovery is the rule in most HAV& HEV infections, also in more than 90% of HBV inctions in adults . Only 20% of HCV patients recover spontaneously. 2- Relapse: - Especially in HAV infections. The attack is usually milder attack, and is marked by elevation of transaminases and serum bilirubin. Recovery usually follows.

3-Prolonged cholestasis: commonly in HAV infection, the jaundice persists for several months. Pruritus is prominent. Abdominal U/S is usually performed to exclude surgical obstructive jaundice. Complete recovery is the rule .

4-Acute liver failure: Jaundice is deep, drowsiness, confusion , flapping tremors develop and coagulopathy.

5-Persistent infection: - HBV carrier rate depends on age (see later).

- HDV super-infection causes chronic hepatitis in about 70%.



- HCV infections persist in about 80%.

TABLE: SOME OF THE CHARACTERISTICS OF HEPATITIS VIRUSES

	HAV	HBV	HCV	HDV	HEV
Genome	RNA	DNA	RNA	RNA	RNA
Incubation period	2-6 w	2-6M	1-6 M	4-7 w	2-6 w
Transmission	Feco - oral	Blood- sexual - vertical	Blood	Blood - sexual	Feco- oral
Serologic diagnosis	IgM anti HAV	IgM anti HBV HBVDNA HBsAg	Anti HCV HCVRNA	Ig M anti HDV	IgM anti HEV
Prevention	Vaccine	Vaccine	-	Vaccine against HBV	-
Chronicity	-	+	+	+	It does not progress to chronic liver disease except in some immunosuppressed patients in solid - organ transplant recipients
Treatment	Non specific	Non specific + antiviral	Non specific + antiviral	Pegylated IFN - Bulevirtide	Non specific
				-	

Pathology:

- All forms of acute viral hepatitis have a basic pathology.
- Acute inflammation of the entire liver
- Centrizonal necrosis & cholestasis are most marked in zone 3.
- Cellular infiltration (mainly lymphocytes) of portal tracts (zone 1) and bile duct proliferation.
- The reticulin framework is usually preserved & liver cells regenerate from zone 1.
- Other organs: There may be regional lymph nodes enlargement. Splenomegaly and hypoplasia in bone marrow

Investigations:

1- Liver function tests: Marked elevation of ALT, AST (500-5000U/L). Serum bilirubin is increased (both direct and indirect). Alkaline phosphatase and prothrombine time are normal or mildly increased. Albumin is normal. γ globulins are increased.

2-Blood changes: - Leucopenia with relative lymphocytosis.- ESR is increased.



3-Urine and stool : Bilirubin appears early (dark urine). - Urobilinogen is variable.

the Stools appears pale due to decreased sterchobilinogen.

5- Viral markers:

- **HAV:**

HAV IgM: indicates recent infection and is positive for 3-6 months thereafter.

HAV IgG: indicates old infection and immunity.

- **HBV:**

HBsAg: appears 6 weeks after infection & disappear in 3 months. Persistence for more than six months indicates the development of a carrier state.

Anti-HBs: appear after 3months. It means recovery & immunity.

HBc IgM: is present in acute phase & in severe chronic hepatitis.

HBc IgG: Past exposure.

HBeAg: viral replication and infectivity.

Anti-HBe: low infectivity.

HBV-DNA: most sensitive marker of viral replication, measured by PCR.

- **HDV:**

HDV IgM and HDV RNA in acute infections. HDV IgG in chronic phase.

- **HCV:**

HCV antibodies become positive after 6-8 weeks of infection. HCVRNA can be detected 1-2 weeks after infection by PCR.

-

HEV:

HEV IgM: indicates recent infection. HEV IgG: old infection.

Treatment (acute attack):

Normal activity as tolerated, patient should be adequately hydrated and advised for nutritional support and avoidance of hepatotoxic medications

: Traditionally the patient was given a low fat, high CHO diet as it is said to be the most appetizing.

Prevention :

Pre-exposure: HAV vaccine (inactivated vaccine) , (previously discussed)

HEV : Efficiency of immunoglobulins is uncertain. An effective vaccine has been developed.

HBV : Pre-exposure HBV recombinant vaccines (or Plasma derived) are both effective & safe. - They contain HBsAg as the immunogen. For the recombinant vaccine: the dose is 1ml IM in the arm at 0-1-6 months. Indications: - Surgical, dental, laboratory staff & medical students.



- Patients: dialysis, oncology, CLD, mental patients.
- Neonates of HBsAg mothers.
- Accidental exposure.
- Drug users & homosexuals.

Post-exposure: both HBV vaccine and HBimmunoglobulin (HBIG) are given in different arms at the same time to: neonates of HBsAg positive -mothers, accidental exposure

HBsAg +ve blood, to sexual contacts of acute sufferers and to prevent reinfection of a donor liver inserted into an HBV DNA positive patient.

■ HCV

Work on HCV vaccine is in progress. Screening of blood donors and blood banks for

HCV antibodies.

HERPES VIRUS HEPATITIS

- A large family of enveloped DNA viruses including cytomegalovirus (CMV), Epstein-Barr virus (EBV), herpes simplex virus (HSV).
- Very common infections ,90 to 95% of adults are (+) for EBV, and CMV serology, and up to 80% for HSV



- Unlike hepatitis viruses A, B, C, D and E, herpes viruses are not hepatotropic, and do not replicate in liver cells but can cause hepatitis by proliferation of infected cells and release of cytokines. They do not cause chronic hepatitis or cirrhosis. EBV and CMV hepatitis are very common and usually mild, subclinical and self-limited. Herpesvirus hepatitis can present as a primary infection or reactivation of latent infection

EBV and CMV transmission : Person to person transmission, uses of contaminated objects .Blood transfusion, and organ transplantations have also been reported as rare modes of transmission.

EBV symptoms : Non-specific: malaise, anorexia, right sided abdominal pain ,characteristic: Infectious mononucleosis (IM) syndrome - sore throat with exudative pharyngitis, enlarged lymph nodes. EBV is especially associated with left sided abdominal pain with splenomegaly.

Diagnosis by Serology : eg viral capsid antibody (VCA IgM) - Heterophil Ab (Specific but not very sensitive) EBV DNA by PCR .Lymphocytosis, atypical lymphocytes (not specific)

CMV Symptoms : similar to EBV (infectious mononucleosis), sore throat, enlarged lymph nodes. Exudative pharyngitis and splenomegaly are less common than with EBV. Diagnosis by serology , IgM Ab to envelope (+) ,Heterophile (-) ,CMV DNA by PCR ,Lymphocytosis, atypical lymphocytes – not specific

EBV and CMV biochemistry : usually only mildly elevated ALT and usually ALT >AST. Elevated alkaline phosphatase in 60%, hyperbilirubinemia in 45%, but jaundice in usually <5% of cases .The laboratory abnormalities typically normalize in a few weeks, but can rise again during reactivation. Severe cholestatic hepatitis, granulomatous hepatitis, and rarely, acute fulminant hepatic failure - possible especially in immunocompromised.

Most cases of EBV, and CMV hepatitis are mild, resolve spontaneously - only supportive treatment is needed. In cases of severe hepatitis, valganciclovir or ganciclovir can be used

HSV Hepatitis :

Rare, usually a result of disseminated herpes sepsis. Recipients of organ transplants on immunosuppressive therapy, chemotherapy and persons with immunodeficiency diseases like HIV – high risk.

Pregnancy, 2nd or 3rd trimester – high risk

- Common cause of fulminant hepatic failure even in immunocompetent patients. Herpetic cutaneous, muco-cutaneous, mucosal lesions.(cold sores) patient presents with severe hepatitis - AST, ALT – very high, >10 X ULN . Fulminant hepatic failure frequent especially, but not only, in immunocompromised Because of potential rapid deterioration and high mortality of HSV hepatitis ,empiric treatment with acyclovir or Foscarnet is recommended as soon as possible after diagnosis to decrease mortality and the need for liver transplant



CHRONIC HEPATITIS

Chronic hepatitis is a group of liver diseases characterized by persistent hepatic inflammation with hepatocellular injury lasting for more than six months. If left untreated, chronic inflammation may progress to hepatic fibrosis, cirrhosis, portal hypertension, liver failure, and hepatocellular carcinoma, depending on the underlying cause.

Although the causes are diverse, the histopathological features of chronic hepatitis share many common characteristics.

Etiology

- Chronic viral hepatitis
 - Hepatitis B virus (HBV), with or without hepatitis D virus (HDV) coinfection
 - Hepatitis C virus (HCV)
- Autoimmune hepatitis
- Drug-induced chronic hepatitis
 - Methyldopa
 - Isoniazid
 - Methotrexate
 - Nitrofurantoin
- Alcohol-associated steatohepatitis (ASH)
- Metabolic liver diseases
 - Metabolic dysfunction-associated steatohepatitis (MASH) (formerly non-alcoholic steatohepatitis, NASH)
- Wilson disease
- Hereditary hemochromatosis
- Alpha-1 antitrypsin deficiency

With the availability of highly effective direct-acting antiviral therapy, chronic hepatitis C has become curable in most patients.

Pathology

The characteristic histological findings include varying degrees of portal inflammation, interface hepatitis, lobular inflammation, and fibrosis.

In mild disease:

- Expansion of the portal tracts by inflammatory infiltrates composed mainly of lymphocytes, with variable numbers of plasma cells (particularly prominent in autoimmune hepatitis).



- Interface hepatitis (formerly called piecemeal necrosis) characterized by inflammatory destruction of hepatocytes at the junction between the portal tract and hepatic lobule.
- Mild lobular hepatitis, with scattered foci of hepatocyte necrosis (spotty necrosis) and inflammatory cell infiltration.

In more advanced disease:

- More extensive interface hepatitis.
- Confluent hepatocyte necrosis may occur.
- Progressive deposition of collagen leading to fibrous septa connecting portal tracts and/or central veins (bridging fibrosis).
- Regenerating hepatocytes may form regenerative nodules.
- Continued fibrosis eventually results in cirrhosis.

Histological Staging

Liver biopsy may be graded and staged using the METAVIR scoring system, particularly in chronic viral hepatitis.

- Activity (A) reflects the degree of necroinflammatory activity:
 - A0: No activity
 - A1: Mild activity
 - A2: Moderate activity
 - A3: Severe activity
- Fibrosis (F) reflects the stage of fibrosis:
 - F0: No fibrosis
 - F1: Portal fibrosis without septa
 - F2: Portal fibrosis with few septa
 - F3: Numerous septa without cirrhosis
 - F4: Cirrhosis

Important note: Although liver biopsy remains the reference standard for histological assessment, it is now performed much less frequently than in the past. In current clinical practice, liver fibrosis is commonly assessed using non-invasive methods, particularly transient elastography (FibroScan) and validated serum fibrosis scores such as FIB-4 and the APRI score.



CHRONIC HEPATITIS B



20



Chronic hepatitis B (CHB) is a major global health problem affecting approximately **250–300 million people worldwide** . It remains highly prevalent in many developing countries. In Egypt, the prevalence has declined considerably following the introduction of universal infant vaccination in 1992 and is currently estimated to be **below 2%** in the general population. **HBV genotype D** is the predominant genotype in Egypt.

Natural History

The likelihood of developing chronic HBV infection depends largely on the **age at infection** and the **host immune response** .

Following acute HBV infection:

- **Neonates:** approximately **90%** develop chronic infection.
- **Children <5 years:** **20–30%** develop chronic infection.
- **Adults:** only **<5%** develop chronic infection because most adults successfully clear the virus.

Most liver injury results from the **host immune response against HBV-infected hepatocytes** , rather than from a direct cytopathic effect of the virus.

Clinical Features

Many patients remain asymptomatic for years.

Symptoms, when present, may include:

- Fatigue
- Malaise
- Right upper quadrant discomfort
- Hepatomegaly

Advanced disease may present with complications of cirrhosis such as:

- Ascites
- Variceal bleeding
- Hepatic encephalopathy
- Hepatocellular carcinoma (HCC)

Phases of Chronic HBV Infection

Current international guidelines classify chronic HBV infection into **five phases** , replacing the older terms “immune tolerant”, “inactive carrier”, etc.

- **“Chronic Infection”** = Chronic Infection = Characterized by low or stable viral replication and normal ALT, though periodic subclinical inflammation can still occur.”
- **“Chronic Hepatitis”** = Elevated ALT (Active liver inflammation).



Phase Breakdown Summary Table

Phase	Old Nomenclature	New Nomenclature	HBeAg	HBV DNA	ALT Level
Phase 1	Immune Tolerant	HBeAg-Positive Chronic Infection	Positive	Very High (>10 ⁷ IU/mL)	Normal
Phase 2	Immune Clearance	HBeAg-Positive Chronic Hepatitis	Positive	High (Variable)	Elevated
Phase 3	Inactive Carrier	HBeAg-Negative Chronic Infection	Negative	Low (<2,000 IU/mL)	Normal
Phase 4	Reactivation	HBeAg-Negative Chronic Hepatitis	Negative	Moderate (>2,000 IU/mL)	Elevated

1. HBeAg-positive chronic HBV infection

(formerly **immune-tolerant phase**)

- HBsAg positive
- HBeAg positive
- Very high HBV DNA levels (usually >10⁷ IU/mL)
- Normal ALT
- Minimal hepatic inflammation or fibrosis

This phase is most common in individuals infected at birth or during early childhood.

Most patients are monitored without treatment, although treatment may be considered in selected older patients with significant fibrosis or other risk factors.

2. HBeAg-positive chronic hepatitis

- HBeAg positive
- Elevated ALT
- High HBV DNA
- Active hepatic inflammation
- Progressive fibrosis may occur

These patients are usually candidates for antiviral therapy.



3. HBeAg-negative chronic infection

(formerly **inactive carrier state**)

- HBsAg positive
- Anti-HBe positive
- HBV DNA usually <2,000 IU/mL
- Persistently normal ALT
- No significant liver inflammation

Most patients remain stable, although lifelong follow-up is necessary because reactivation may occur, especially during immunosuppressive therapy.

Spontaneous HBsAg clearance occurs in approximately **1% per year** .

4. HBeAg-negative chronic hepatitis

- HBeAg negative
- Anti-HBe positive
- HBV DNA usually >2,000 IU/mL
- Elevated ALT
- Active hepatic inflammation

This phase usually results from precore or basal core promoter mutations preventing HBeAg production.

5. HBsAg-negative phase

(Occult HBV infection)

is defined by undetectable HBsAg, positive anti-HBc, and detectable HBV DNA in the liver or blood.

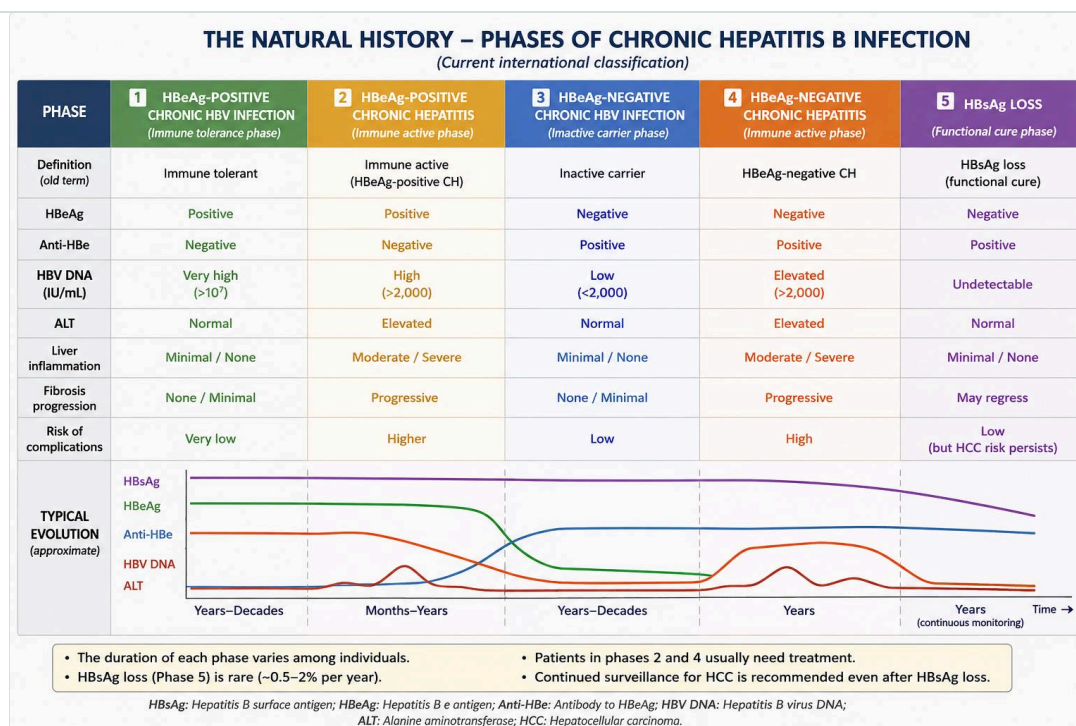


Figure 20.1 — Natural history and phases of chronic hepatitis B infection

Reactivation may occur during chemotherapy or immunosuppressive therapy.

Investigations

Diagnosis is established by detection of **HBsAg** for more than **6 months**.

Routine investigations include:

- Liver function tests (ALT, AST, bilirubin, albumin)
- Complete blood count
- Prothrombin time/INR
- HBeAg and anti-HBe
- Quantitative HBV DNA by PCR
- Abdominal ultrasound
- Serum alpha-fetoprotein (AFP) for HCC surveillance when indicated
- *Abdominal ultrasound every 6 months, with or without serum alphafetoprotein (AFP), for hepatocellular carcinoma (HCC) surveillance in high-risk patients.*

Assessment of Fibrosis

Transient elastography (FibroScan) has largely replaced liver biopsy for fibrosis assessment.

Liver biopsy is now reserved for selected patients in whom the diagnosis or fibrosis stage remains uncertain.

Typical histological findings include:

- Chronic hepatitis



- Interface hepatitis
- Fibrosis
- **Ground-glass hepatocytes** , “Histology may reveal ground-glass hepatocytes, which reflect the accumulation of HBsAg within the cytoplasm.”

Indications for Treatment

Treatment decisions are based on several factors:

- HBV DNA level
- ALT level
- Severity of liver fibrosis
- Presence of cirrhosis
- Age
- Family history of hepatocellular carcinoma
- Extrahepatic manifestations
- Planned immunosuppressive therapy

Treatment is recommended in:

- Patients with chronic hepatitis and:
 - HBV DNA >2,000 IU/mL
 - Elevated ALT
 - Significant fibrosis
- **All patients with compensated or decompensated cirrhosis** and detectable HBV DNA, regardless of ALT level.
- Selected patients with normal ALT but significant fibrosis or a strong family history of HCC.

Goals of Therapy

The aims of treatment are to:

- Suppress HBV replication
- Prevent progression to cirrhosis
- Prevent hepatic decompensation
- Reduce the risk of hepatocellular carcinoma
- Improve survival

The ideal endpoint is **loss of HBsAg** , with or without development of anti-HBs antibodies (functional cure). However, this is achieved in only a small proportion of patients.

In HBeAg-positive patients, **HBeAg seroconversion** is an important treatment milestone.



Antiviral Therapy

First-line Oral Antiviral Agents

■ Tenofovir disoproxil fumarate (TDF)

- 300 mg orally once daily
- Highly potent
- Extremely low resistance

Long-term therapy requires monitoring of:

- Renal function
- Serum phosphate
- Bone mineral density in at-risk patients

■ Tenofovir alafenamide (TAF)

- 25 mg orally once daily

TAF has similar antiviral efficacy to TDF but causes less renal and bone toxicity, making it preferable in patients with chronic kidney disease or osteoporosis.

■ Entecavir

- 0.5 mg once daily
- 1 mg daily in patients with previous lamivudine resistance

Highly effective with a very low resistance rate in nucleos(t)ide-naïve patients.

Drugs No Longer Recommended as First-Line Therapy

The following agents are no longer recommended because of high resistance rates:

- Lamivudine ,Adefovir Telbivudine

They have largely been replaced by tenofovir or entecavir.

Duration of Therapy

Treatment with nucleos(t)ide analogues is usually **long-term** , and many patients require lifelong therapy, especially those with cirrhosis.

Treatment discontinuation may be considered in carefully selected non-cirrhotic patients under close specialist follow-up.

Pegylated interferon- α 2a:

- **180 μ g subcutaneously once weekly**
- Duration: **48 weeks**

Advantages:

- Finite treatment duration
- Higher likelihood of HBsAg loss than oral therapy



Best candidates include:

- Younger patients
- Elevated ALT
- Lower HBV DNA levels
- Genotypes A and B respond better than genotype D

Common adverse effects include:

- Influenza-like symptoms
- Depression
- Bone marrow suppression
- Thyroid dysfunction
- HBV genotype D is the predominant genotype in Egypt. Because Genotype D responds poorly to interferon, so oral nucleos(t)ide analogues (TDF/TAF/Entecavir) are the preferred first-line choice in the Egyptian clinical context.

Hepatitis D (HDV)

Patients with HBV should be screened for HDV when clinically indicated.

Treatment options include:

- **Pegylated interferon-α**
- **Bulevirtide**, the first approved entry inhibitor for chronic hepatitis D, which has significantly improved treatment outcomes in patients with compensated liver disease.

CHRONIC HEPATITIS C

Clinical features

Acute HCV infection is usually asymptomatic. Approximately 20–30% of infected individuals clear the virus spontaneously, whereas 55–85% develop chronic infection.

- Most acute infections are asymptomatic.
- Patients are identified years later and usually accidentally (elevated liver enzymes or with positive HCV antibody).
- HCV Infection has the propensity to become a chronic hepatitis in 80% of cases.
- Patients may be asymptomatic or present with fatigue or upper abdominal pain; they may also present with manifestations of liver cirrhosis and its complications. There is a long period between infection and cirrhosis (usually 20 years). Regression is usually slow. Cirrhosis develops in approximately 15–30% of untreated patients over 20–30 years, depending on age, alcohol consumption, obesity, diabetes and other cofactors.



- Hepatocellular carcinoma occurs in about 15% of cirrhotic patients.
- Extrahepatic manifestations may be due to deposition of immune complexes:
 - Mixed cryoglobulinemia (most common)
 - Membranoproliferative glomerulonephritis
 - Porphyria cutanea tarda
 - B-cell non-Hodgkin lymphoma
 - Vasculitis
 - Insulin resistance and type 2 diabetes mellitus

Diagnosis

HCV antibody alone is NOT diagnostic of active infection.

Current approach:

1. Anti-HCV antibody
2. HCV RNA PCR

Quantitative HCV RNA before treatment

4. Fibrosis assessment

Liver biopsy is now rarely required because liver fibrosis can usually be assessed non-invasively using transient elastography (FibroScan) and serum fibrosis scores (FIB-4 or APRI).

-Viral markers: HCV Ab by ELISA to indicate infection or exposure.

- HCV RNA by PCR to indicate viral replication.
- Liver biopsy is variable: and may show minor pathological changes or show cirrhotic changes . Fatty changes and lymphoid follicles in the portal tracts are frequently present.
- ALT may be normal/ elevated or fluctuating.

Treatment

Direct-acting antiviral agents (DAAs) have revolutionized the treatment of chronic hepatitis C. They are orally administered, well tolerated, and achieve sustained virological response (SVR) rates exceeding **95%** in most patients.

Current first-line pangenotypic regimen :

■ Sofosbuvir/Velpatasvir for 12 weeks

This regimens is effective against all HCV genotypes.

Earlier combinations such as Sofosbuvir/Daclatasvir and Sofosbuvir/Ledipasvir played an important role in the initial DAA era but have largely been replaced by newer pangenotypic regimens in many countries.



Ribavirin

Ribavirin is now used only in selected difficult-to-treat patients such as those with decompensated cirrhosis or previous DAA failure.

Pegylated interferon

Pegylated interferon is no longer recommended for routine treatment of chronic hepatitis C because highly effective interferon-free DAA regimens are available..

Cure

Sustained virological response (SVR12) is defined as undetectable HCV RNA **12 weeks after completion of therapy** and is considered a virological cure.

Vaccination

Patients with chronic HCV should be vaccinated against hepatitis A and hepatitis B if they are not already immune.

Egypt and HCV

Egypt previously had the highest prevalence of hepatitis C worldwide, predominantly genotype 4. Much of this epidemic resulted from the use of inadequately sterilized needles during the national parenteral antischistosomal therapy campaigns conducted between the 1950s and 1980s.

Beginning in 2014, Egypt implemented one of the world's largest hepatitis C elimination programs using locally produced direct-acting antivirals.

Starting in 2018–2019, Egypt launched the nationwide “**100 Million Healthy Lives**” initiative, which included large-scale screening of adults for HCV infection. Individuals diagnosed with HCV were offered highly effective directacting antiviral (DAA) therapy free of charge through the national program. To date, **approximately 60 million people have been screened for HCV**, making Egypt's program one of the largest population-based HCV screening and treatment initiatives worldwide

As a result, Egypt became the **first country to achieve the WHO Gold Tier status on the path toward hepatitis C elimination**, with a dramatic reduction in the national prevalence of active HCV infection.



Autoimmune hepatitis



21



It is an immune-mediated chronic inflammatory liver disease of unknown exact etiology occurring in genetically susceptible individuals. It is characterized by the presence of circulating autoantibodies, increased gammaglobulins and interface hepatitis.

Near 50% increase in incidence over the last few decades

- Occurs at all ages and within all ethnic groups around the world

Types of autoimmune hepatitis (based on serological markers):

Type 1: Characterized by detection of serum anti-nuclear antibody (ANA), anti-smooth muscle antibody (ASMA). It accounts for approximately 80–90% of cases

Type 2: Characterised by detection of serum anti-liver-kidney microsomal antibody (LKM-1), anti-liver cytosole antibody-1 (LC I)

Clinical Picture and diagnosis:

Autoimmune hepatitis (AIH) usually occurs in females (70%) mostly between 15–25 years or at menopause.

- Broad range from asymptomatic to acute/severe disease
- Diagnosis of AIH relies particularly on the presence of autoantibodies, hypergammaglobulinaemia and typical or compatible histology
- Elevated IgG levels, especially in the absence of cirrhosis, are a distinctive feature of AIH
- AIH should be considered in any patient with acute or chronic liver disease particularly if hypergammaglobulinaemia is present and if the patient has features of other autoimmune diseases

The diagnosis is based upon

- Autoantibodies
- Elevated IgG
- Typical biopsy
- Exclusion of viral hepatitis

Typical symptoms are fatigue, joint pains, hepatomegaly and jaundice.

Associated autoimmune manifestations: polyarthritis, hemolytic anemia, purpura, DM, autoimmune thyroiditis.

γ globulins are increased mostly IgG (>1.5 folds ULN).

- AST & ALT are elevated. Alkaline phosphatase is not markedly increased
- Serum autoantibodies: ANA, SMA, LKM-1, LC-1.
- Liver biopsy: interface hepatitis with a lymphoplasmacytic infiltrate rich in plasma cells is the characteristic histological finding, no biliary lesions.



- Other causes of liver disease must be excluded : viral hepatitis, drug induced liver diseases (DILI) ,Wilson's disease, consumption of alcohol ,NASH , alpha- 1 antitrypsin deficiency, hepatotoxic drugs, PBC, PSC, genetic hemochromatosis must be excluded .

Treatment :

Aims:

-To obtain complete remission of the disease and to prevent its further progression , this requires complete normalization of aminotransferases and IgG levels . Generally the treatment requires permanent maintenance therapy

Absolute treatment indications :

- **Clinical:**
 - Incapacitating symptoms
 - Progression of disease
- **Laboratory**
 - $AST \geq 10 \times ULN$
 - $AST \geq 5 \times ULN + GG \geq 2 \times ULN$
- **Histology**
 - Bridging necrosis
 - Multilobular necrosis

* Corticosteroids improve symptoms, signs, LFTs & histology. Dose: prednisone 0.5 – 1 mg/ Kg Bwt or 20-40 mg / day for 2 weeks then reduce to a maintenance dose for 4weeks in combination with azathioprine (1-2 mg /Kg Bwt or 50-150mg/day) . Budesonide (9 mg/day) may be used instead of prednisone in selected non-cirrhotic patients because it causes fewer systemic adverse effects
Azathioprine , cyclosporine and other immunosuppressive drugs have been tried alone or with steroids as steroid sparing agent

Disease remission :

Disappearance of symptoms GGT and bilirubin: normal

65% achieve remission within 18 months, and 80% within 3 yrs of initiation of Rx

10 year survival: 90%

Liver transplantation: is another option , however the condition may recur.

Treatment endpoint : Instead of $AST < 2 \times ULN$



Current guidelines recommend:

Complete biochemical remission is defined by normal ALT, AST and IgG levels.

Second-line therapy

Patients intolerant of azathioprine or with inadequate response may receive:

- Mycophenolate mofetil
- Tacrolimus
- Cyclosporine
- Other immunosuppressive agents in specialized centers

Prognosis

Untreated autoimmune hepatitis has a poor prognosis, whereas appropriate immunosuppressive therapy results in excellent long-term survival, with a 10-year survival exceeding 80–90%.

Metabolic dysfunction-associated steatotic liver disease (MASLD) and metabolic dysfunction-associated steatohepatitis (MASH)

Metabolic dysfunction-associated steatotic liver disease (MASLD) is defined as hepatic steatosis in individuals with at least one cardiometabolic risk factor and without another predominant cause of hepatic steatosis such as significant alcohol intake or chronic liver diseases.

Metabolic dysfunction-associated steatohepatitis (MASH) is characterized by hepatic steatosis associated with hepatocyte injury (ballooning) and lobular inflammation, with or without fibrosis

MASLD and MASH are the current internationally accepted terms that replaced NAFLD (Non alcoholic fatty liver disease) and NASH (Nonalcoholic steatohepatitis) in 2023.

- Steatosis: accumulation of fat in the liver (mainly triglycerides). Simple steatosis usually has a favorable prognosis, although some patients may progress to steatohepatitis and fibrosis. Hepatic steatosis results mainly from insulin resistance, which increases the delivery and synthesis of fatty acids in the liver.

Inflammation and necrosis of liver cells may be caused by the toxicity of excess free fatty acids, oxidant stress, or mitochondrial dysfunction. This may lead finally to fibrosis, cirrhosis and even hepatocellular carcinoma.

MASLD is the most common chronic liver disease worldwide as it is associated with the global epidemic of obesity. **MASLD is strongly associated with metabolic syndrome, which includes:**

- Central obesity



- Type 2 diabetes mellitus
- Insulin resistance
- Hypertension
- Dyslipidemia

Clinical picture

- Typically asymptomatic. Most patients are diagnosed incidentally during ultrasound or routine liver function tests.
- Occurs in both sexes
- Fatigue/ right upper quadrant pain
- Hepatomegaly: ALT, AST: normal or increased. ALT is often greater than AST in early disease.

U/S: Increase hepatic echogenicity (bright liver).

- Assess fasting blood glucose ,HbA1c and a lipid profile..
- -Exclude viral, drug, autoimmune & other causes.

Diagnosis

Fibroscan device Transient elastography measures liver stiffness to estimate fibrosis. It also estimates hepatic steatosis using the Controlled Attenuation Parameter (CAP).

- can distinguish between advanced cirrhosis and no cirrhosis



Liver biopsy - Liver biopsy remains the reference standard for diagnosing MASH and grading fibrosis but is reserved for selected patients because it is invasive.

Non-invasive fibrosis assessment using FIB-4 followed by transient elastography is recommended before considering liver biopsy.

Management:

Lifestyle modification and diet manipulation are the cornerstone of treatment and includes:

- Weight loss of 7–10%
- Mediterranean diet
- Regular aerobic and resistance exercise

Gradual and sustained weight loss of 7–10% of body weight improves steatosis, inflammation, and fibrosis.

Increase vegetables& fruits/ decrease fat. Mediterranean diet

- Rapid weight loss should be avoided because it may worsen hepatic inflammation
- Pioglitazone may be considered in selected patients, particularly those with type 2 diabetes.
- GLP-1 receptor agonists (e.g. semaglutide) improve weight loss and liver steatosis and are increasingly used in obese or diabetic patients.
- New therapies for MASH continue to emerge. Metformin should be used to treat diabetes but is not recommended specifically for treating MASH
- Management of co-morbid factors: DM, hypertension, hyperlipidemia
- Bariatric surgery for morbidly obese patients (BMI>35-40). According to current obesity guidelines after multidisciplinary assessment.

Surveillance : Patients with cirrhosis require

- HCC surveillance
- screening for esophageal varices

Prognosis

Most patients with simple steatosis have a favorable prognosis. Patients with MASH and advanced fibrosis are at increased risk of cirrhosis, liver failure, and hepatocellular carcinoma. Cardiovascular disease remains the leading cause of death in patients with MASLD.

LIVER CIRRHOSIS



22



It is a diffuse process characterized by necrosis of liver cells, followed by fibrosis, regeneration and nodule formation. The liver architecture is diffusely abnormal and this interferes with liver blood flow and function. This finally leads to impaired liver function.

Liver cirrhosis is the end stage of chronic liver disease, characterized by diffuse hepatic fibrosis associated with the formation of regenerative nodules, resulting in distortion of the normal hepatic architecture and impairment of liver function and intrahepatic blood flow.

Etiology :It is caused by chronic non self-limited injury of the liver :

-Viral hepatitis (chronic hepatitis C, Chronic hepatitis B±D).

- Metabolic dysfunction-associated steatohepatitis (MASH) (formerly NASH) - Alcohol-associated liver disease (ALD): Prolonged (>10 years) and regular alcohol intake (>30 ml/day).

- Metabolic : Wilson's disease, hemochromatosis, α 1 antitrypsin deficiency.

Primary biliary cholangitis (PBC).

- Chronic right-sided heart failure and hepatic venous outflow obstruction (congestive hepatopathy) -Autoimmune hepatitis. -Toxins & drugs: Methotrexate- Isoniazide- methyl dopa

Morphological classification:

Morphological classification has largely been replaced by etiological and clinical classifications because mixed patterns are common in advanced disease.

1- Micronodular cirrhosis: small nodules, nearly same size (<3mm) usually caused by ongoing alcohol damage .

2- Macronodular cirrhosis: big nodules of variable size (>3mm), usually follow chronic hepatitis

3- Mixed type.

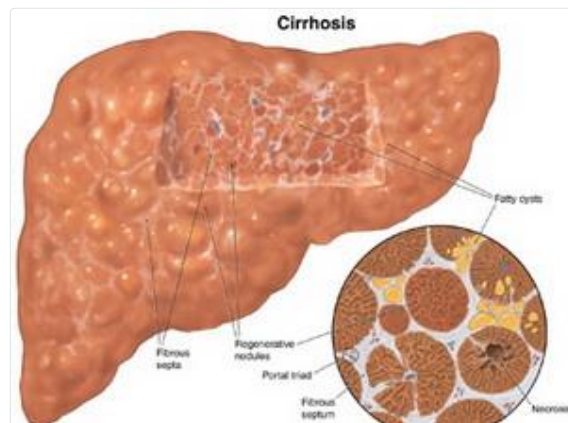


Figure 22.1 — Gross and microscopic picture of liver cirrhosis



Gross and microscopic picture of liver cirrhosis

Clinical picture

1- Compensated cirrhosis:

The disease is discovered accidentally on routine examination and by screening. Examination may reveal an enlarged firm liver, splenomegaly, palmer erythema, spiders. Liver function tests are nearly normal.

1- Decompensated cirrhosis: There are several clinical manifestations of cirrhosis that include liver cell failure: ascites, jaundice, encephalopathy... The transition from compensated to decompensated cirrhosis is marked by the development of ascites, variceal bleeding, hepatic encephalopathy or jaundice.

Symptoms: There are usually three modes of presentation:

1-Asymptomatic: The patient may remain non symptomatizing for years to be discovered by chance during routine check-up, pre-employment or pre-operative assessment

2-Symptoms of conditions commonly associated with chronic liver disease such as gall stones, parotid enlargement, Dupuytren's contractures and muscle cramps

3- Symptoms of complications e.g. ascites, bleeding, encephalopathy, recurrent lower limb edema, spontaneous bacterial peritonitis or renal impairment. Fever due to infections: UTI, SBP, chest and skin infections.

Signs:

1- Protein-calorie malnutrition : found in 20-40%. of patients as shown by muscle wasting and decreased fat stores

2-Skin: Spider naevi and Palmer erythema may be detected .

3- Nail changes:

a. Disappearance of nail lacunae

b. White opaque nails (leuconychia)

c. Clubbing of fingers is more commonly found in biliary cirrhosis and in cirrhosis associated with large pulmonary arterio-venous shunt

4-Manifestations of hyperdynamic circulation due to the presence of excess vasodilator materials (VDM) leading to warm hands, high volume pulse, tachycardia

2- Manifestations of hormonal disturbance: manifestations of hypogonadism and hyperestrogenism including gynecomastia, testicular atrophy and loss of body

3- Sarcopenia (progressive loss of skeletal muscle mass and function) as one of the most common and clinically vital manifestations of advanced chronic liver disease. It affects between 30% to 70% of patients with cirrhosis.



Abdominal examination:

- 1- Abdominal wall hernias are common particularly umbilical hernia.
- 2- Liver is usually palpable with a sharp edge and firm consistency. The surface is usually smooth except in cases of macronodular cirrhosis or carcinoma. The liver size may become shrunken in late cases.
- 3- Spleen is usually palpable, it varies in size, and its lower end may reach to iliac fossa in cases of hepatosplenic schistosomiasis.
- 4- Ascites may be detected.

Chest examination

- Pleural effusion in association with cirrhosis and in the absence of primary pulmonary or cardiac disease. It is usually right sided (70%) with ascites but sometimes without ascites.
 - The hepatopulmonary syndrome may be present
Central cyanosis, clubbing, and dyspnea due to:
Pulmonary AV shunts Porto- pulmonary shunts basal lung collapse due to tense ascites.
 - Fetor hepaticus: Fetor hepaticus is a sweet, musty odor caused by volatile sulfur-containing compounds .

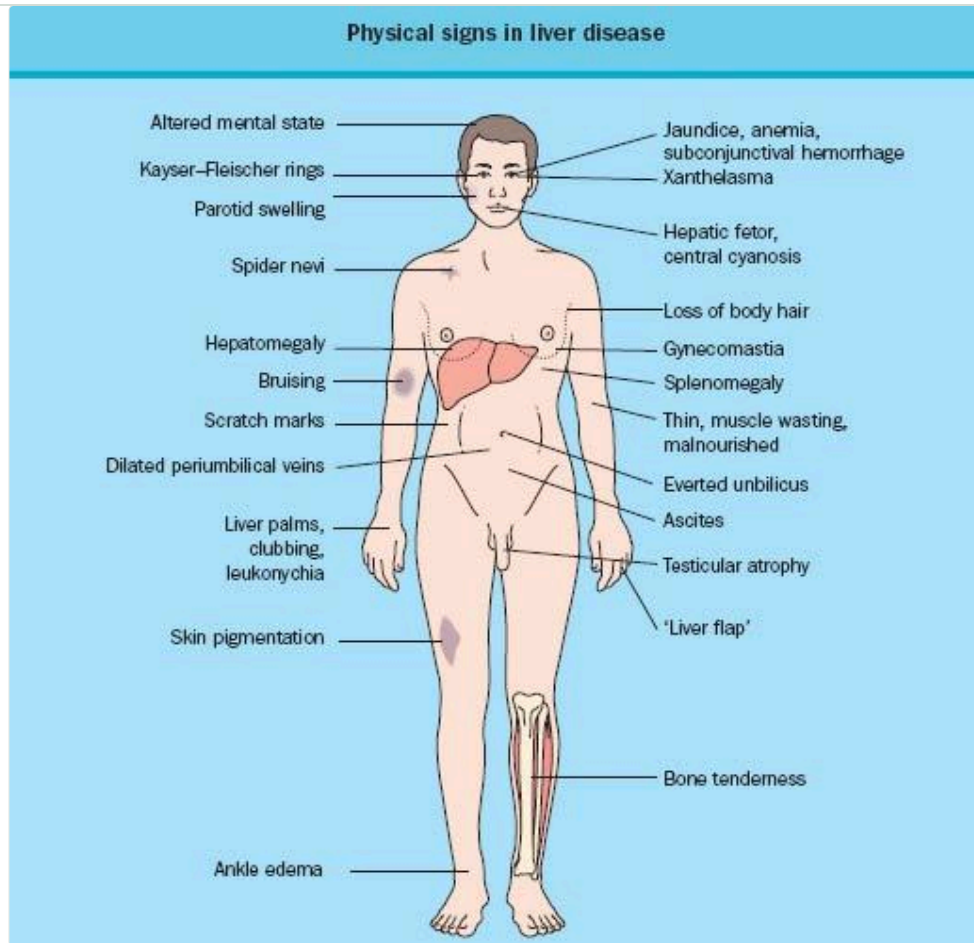


Figure 22.2 — Physical signs in liver cirrhosis

Physical signs in liver cirrhosis

Investigations:

A- Laboratory investigations:

1- Blood picture:

-Microcytic hypochromic anemia is common in patients having gastrointestinal hemorrhage.

- Normochromic macrocytic anemia may be found. -Leucopenia and thrombocytopenia in patients with hypersplenism. -Prolonged prothrombin time that is not corrected by vitamin K injections indicating failure of the liver to synthesize prothrombin complex.

2- Serum albumin is decreased, serum gamma globulins are increased, and bilirubin is increased..

3-Serum aminotransferases (ALT and AST) are mildly elevated.

4-Blood glucose: impaired glucose tolerance may be found in 50% of patients and frank diabetes in 15-30% of patients, particularly in post-hepatitis C cirrhosis .



B- Imaging

-Abdominal ultrasound

The liver may show a coarse echogenic pattern. The edge may be irregular and in advanced cases, the liver may be shrunken and portal hypertension may be shown by the dilated portal vein. Splenomegaly and ascites may be detected. Ultrasonography is best combined with Doppler evaluation of the hepatic vasculature

- Elastography (FibroScan)

C- Upper endoscopy for detection of varices screening endoscopy is guided by Baveno VII criteria, and some patients can safely avoid endoscopy.

D- Investigations for the cause:

Viral markers, serum autoantibodies, serum iron-and ferritin, copper and caeruloplasmin, liver biopsy could be done to reach the possible cause of cirrhosis

Non invasive tests for the diagnosis of hepatic fibrosis, cirrhosis and steatosis

Noninvasive tests has now replaced the role of liver biopsy in the diagnosis of fibrosis. These approaches can overcome the limitations of liver biopsy and have become more widely used in routine clinical practice Imaging-based elastography and serum biomarkers are currently the two main noninvasive tests for liver fibrosis staging. -Liver stiffness measurement by imaging techniques such as transient elastography (Fibroscan). -Simple biomarker blood tests for fibrosis, including the fibrosis-4 index (FIB-4) score and aspartate aminotransferase-platelet ratio index (APRI)

Treatment of cirrhosis:

Actually the condition of management of hepatic fibrosis and portal hypertension has changed with the development of 2 new definitions - Compensated advanced chronic liver disease (cACLD)

- Clinically significant portal Hypertension (CSPH).

. Transition from the compensated to the decompensated stage in chronic liver disease is clinically marked by the development of complications such as ascites, variceal hemorrhage and overt hepatic encephalopathy.

Among patients with cACLD, at least two different stages have been identified based on the presence or absence of clinically significant portal hypertension (CSPH)

The use of Fibroscan (elastography) in clinical practice has enabled the early identification of patients who are at risk of having CSPH and decompensation to be treated by non selective beta blocker like Carvedilol



Although the development of cirrhosis is an irreversible process, many lines of treatment can be provided to the patient ensuring a better quality of life and may promote regeneration of normally functioning hepatocytes.

Nutrition:

Protein restriction is no longer recommended.

Current guidelines recommend Protein 1.2–1.5 g/kg/day even in patients with hepatic encephalopathy.

Temporary reduction may occasionally be necessary during severe overt encephalopathy, but prolonged protein restriction should be avoided because it worsens sarcopenia

- Calories: caloric requirements are supplied as normal individuals. Good intake of carbohydrates and moderate amounts of fat should be supplied.

-Salt restriction in ascitic patients.

Late evening carbohydrate snack.

- Treatment of underlying cause: -Antiviral drugs in early cases with active inflammation. -Abstinence of alcohol intake.
- Chelation of copper and iron in Wilson's disease and hemochromatosis respectively.
- Corticosteroids in autoimmune hepatitis.
- Treatment of complications: ascites, encephalopathy, bleeding, spontaneous bacterial peritonitis and renal impairment.
- Liver transplantation is the treatment of choice if available.

Course and Prognosis: This is extremely variable and depends on several factors such as the etiology, and the presence or absence of complications. The Child-Turcotte Pugh (CTP) Score is used to assess the prognosis. It is proposed to assess survival, risk of general anesthesia, and operative mortality. Patients are classified into A, B & C according to 5 parameters: The level of albumin, bilirubin, prothrombin time & the presence of ascites& encephalopathy

TABLE (2). CHILD-TURCOTTE PUGH (CTP) SCORE

Parameter	1 point	2 points	3 points
Encephalopathy	None	Grade 1-2	Grade 3-4
Ascites	None	Medically controlled	Uncontrolled
Albumin	>3.5	2.8-3.5 g/dl	< 2.8 g/dl
Bilirubin	<2	2-3 mg/dl	>3mg/dl
International normalized ratio (INR)	< 1.7	1.7-2.3	> 2.3

A patient is considered to be :

Child A = 5–6



Child B = 7–9

Child C = 10–15

. The Model for End-Stage Liver Disease incorporating serum sodium (MELD- Na) is a prognostic scoring system based on serum bilirubin, serum creatinine, INR, and serum sodium. It is used to assess disease severity and prioritize patients for liver transplantation.

PRIMARY BILIARY CHOLANGITIS (PBC)

Primary biliary cholangitis (PBC) is a chronic autoimmune cholestatic liver disease characterized by progressive immune-mediated destruction of the small intrahepatic bile ducts, leading to chronic cholestasis, fibrosis and, if untreated, cirrhosis.

The cause is not yet identified but it is said to be an immunological disease also there is a familial background The exact cause remains unknown. It is believed to result from an interaction between genetic susceptibility and environmental triggers, leading to autoimmune destruction of the small intrahepatic bile ducts.

It is sometimes associated with autoimmune manifestations in the form of rheumatoid arthritis, and Sjogren's syndrome.

- Sjögren syndrome
- Autoimmune thyroid disease
- Systemic sclerosis
- Rheumatoid arthritis
- Celiac disease

Pathogenesis: - Autoreactive T lymphocytes target cholangiocytes in genetically susceptible individuals after exposure to environmental triggers. It is proposed that an environmental factor acts on a genetically predisposed host. Infective agents have been accused as the triggers of the autoimmune process e.g. E.coli, enterococci and retroviruses. There may be a defect in T suppressor cells or a defect in immunoregulation producing damage to the duct system.

Clinical picture: Females are more affected than males especially in their fourth to sixth decades of life. Pruritus is the usual presenting symptom and is present in 50% of symptomatic patients , fatigue is usually present . However, the condition may be asymptomatic. Hepatomegaly and jaundice then develop. The symptoms may resemble those of obstructive jaundice. In later stages the patient may have xanthelasmas on the eyelids due to cholesterol deposition.

Malabsorption of fat soluble vitamins and steatorrhea due to deficiency of bile salts may occur in advanced stages. Osteoporosis is one of the complications.

Many patients are now diagnosed incidentally because of isolated elevation of alkaline phosphatase.



Diagnosis :

Current diagnosis requires two of three:

1. Persistent cholestatic liver enzymes (especially ALP)
2. Positive AMA (or PBC-specific ANA if AMA negative)
3. Compatible liver biopsy

Laboratory tests:

Elevated alkaline phosphatase is often the only and the early abnormality in liver biochemistry. Levels of aminotransferases and GGT are also raised. Specific autoantibodies are present: Anti Mitochondria' antibodies (AMA) especially M2 are specific and are present in over 95% of patients. Antinuclear antibodies, smooth muscle antibodies may be present and are not specific. Serum cholesterol is usually raised.

Imaging:

- Abdominal ultrasound is indicated to exclude extrahepatic biliary obstruction and if there is doubt. MRCP can be done .

Liver biopsy:

Damage and loss of the small bile ducts with Florid duct lesion are characteristic of the disease.

N.B.: Other causes of hepatic granulomas include; sarcoidosis, tuberculosis, schistosomiasis, and drug reactions.

Treatment

Ursodeoxycholic acid (UDCA) 13–15 mg/kg/day

It is considered the first-line treatment. It improves liver biochemistry , slows fibrosis progression and improves survival

Patients are evaluated after one year.

If response inadequate

Second-line therapy

Obeticholic acid or Fibrates (bezafibrate/fenofibrate)

depending on availability.

- Pruritus is difficult to treat; First-line is Cholestyramine or second-line drugs as Rifampicin , Naltrexone , Sertraline

Cholestyramine and naloxone have been tried with mild improvement.

- Control of signs and symptoms of liver cirrhosis and liver cell failure in later stages.
- Liver transplantation is the definite treatment.



N.B.: Secondary biliary cholangitis can result from prolonged (months to years) large duct obstruction by stones or strictures

PRIMARY SCLEROSING CHOLANGITIS (PSC)

Primary sclerosing cholangitis (PSC) is a chronic progressive cholestatic liver disease characterized by immune-mediated inflammation and fibrosis of the intrahepatic and/or extrahepatic bile ducts, resulting in multifocal biliary strictures, cholestasis, and eventually biliary cirrhosis and liver failure.

Pathogenesis:

Chronic inflammation of the bile ducts leads to concentric periductal fibrosis and the formation of multiple short strictures alternating with mildly dilated segments, producing the characteristic "beaded" appearance of the biliary tree

- The exact cause is unknown. Genetic predisposition has been proposed - -These changes are characteristic and are seen on MRCP and ERCP.
- The changes may range from mild inflammation to established cirrhosis.

Clinical picture:

- Males are commonly affected. Age of onset is usually the forties.
- Manifestations of cholestasis are usually present e.g. pruritus and jaundice . The course of the disease is usually fluctuating.
- Approximately 70–80% of patients have inflammatory bowel disease, most commonly ulcerative colitis. .
- Later on, picture of end stage liver disease eventually takes place.

Many patients are discovered incidentally because of isolated elevation of alkaline phosphatase.

Patients may experience fatigue and recurrent bacterial cholangitis

Investigations:

-Elevated bilirubin and alkaline phosphatase are the earliest abnormalities. - The autoantibody p-ANCA is found in the serum of 60% of patients. p-ANCA is not diagnostic because of poor specificity. Magnetic resonance cholangiopancreatography (MRCP) is the preferred diagnostic imaging modality and demonstrates multiple short strictures alternating with dilated segments ("beading"). ERCP is reserved for therapeutic intervention or evaluation of dominant strictures. -However, the diagnosis is mainly based on



imaging studies; MRCP and or ERCP show attenuation and the characteristic beading of the extra and intra hepatic biliary radicals. -Confirmation may need liver biopsy which shows the changes mentioned above.

Biopsy is not routinely required.

It is useful when: small-duct PSC is suspected also when PSC-AIH overlap syndrome is suspected

Characteristic histology

"Onion-skin" periductal fibrosis

PSC may overlap with autoimmune hepatitis, particularly in younger patients.

Treatment:

- The only proven treatment is liver transplantation. Five-year survival exceeds 80%.
- Ursodeoxycholic acid (UDCA) Supplementary treatment as in cases of obstructive jaundice. Ursodeoxycholic acid may improve liver biochemistry in some patients, but it has not been shown to alter disease progression and is not routinely recommended solely to improve prognosis.
- Control of manifestations of end stage liver disease.

ERCP may benefit in balloon dilatation or temporary stenting to relieve cholestasis , pruritus or recurrent cholangitis.

Surveillance

Patients require surveillance for:

- Cholangiocarcinoma
- Gallbladder carcinoma
- Hepatocellular carcinoma (if cirrhotic)

Patients with PSC and ulcerative colitis require annual colonoscopic surveillance because of the markedly increased risk of colorectal cancer.



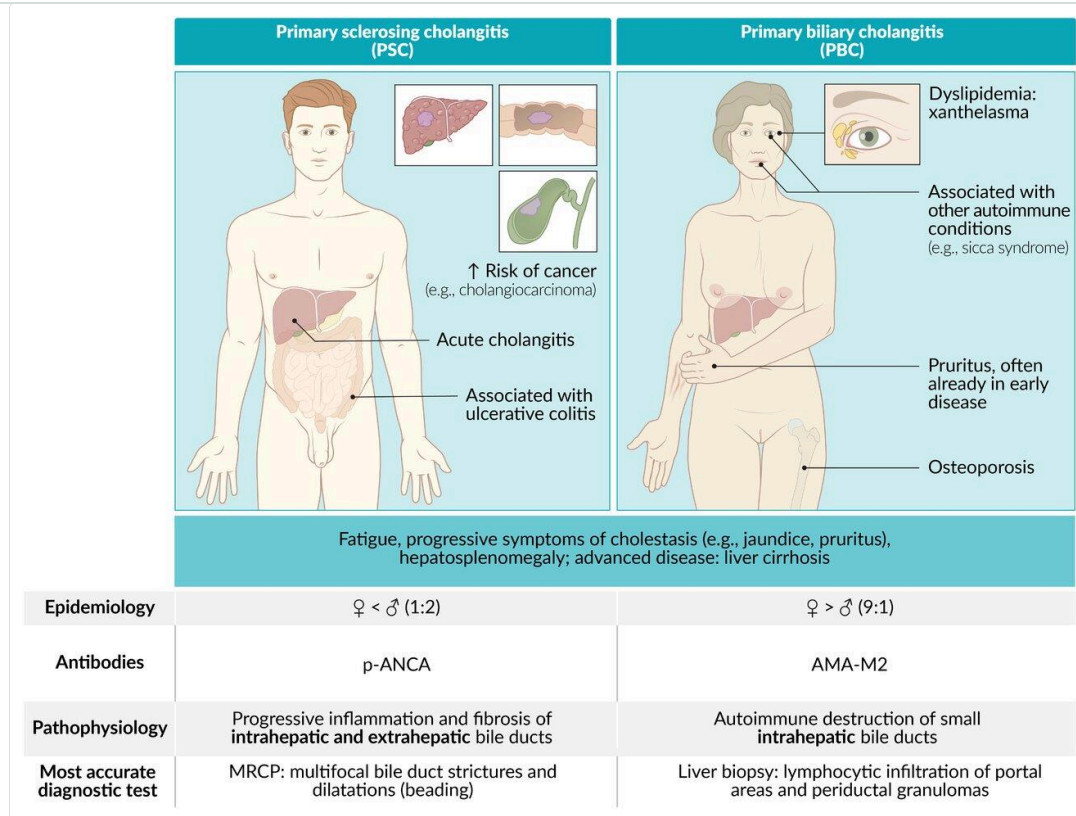


Figure 22.3 — Primary sclerosing cholangitis and primary biliary cholangitis

HEREDITARY HEMOCHROMATOSIS

Definition:

This is an inherited autosomal recessive disease characterized by iron deposition in various organs leading to eventual fibrosis and functional organ failure. It is due to mutation of HFE gene on the short arm of chromosome 6.

Pathophysiology:

The HFE gene mutation results in decreased hepatic production of hepcidin, the principal hormone regulating iron homeostasis.

Low hepcidin levels increase the activity of ferroportin, leading to excessive intestinal iron absorption and increased iron release from macrophages.

Because the body has no physiological mechanism for iron excretion, excess iron progressively accumulates in tissues.

Initially, iron is deposited predominantly in periportal hepatocytes. With disease progression, iron deposition extends throughout the liver and later involves:

- Kupffer cells
- Biliary epithelium



- Fibrous tissue
- Pancreas
- Heart
- Pituitary gland
- Joints
- Skin

Persistent iron overload causes oxidative stress, hepatocyte injury, fibrosis, cirrhosis, and increases the risk of hepatocellular carcinoma (HCC)

Clinical Features

Many patients are now diagnosed before symptoms develop through family screening or incidental discovery of elevated serum ferritin and transferrin saturation.

Symptomatic disease usually appears after the age of 40–50 years in men and later in women because menstruation and pregnancy delay iron accumulation.

Clinical manifestations include:

- Fatigue and weakness (most common early symptoms)
- Hepatomegaly
- Chronic liver disease and cirrhosis
- Skin hyperpigmentation ("bronze diabetes")
- Diabetes mellitus due to pancreatic iron deposition
- Arthropathy, especially involving the second and third metacarpophalangeal joints
- Hypogonadotropic hypogonadism
- Loss of libido and erectile dysfunction
- Cardiomyopathy
- Cardiac arrhythmias
- Heart failure

Patients with established cirrhosis remain at increased risk of hepatocellular carcinoma, even after successful iron depletion.

Investigations

Laboratory studies

- Increased transferrin saturation (>45%) (earliest biochemical abnormality)
 - Elevated serum ferritin
 - Elevated serum iron
 - Reduced or normal total iron-binding capacity (TIBC)



- Mild elevation of liver enzymes

Genetic testing

Testing for HFE mutations (C282Y and H63D) confirms the diagnosis in most patients.

Imaging

MRI is the preferred non-invasive method for quantifying hepatic iron concentration and assessing extrahepatic iron deposition.

Liver biopsy

Liver biopsy is no longer routinely required for diagnosis. It is reserved for patients in whom:

- The diagnosis remains uncertain
- Non-HFE hemochromatosis is suspected
- The degree of fibrosis requires assessment
- Another liver disease is suspected

Treatment:

Therapeutic phlebotomy (Venesection)

This is the treatment of choice.

- Approximately 500 mL of blood is removed weekly or every two weeks during the induction phase.
- Treatment continues until serum ferritin reaches 50–100 ng/mL.
- Maintenance phlebotomy is then performed every 2–4 months according to iron status.

Iron chelation

Iron chelators (e.g., deferoxamine, deferasirox, or deferiprone) are reserved for patients who cannot undergo phlebotomy, such as those with severe anemia or transfusion-dependent conditions. They are not first-line therapy in hereditary hemochromatosis.

Lifestyle measures

Patients should be advised to:

- Avoid iron supplements
- Avoid vitamin C supplements in large doses
- Limit alcohol consumption
- Avoid raw shellfish because of the risk of *Vibrio vulnificus* infection



Liver transplantation Liver transplantation is indicated for patients with decompensated cirrhosis or HCC meeting transplant criteria. It corrects liver failure but does not correct the underlying genetic defect.

Family Screening

First-degree relatives should undergo screening with:

- Transferrin saturation
- Serum ferritin
- HFE genetic testing when appropriate

WILSON DISEASE (Hepatolenticular Degeneration)

Definition

Wilson disease is a rare autosomal recessive disorder of copper metabolism caused by mutations in the ATP7B gene on chromosome 13. The disease results in impaired biliary copper excretion and defective incorporation of copper into ceruloplasmin, leading to progressive copper accumulation, mainly in the liver, brain, cornea, kidneys, and other organs.

A history of parental consanguinity may be present.

Pathophysiology

Dietary copper is absorbed mainly from the stomach and proximal small intestine and is transported to the liver bound to albumin.

Within hepatocytes, copper normally has two major fates:

- Incorporation into ceruloplasmin, which is released into the circulation.
- Excretion into bile, which is the major route of copper elimination.

In Wilson disease, mutations of the ATP7B gene impair both:

- Incorporation of copper into ceruloplasmin.
- Biliary excretion of copper.

As a result, copper accumulates initially in the liver. Once hepatic storage capacity is exceeded, free copper is released into the bloodstream and deposited in other organs, particularly:

- Brain (especially the basal ganglia)
- Cornea
- Kidneys
- Bones and joints





Copper-induced oxidative injury leads to progressive hepatic inflammation, fibrosis, cirrhosis, and neurological damage.

Clinical Features

The disease usually presents between 5 and 35 years of age, although presentation outside this age range may occur.

Clinical manifestations vary widely.

Hepatic manifestations

- Asymptomatic elevation of liver enzymes
 - Acute hepatitis
 - Chronic hepatitis
 - Compensated or decompensated cirrhosis
 - Acute liver failure (particularly in children and young adults)

Neurological manifestations

- Tremor
 - Dysarthria
 - Dystonia
 - Parkinsonian features
 - Chorea
 - Incoordination
 - Drooling (unintentional loss of saliva)
 - Dysphagia

Psychiatric manifestations

- Personality changes
 - Depression
 - Anxiety
 - Behavioral disturbances
 - Cognitive impairment

Ophthalmic manifestations

Kayser-Fleischer rings are caused by copper deposition in Descemet's membrane of the cornea. They are best detected by slit-lamp examination and are present in nearly all patients with neurological disease.

Other manifestations

- Coombs-negative hemolytic anemia



- Renal tubular dysfunction
- Osteopenia or osteoporosis
- Premature arthritis
- Cardiomyopathy (uncommon)

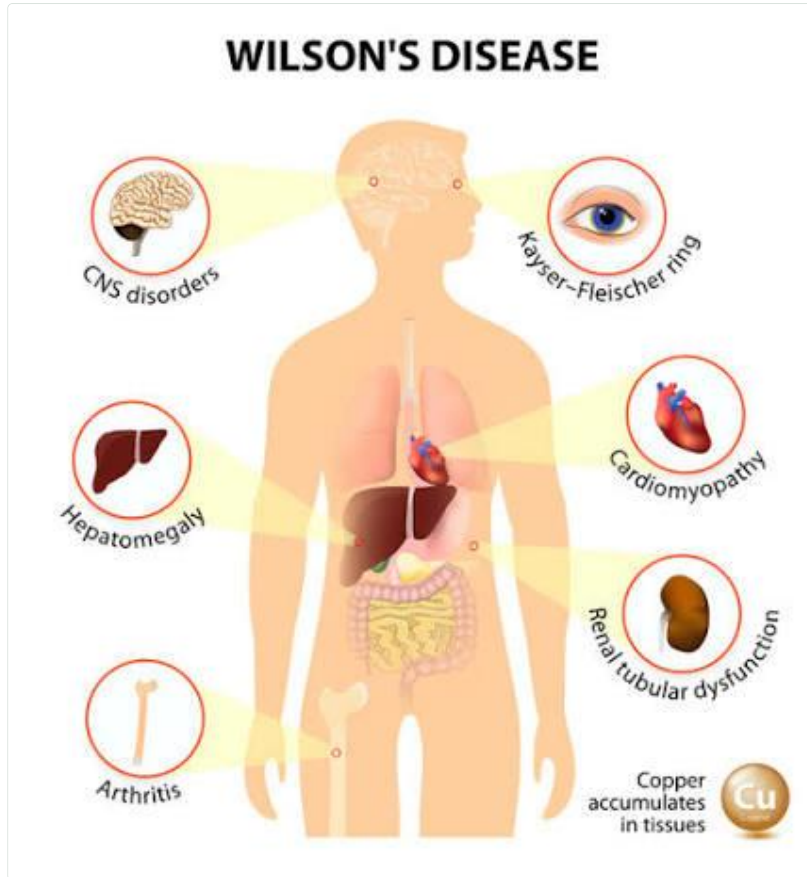


Figure 22.4 — Wilson disease

Investigations

Laboratory studies

- Low serum ceruloplasmin (usually <20 mg/dL)
 - Increased 24-hour urinary copper excretion (>100 $\mu\text{g/day}$ in symptomatic adults)
 - Elevated non-ceruloplasmin ("free") copper
 - Liver function tests may be abnormal

Ophthalmologic examination

- Slit-lamp examination for Kayser-Fleischer rings

Imaging

- Brain MRI may show characteristic abnormalities of the basal ganglia in neurological disease.



Liver biopsy

Liver biopsy is reserved for selected cases when the diagnosis remains uncertain. Quantitative hepatic copper measurement ($>250 \mu\text{g/g}$ dry weight) supports the diagnosis.

Genetic testing

Identification of ATP7B mutations confirms the diagnosis and is increasingly used, especially for family screening.

Diagnosis

The diagnosis is based on a combination of:

- Clinical findings
- Serum ceruloplasmin
- 24-hour urinary copper excretion
- Kayser-Fleischer rings
- Hepatic copper quantification (when needed)
- ATP7B genetic testing

Treatment

Treatment is lifelong.

Copper chelation

- D-Penicillamine (first-line in many patients)
 - Trientine (alternative with fewer adverse effects)

Zinc salts

Zinc reduces intestinal copper absorption and is commonly used for:

- Presymptomatic patients
- Maintenance therapy
- Patients intolerant of chelators

Dietary measures

Patients should initially avoid foods rich in copper, including:

- Liver
- Shellfish
- Nuts
- Chocolate
- Mushrooms





Liver transplantation

Indicated for:

- Acute liver failure
- Decompensated cirrhosis unresponsive to medical therapy

Liver transplantation corrects the underlying metabolic defect because the transplanted liver contains normal ATP7B function.

Family Screening

All first-degree relatives, particularly siblings, should be screened by:

- Clinical assessment
- Serum ceruloplasmin
- 24-hour urinary copper excretion
- Liver function tests
- ATP7B genetic testing when available

Patients with confirmed Wilson disease should receive lifelong treatment even if asymptomatic.

Heterozygous carriers do not develop Wilson disease and do not require treatment.

Alcohol-Associated Cirrhosis

Etiology

Alcohol-associated cirrhosis develops after prolonged excessive alcohol consumption, usually for more than 10 years. The risk increases with both the amount and duration of alcohol intake. Daily drinking is more harmful than intermittent drinking.

Alcohol may also enhance the hepatotoxicity of certain drugs, particularly paracetamol (acetaminophen).

Pathology

- Diffuse micronodular cirrhosis that may become mixed micro- and macronodular in advanced disease.
- Macrovesicular hepatic steatosis.
- Hepatocyte ballooning with Mallory–Denk bodies.

Clinical Features

Alcohol-associated liver disease includes a spectrum of:



- Hepatic steatosis (fatty liver)
- Alcohol-associated hepatitis

Alcohol-associated cirrhosis

Patients with chronic alcohol misuse may also have:

- Gastrointestinal: gastritis, peptic ulcer disease, pancreatitis, malabsorption.
- Cardiovascular: alcoholic cardiomyopathy.
- Neurological: peripheral neuropathy, cerebellar ataxia, myopathy.
- Hematological: macrocytic anemia, leukopenia, thrombocytopenia.
- General: parotid enlargement and Dupuytren's contracture.

Investigations

- Mild to moderate elevation of liver enzymes with AST typically higher than ALT (often AST/ALT ratio >2).
- Elevated γ -glutamyl transferase (GGT).
- Abdominal ultrasonography to assess liver morphology and portal hypertension.
- CT or MRI and liver biopsy are reserved for selected cases.

Treatment

- Complete abstinence from alcohol (the most important treatment).
- Adequate nutritional support, including vitamin supplementation (especially thiamine).
- Management of cirrhosis and its complications.

Alpha-1 Antitrypsin Deficiency

Definition

Alpha-1 antitrypsin (AAT) deficiency is an autosomal recessive inherited metabolic disorder caused by mutations in the SERPINA1 gene (formerly known as the Pi gene). It results in reduced circulating levels of alpha-1 antitrypsin and accumulation of abnormal AAT protein within hepatocytes.

Pathophysiology

Alpha-1 antitrypsin is synthesized mainly by the liver and acts as an inhibitor of several proteolytic enzymes, particularly neutrophil elastase.



Deficiency of circulating AAT allows unopposed neutrophil elastase activity, resulting in progressive destruction of alveolar walls and early-onset emphysema.

The abnormal AAT protein is retained within hepatocytes, leading to hepatocellular injury, chronic hepatitis, fibrosis, cirrhosis, and occasionally hepatocellular carcinoma.

Clinical Features

The clinical manifestations involve mainly the lungs and liver.

Pulmonary manifestations

- Early-onset emphysema (usually in the third or fourth decade, especially in smokers)
 - Chronic obstructive pulmonary disease (COPD)

Hepatic manifestations

- Neonatal cholestatic jaundice (in some infants)
 - Chronic hepatitis
 - Cirrhosis
 - Portal hypertension
 - Hepatocellular carcinoma (in patients with cirrhosis)

Diagnosis

- Low serum alpha-1 antitrypsin level.
- Phenotyping or genotyping of the SERPINA1 gene (the PiZZ genotype is the most severe and clinically important).

Treatment

- Smoking cessation.
- Standard treatment of COPD.
- Intravenous alpha-1 antitrypsin augmentation therapy (plasma-derived) for selected patients with emphysema.
- Management of chronic liver disease and cirrhosis.
- Liver transplantation for end-stage liver disease.

Hepatic Venous Outflow Disorders

Hepatic venous outflow disorders are conditions that impair the drainage of blood from the liver.



The main conditions are:

- **Congestive hepatopathy** – due to increased venous pressure from cardiac disease
- **Budd–Chiari syndrome** – due to obstruction of the hepatic veins
- **Sinusoidal obstruction syndrome (SOS)** – injury and obstruction of hepatic sinusoids
- **Inferior vena cava (IVC) obstruction** – obstruction of the IVC affecting hepatic venous drainage

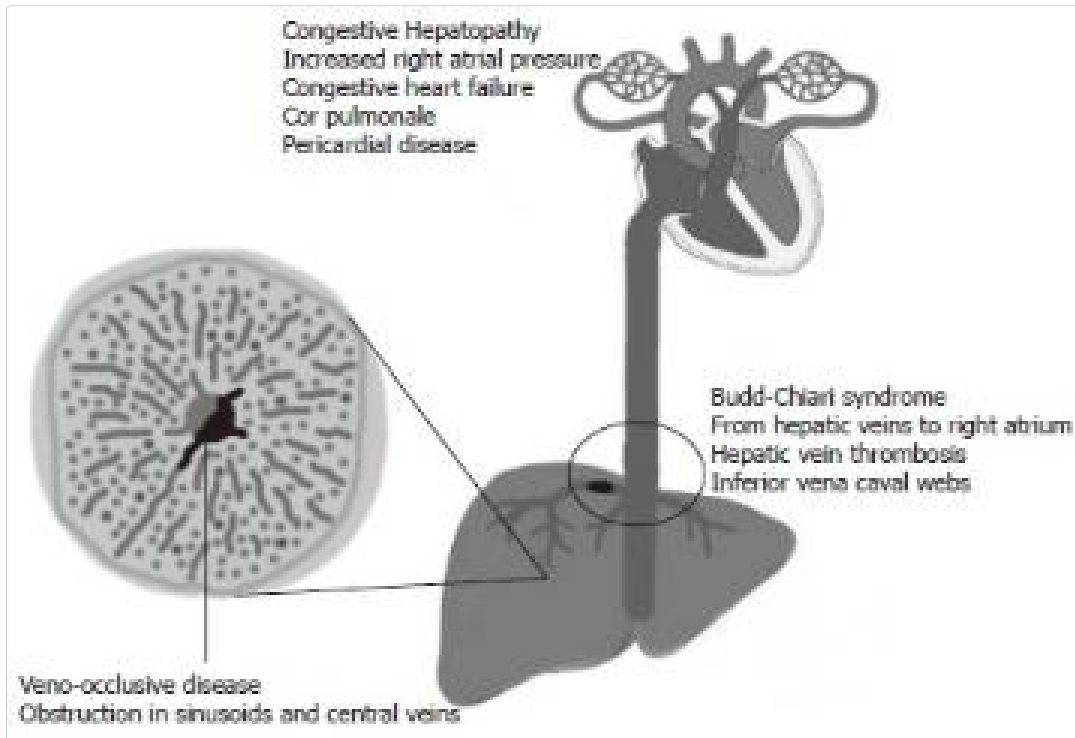


Figure 22.5 — Hepatic venous outflow disorders

1. Congestive Hepatopathy

Definition

Congestive hepatopathy is chronic liver congestion caused by increased central venous pressure, most commonly due to **right-sided heart failure**.

Causes

- Right-sided heart failure
- Severe tricuspid regurgitation
- Constrictive pericarditis



Pathophysiology

Increased right-sided cardiac pressure is transmitted backward to the hepatic veins and sinusoids, causing **hepatic congestion**, particularly around the central veins.

Clinical Features

- Hepatomegaly, often tender
- Right upper quadrant discomfort
- Peripheral edema
- Ascites, usually in advanced disease
- Clinical features of right-sided heart failure

Diagnosis

- Liver function tests may show mild abnormalities
- **Echocardiography** to assess the underlying cardiac disease
- **Doppler ultrasound** can demonstrate hepatic venous congestion

Treatment

The main treatment is **treatment of the underlying cardiac disease**.

- Optimize treatment of heart failure
- Diuretics when appropriate
- Treat significant valvular disease

Key concept:

Congestive hepatopathy = the heart cannot pump effectively → blood backs up into the liver.

2. Budd–Chiari Syndrome

Definition

Budd–Chiari syndrome is **obstruction of hepatic venous outflow**, usually involving the hepatic veins, in the absence of cardiac disease as the primary cause.

Causes

Most cases are related to **thrombosis or a prothrombotic state**.

Important associated conditions include:

- Myeloproliferative disorders
- Other thrombophilic disorders
- Pregnancy and oral contraceptive use



- Malignancy

Clinical Presentation

Budd–Chiari syndrome may be acute, subacute, or chronic.

The classic clinical triad is:

■ Abdominal pain + hepatomegaly + ascites

Chronic disease may lead to:

- Portal hypertension
- Splenomegaly
- Esophageal varices
- Chronic liver disease/cirrhosis

Diagnosis

Doppler ultrasonography is the first-line investigation.

CT or MRI may be required for further assessment.

Typical Imaging Findings

- Reduced or absent hepatic venous flow
- Hepatic vein obstruction
- Collateral venous circulation
- Caudate lobe enlargement

Treatment

Treatment depends on the severity and underlying cause and may include:

- **Anticoagulation**
- Treatment of the underlying prothrombotic condition
- Angioplasty/stenting in selected patients
- TIPS in patients with persistent portal hypertension
- Liver transplantation in severe cases

■ Key concept:

Budd–Chiari syndrome = the liver cannot drain → hepatic veins are obstructed.

3. Sinusoidal Obstruction Syndrome

Sinusoidal obstruction syndrome (SOS), formerly called **veno-occlusive disease**, results from injury to the hepatic sinusoids and small hepatic venules.

It occurs mainly after **hematopoietic stem-cell transplantation or certain chemotherapy regimens**.



Clinical Features

- Painful hepatomegaly
- Jaundice
- Ascites
- Rapid weight gain due to fluid retention

Treatment

Management is mainly supportive. **Defibrotide** is used in severe cases.

Remember: SOS is mainly associated with **hematopoietic stem-cell transplantation/chemotherapy**.

4. Inferior Vena Cava Obstruction

IVC obstruction can impair hepatic venous drainage and cause hepatic congestion.

Causes

- Thrombosis
- Membranous webs
- Tumor compression or invasion

Clinical Features

- Hepatomegaly
- Ascites
- Lower-limb edema
- Dilated abdominal wall veins

Diagnosis

- Doppler ultrasound
- CT or MRI when required

Treatment

Treatment depends on the cause and may include **anticoagulation or angioplasty/stenting**.

Feature	Congestive hepatopathy	Budd–Chiari syndrome
Basic problem	Increased venous pressure from the heart	Obstruction of hepatic venous drainage
Main cause	Right-sided heart disease	Thrombosis/prothrombotic disorders



Feature	Congestive hepatopathy	Budd–Chiari syndrome
Important clinical clue	Raised JVP + peripheral edema	Abdominal pain + hepatomegaly + ascites
Hepatic veins	Congested/dilated	Obstructed
Main treatment	Treat cardiac disease	Anticoagulation ± restoration of venous drainage

■ **Congestive hepatopathy:**

Heart → pressure backs up → liver becomes congested

Budd–Chiari:

Liver → hepatic veins blocked → blood cannot leave

Drug-Induced Liver Injury (DILI)

Definition

Drug-induced liver injury (DILI) is liver injury caused by a **medication, herbal product, or dietary supplement**. It may range from mild asymptomatic elevation of liver enzymes to **acute liver failure**.

DILI should be suspected when liver injury develops after exposure to a potentially hepatotoxic drug and other common causes have been excluded.

Types of DILI

1. Intrinsic DILI

- Usually **dose-dependent and predictable**
- Can occur in most individuals at sufficiently high doses
- **Paracetamol overdose** is the classic example

2. Idiosyncratic DILI

- Usually **unpredictable**
- Occurs only in susceptible individuals
- Most clinically important DILI is idiosyncratic

Patterns of Liver Injury

DILI may have three biochemical patterns:

Hepatocellular

- Predominant elevation of **ALT/AST**



- May present with hepatitis and jaundice

Cholestatic

- Predominant elevation of **ALP and bilirubin**
- May cause jaundice and pruritus

Mixed

- Features of both hepatocellular and cholestatic injury

The R-ratio can be used to classify the pattern of liver injury.

Clinical Manifestations

DILI may present with:

- Asymptomatic elevation of liver enzymes
- Acute hepatitis
- Cholestatic jaundice
- Hypersensitivity features such as fever, rash and eosinophilia
- Acute liver failure

Diagnosis

There is **no single diagnostic test** for DILI.

Diagnosis is based on:

- Detailed history of medications, herbal products and supplements
- Appropriate temporal relationship between drug exposure and liver injury
- Identification of the biochemical pattern
- Exclusion of other causes of liver disease
- Improvement after withdrawal of the suspected drug (**dechallenge**)

Liver biopsy is not routinely required.

Paracetamol Hepatotoxicity

Paracetamol overdose is the classic example of **intrinsic, dose-dependent DILI** .

In overdose, increased formation of the toxic metabolite **NAPQI** leads to depletion of glutathione and hepatocellular necrosis.

Treatment : N-acetylcysteine (NAC) is the specific antidote.

Treatment of DILI

The most important treatment: Stop the suspected offending drug , when clinically appropriate.

Other measures include:



- Monitor liver enzymes, bilirubin and INR
- Provide supportive treatment
- Avoid unnecessary hepatotoxic medications
- Treat complications

Patients who develop **acute liver failure** require specialist management and may require **liver transplantation** .

Important Examples

Drug	Typical association
Paracetamol	Dose-dependent hepatocellular injury
Isoniazid	Hepatocellular injury
Amoxicillin-clavulanate	Cholestatic/mixed injury
Diclofenac	Idiosyncratic DILI
Methotrexate	Chronic liver injury/fibrosis
Amiodarone	Hepatocellular/chronic liver injury

Key Points

DILI may be **intrinsic or idiosyncratic** .

- The three patterns are **hepatocellular, cholestatic and mixed** .
- DILI is mainly a **clinical diagnosis and diagnosis of exclusion** .
- Always ask about **medications, herbal products and supplements** .
- **Withdrawal of the offending drug is the main treatment.**
- **Paracetamol overdose → NAC.**
- Severe DILI can cause **acute liver failure** and may require transplantation.

PORTAL HYPERTENSION



23



- The portal vein is formed by the union of the superior mesenteric vein and splenic vein.

It enters the liver at the porta hepatitis and divides into right and left branches supplying the corresponding hepatic lobes.

- Normal portal pressure: 5–10 mmHg.
- Clinically significant portal hypertension develops when portal pressure exceeds approximately 10–12 mmHg.
- Increased resistance to portal blood flow results in the development of portosystemic collaterals, which divert portal blood into the systemic circulation, bypassing the liver.

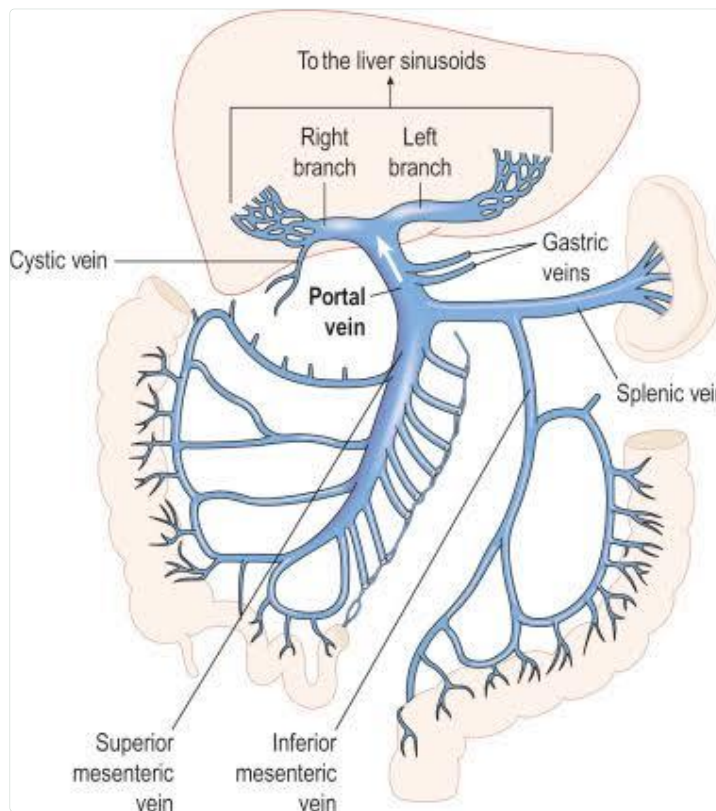


Figure 23.1 — Portal venous circulation

Causes and Classification

- Portal hypertension is classified according to the site of obstruction into prehepatic, intrahepatic, and posthepatic causes.

1. Prehepatic Portal Hypertension

- Caused by obstruction of portal blood flow before entering the liver, most commonly due to portal vein thrombosis.

Causes of portal vein thrombosis include:

- Idiopathic causes.



- Neonatal umbilical sepsis.
- Intra-abdominal infection.
- Increased blood viscosity, e.g. dehydration.
- Malignancy, e.g. hepatocellular carcinoma.
- Congenital portal vein stenosis or narrowing.

2. Intrahepatic Portal Hypertension

- Results from increased resistance within the liver due to distortion or obstruction of the intrahepatic portal circulation.

A. Presinusoidal

- Schistosomiasis with periportal fibrosis.
- Hodgkin's disease.
- Leukemia.
- Sarcoidosis.
- Congenital hepatic fibrosis.
- Toxins, e.g. arsenic.

B. Sinusoidal and Postsinusoidal

- Cirrhosis — the most common cause.
- Alcoholic hepatitis.
- Hepatic veno-occlusive disease.

3. Posthepatic Portal Hypertension

- Caused by obstruction of venous outflow after the liver, including:
 - Budd–Chiari syndrome.
 - Right-sided heart failure, particularly with tricuspid incompetence.
 - Constrictive pericarditis.

Cirrhosis is the most common cause of portal hypertension worldwide, while schistosomiasis is an important cause in endemic areas such as Egypt.

Portosystemic Collaterals

- When portal circulation is obstructed and portal pressure rises, particularly above 10–12 mmHg, extensive collateral circulation develops to divert portal blood into the systemic venous circulation.

1. Gastroesophageal junction

- Esophageal varices arise from anastomoses between the left and short gastric veins (portal system) and the azygos system (systemic circulation).



- Gastric varices may also occur, particularly in the fundus.
- Esophageal varices are superficial and therefore prone to rupture and life-threatening hemorrhage.

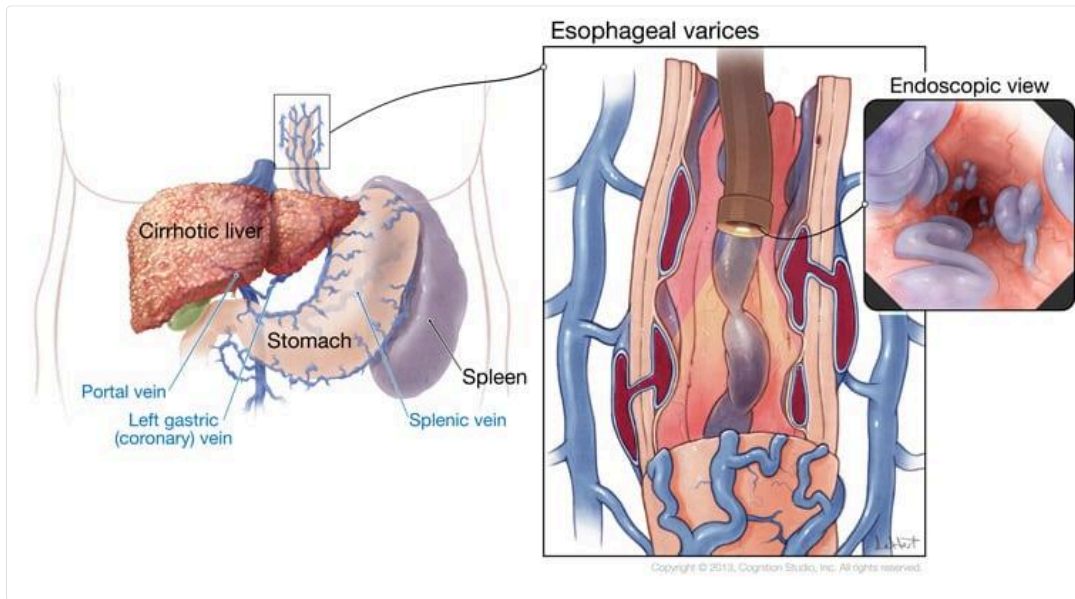


Figure 23.2 — Esophageal varices

2. Rectum

- Rectal portosystemic collaterals occur between the superior hemorrhoidal vein (portal) and the middle and inferior hemorrhoidal veins (systemic).

3. Anterior abdominal wall

- Results from recanalization of the umbilical vein with communication with the superior and inferior epigastric veins.
 - Prominent abdominal wall veins may produce caput medusa.

4. Other sites

Collateral vessels may develop between:

- Liver and diaphragm.
- Spleen and diaphragm.
- Spleen and kidney, producing lienorenal collaterals.
- Other retroperitoneal sites.

Clinical Features

Patients may remain asymptomatic, and splenomegaly may be the only clinical finding, particularly in presinusoidal portal hypertension.

Clinical manifestations:

- Splenomegaly.



- Hypersplenism, particularly thrombocytopenia.
- Hematemesis and melena due to rupture of gastroesophageal varices or bleeding from portal hypertensive/congestive gastropathy.
- Hepatic encephalopathy due to passage of gut-derived toxins into the systemic circulation through portosystemic shunts.
- Ascites.

Investigations

1. Upper Gastrointestinal Endoscopy

- Esophageal varices.
 - Gastric varices.
 - Grade/size of varices.
 - Stigmata or active bleeding.
 - Portal hypertensive gastropathy.

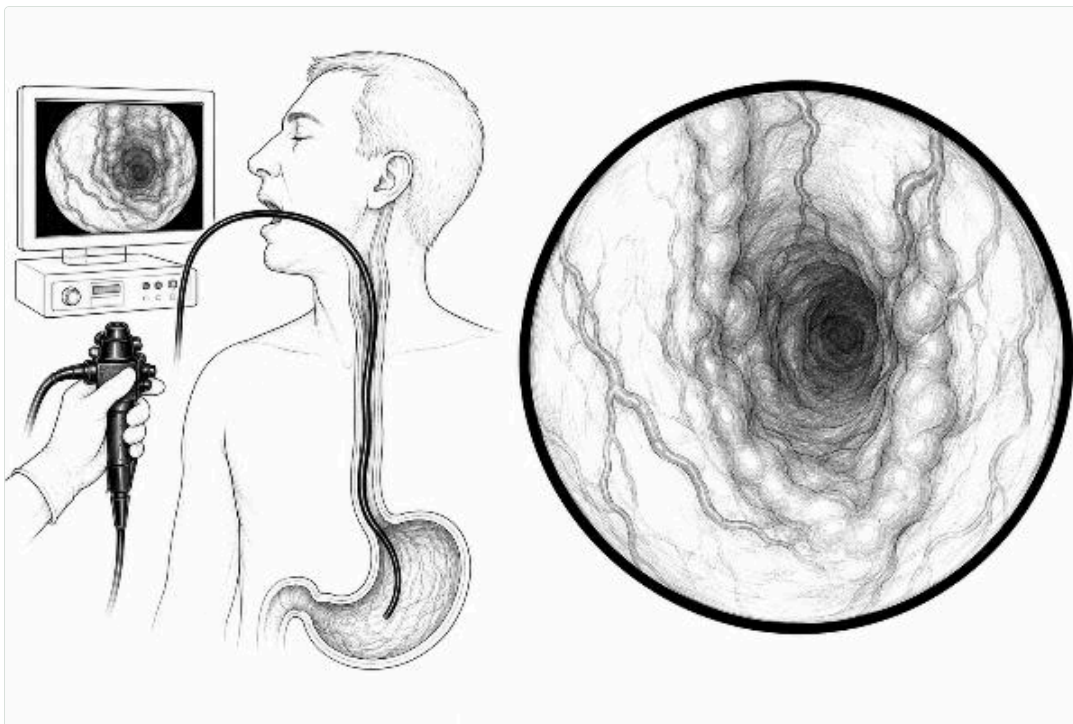


Figure 23.3 — Esophageal varices on endoscopy

2. Abdominal Ultrasonography and Doppler

- A non-invasive investigation used to:
 - Assess the portal and splenic veins.
 - Measure portal and splenic vein diameter.
 - Detect portal vein thrombosis or other vascular abnormalities.



- Identify portosystemic collaterals, including lienorenal collaterals.
- Assess the liver and spleen.
- Detect ascites.
- Help determine the underlying cause of portal hypertension.

3. Laboratory and Histological Assessment

- Liver function tests.
 - Other relevant laboratory investigations according to the suspected cause.
 - Liver biopsy may be required to determine the etiology and assess hepatic pathology in selected cases.

Variceal Hemorrhage

Approximately 90% of patients with cirrhosis may develop gastroesophageal varices over 10 years, although only about one-third develop variceal bleeding.

Esophageal variceal hemorrhage accounts for a significant proportion of deaths in patients with cirrhosis and portal hypertension.

Mortality depends on the patient's underlying hepatic function, general clinical status, and severity of the bleeding episode.

Management is divided into:

1. **Primary prophylaxis** — prevention of the first bleeding episode.
2. Management of **acute variceal bleeding**.
3. **Secondary prophylaxis** — prevention of recurrent bleeding.

Prevention of First Variceal Bleeding

Patients with cirrhosis should undergo assessment for clinically significant portal hypertension and varices using appropriate non-invasive criteria and/or upper GI endoscopy.

For patients with significant esophageal varices who have not previously bled, prophylactic treatment includes:

1. Non-selective β -Blockers

- Carvedilol is a non-selective β -blocker with additional α_1 -blocking activity and is commonly used for prevention of decompensation and variceal bleeding, 6.25 mg twice daily.
 - They reduce the risk of variceal bleeding and may also reduce bleeding from portal hypertensive gastropathy.



2. Endoscopic band Ligation

- used for patients who have an indication for endoscopic prophylaxis, particularly when β -blockers are contraindicated or not tolerated, such as in:
 - Bronchial asthma.
 - Significant heart block.
 - Hypotension.
 - The varix is suctioned into the endoscope cap and a rubber band is deployed around it, producing thrombosis, necrosis, and subsequent fibrosis.

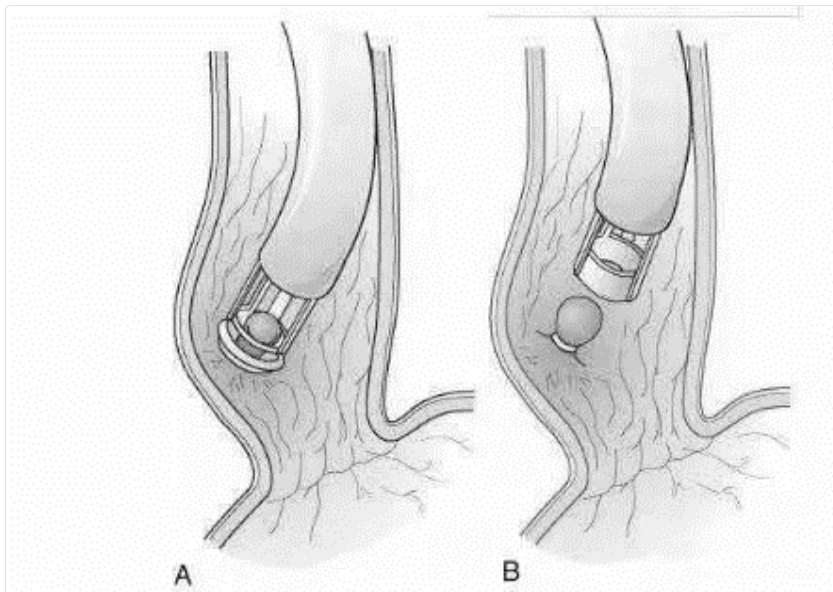


Figure 23.4 — Endoscopic variceal band ligation

Management of Acute Variceal Bleeding

1. Initial Resuscitation:

- Assess airway, breathing, circulation and general condition.
 - Monitor pulse and blood pressure.
 - Establish adequate intravenous access.
 - Obtain blood for:
 - CBC.
 - Blood grouping and cross-matching.
 - PT/INR.
 - Liver function tests.
 - Renal function.
 - Electrolytes.



- Blood glucose.
- Blood transfusion should be restrictive, with a target hemoglobin generally around 7–8 g/dL, individualized according to the patient's clinical condition.
- Avoid over-transfusion because excessive volume replacement may increase portal pressure and precipitate rebleeding.
- A nasogastric tube may be considered selectively for aspiration or gastric lavage, but it is not routinely required in every patient.

2. Urgent Upper GI Endoscopy:

- Endoscopy should be performed after initial stabilization, preferably within 12 hours in suspected acute variceal bleeding.
 - It is performed to:
 - Confirm variceal bleeding.
 - Exclude other causes of upper GI bleeding, such as peptic ulcer disease.
 - Identify gastric varices.
 - Detect portal hypertensive gastropathy.
 - Provide definitive endoscopic hemostasis.

Endoscopic Treatment

- **Endoscopic variceal ligation** the preferred endoscopic treatment.
- **Sclerotherapy:** Injection of a sclerosant, such as ethanolamine oleate, produces thrombosis and obliteration of the varix.
- **Gastric varices:** Large gastric varices may require injection of tissue adhesives (Histoacryl)

Band ligation and Sclerotherapy are the treatment of choice as they arrest bleeding in 80% of cases

3. Vasoactive Therapy

- Vasoactive drugs should be started as soon as variceal bleeding is suspected, ideally before endoscopy, and continued for several days.
- They reduce portal pressure by decreasing splanchnic arterial blood flow and portal venous inflow.
 - **Terlipressin**
 - Usually continued for 2–5 days.
 - Possible adverse effects include abdominal cramps, pallor, and ischemic complications.
 - Particular caution is required in patients with ischemic heart disease or other vascular disease.
 - **Somatostatin and Octreotide**
 - Reduce splanchnic blood flow and portal pressure.



- Have relatively few adverse effects.
- Octreotide is commonly used when terlipressin is contraindicated or unsuitable.

4. Additional Measures

- Prophylactic antibiotics : oral or IV.
- Measures to prevent encephalopathy.
- Vitamin K IV .
- PPI IV or H2 blockers.
- Sucralfate 1 g/ 6 hours to reduce esophageal ulceration after endoscopic therapy.

Balloon tamponade is now rarely used because of serious complications, particularly aspiration and esophageal injury.

Management of Failure or Early Rebleeding

Patients may rebleed within 10 days and in this case endoscopic therapy is repeated.

Transjugular Intrahepatic Portosystemic Shunt (TIPS)

- TIPS creates a channel between the portal vein and a hepatic vein, thereby decompressing the portal circulation.

Technique

- A guidewire is introduced through the jugular vein into the hepatic venous system.
- The portal vein is accessed through the liver.
- The tract between the hepatic and portal veins is dilated.
- A covered expandable metal stent is deployed to establish the portosystemic shunt.
 - TIPS markedly reduces portal pressure and is highly effective in controlling refractory variceal bleeding.
 - increased risk of hepatic encephalopathy.

Prevention of Recurrent Variceal Bleeding

Following an episode of variceal hemorrhage, the risk of recurrent bleeding is approximately 60–80% within 2 years, making secondary prophylaxis essential.

1. Endoscopic Variceal Ligation

- Repeated EVL sessions are performed, commonly at approximately 2-week intervals, until variceal obliteration.
- Follow-up endoscopy is performed to assess eradication and recurrence.



2. Non-selective β -Blockers

3. TIPS

4. Surgical Portosystemic Shunts

Surgical shunts decompress the portal circulation and reduce rebleeding but may produce significant hepatic encephalopathy.

5. Liver Transplantation

Measurement of Portal Pressure

Patency of the portal vein should be assessed before measurement of portal pressure by **Doppler ultrasound**

- **Direct measurement** of portal pressure is **invasive**, expensive and complicated (percutaneous transhepatic measurement - transjugular measurement)
- **Indirect measurement** of portal pressure is safer and less invasive.

Hepatic vein catheterization: Cannulation of the hepatic vein, balloon occlusion of the hepatic vein and measurement of wedged vein pressure (WHVP). This actually measures the sinusoidal pressure.

Portal venous pressure gradient (PVPG): difference between portal pressure and pressure in the inferior vena cava.

-Hepatic venous pressure gradient (HVPG): is the difference between the wedged (occluded hepatic venous pressure and the free hepatic venous pressure.

- HVPG has a high correlation with PVPG (when portal hypertension is due to sinusoidal post sinusoidal lesion).

HVPG > 12 mmHg is necessary for esophageal varices to form and bleed.

Hydatid cyst



24



A common parasitic disease that is endemic in sheep-raising areas.

Mode of infection

- 1- Humans become infected by ingestion of eggs of the E- Granulosus, either by eating contaminated food or from contact with dogs .
- 2- The ingested embryos invade the intestinal mucosal wall and proceed to the liver via the portal venous system.

Structure of hydatid cyst:

1. Adventitia: layer of fibrous tissue surrounding the cyst = **Pseudocyst or pericyst**
2. Laminated membrane: the outer surrounding true layer = **Ectocyst**
3. Inner germinal cell layer: the most active layer of the cyst = **Endocyst**
4. Hydatid fluid containing hydatid sand, scolices, daughter cysts, and brood cysts – it is highly antigenic fluid

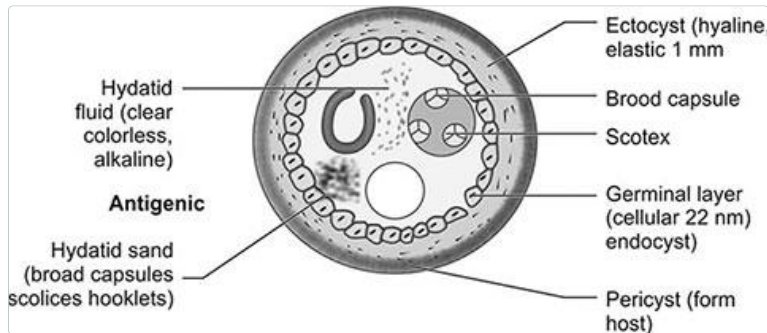


Figure 24.1 — Hydatid cyst

Sequelae of hydatid cyst:

1. The cyst may gradually enlarge and produces symptoms.
2. The parasite may die, fluid is absorbed, and the laminated membrane may calcify. A completely calcified cyst indicate a dead non- reactive cyst
3. May lead **to some complications.**

Clinical picture:

- 1- Most patients may be asymptomatic for years .
- 2- Non- specific picture:
 - a) Right upper quadrant or epigastric pain
 - b) On examination → enlarged liver and a palpable mass
- 3- One third may present with



complications:

as infection, pressure or rupture.

A- Rupture of the cyst:

- 1- Intra-peritoneal rupture (spontaneous or post-traumatic).
- 2- Rarely, intra-thoracic or intra-pericardial rupture.

Clinically:

- Cyst rupture can be associated with allergic reaction as Urticaria, itching and erythema.
- Shock in cases of generalized anaphylactic reaction.
- With biliary rupture the classic triad of jaundice, biliary colic and urticarial rash occur.
- Multiple intra-peritoneal cysts → abdominal pain, peritonitis and ascitis.

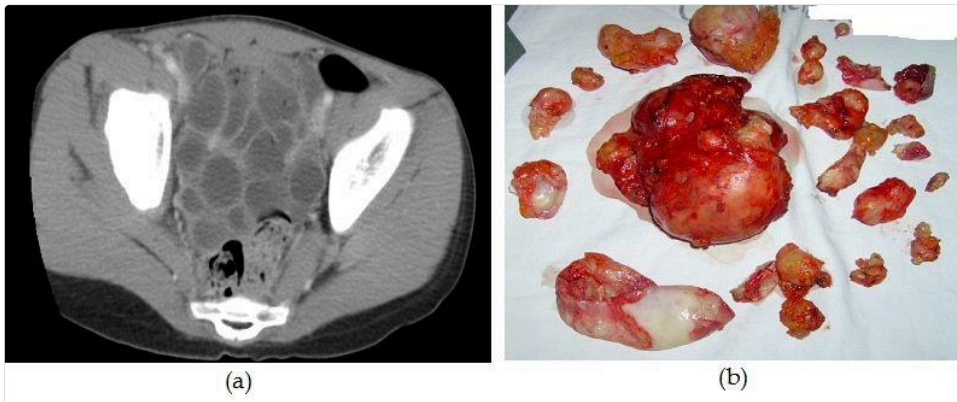


Figure 24.2 — Hydatid cyst complications: source imaging and specimen

B- Secondary infection: infected cyst (picture of abscess)

- Tender hepatomegaly, rigors and fever.

C- Pressure complications:

- Pressure on main biliary ducts causing obstructive Jaundice.

Investigations:

1- **ELISA** has a sensitivity of 90% and is the initial screening tests of choice

Casoni test:

not used now.

2 - **Ultrasound** is the screening method of choice.

It is currently the primary diagnostic technique and has diagnostic accuracy of 90%

3- **C.T. and MRI:**

- A well-defined cyst with thick wall



- Dense wall calcifications are present in 50% of cases.
- Daughter cysts are identified inside.

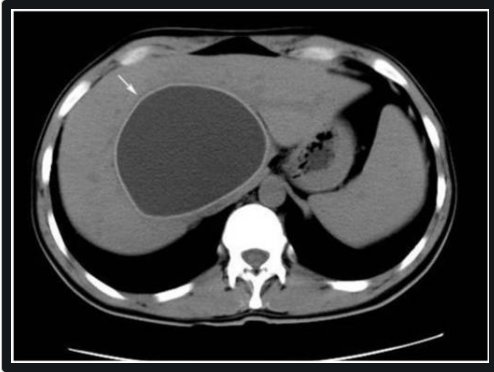


Figure 24.3 — Hydatid cyst: CT view

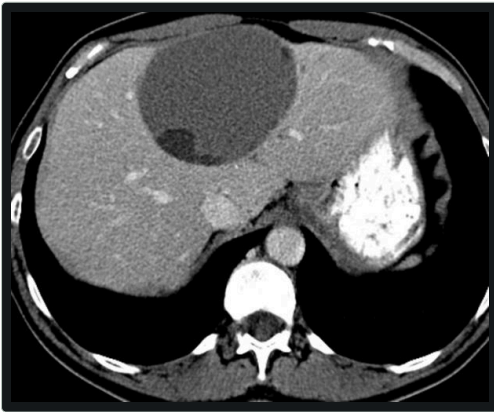


Figure 24.4 — CT: Hydatid cyst

Management of hydatid cyst in the liver:

Lines of treatment:

in symptomatic cases

1- Medical treatment → Albendazole

2- Surgical → is the main line used in symptomatic cases not responding to medications

3- Per-cutaneous aspiration. Needs special precautions.

■ Surgery:

■ Surgery remains the gold standard treatment for hydatid liver disease.

■ Types of surgical procedures:

1- Hydatid cystectomy (removal of the cyst only).

2- Peri-cystectomy (Cyst + part surrounding liver tissue)

3- Liver resection mainly in left lobe cyst



Aim of surgery:

- Injection of scolical solution inside the cyst as hypertonic saline or formaline
- Aspirate the content before removal.

The surgery Comprising removal of:

- Laminar layer
- Germinal layer
- Cyst contents (daughter cysts and brood capsules).
- Preventing spillage of the cyst contents in the peritoneal cavity (Green towels were placed around the liver)
- Obliterate the residual cavity.



Figure 24.5 — Hydatid cyst surgery

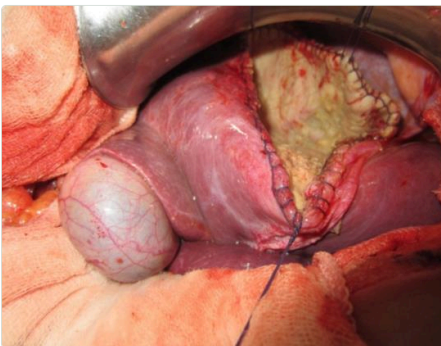


Figure 24.6 — Hydatid cystectomy





■ Percutaneous Treatment of Hydatid CYST:

PAIR:

(puncture, aspiration, injection, and re-aspiration) is a percutaneous treatment technique for hydatid disease.

- This technique has become more common and its safety and efficacy have been reported.
 - Done under radiological guidance as U/S or C.T. scan
 - It has been increasingly accepted as a treatment option for hydatid disease in special centers.
 - Needs well trained persons and precautions to avoid peritoneal rupture
-



Pyogenic liver abscess



25



Sources (routes) of infection:

1- Portal vein :

Infection causes portal pyemia reaching the liver producing abscess.

- Acute suppurative appendicitis and appendicular abscess **(most common)**. •
Acute diverticulitis of the colon.
- Diverticular abscess.
- Infected Meckel 's diverticulum
- Inflammatory bowel disease
- Perforated hollow viscus.

2- Bile duct: (the most common)

Extra-hepatic biliary obstruction leading to ascending cholangitis and abscess formation is the most common cause.

e.g. CBD stones – stricture – malignant obstruction

- Biliary-enteric anastomoses (choledocho-duodenostomy or choledocho-jejunostomy)

3- Hepatic artery:

Infection reaches the liver through systemic bacteremia caused by:

- Bacterial endocarditis
- Renal infection

4- Sub phrenic abscess :

directly reach the liver

5- Umbilical sepsis :

infection reaches liver via para-umbilical veins



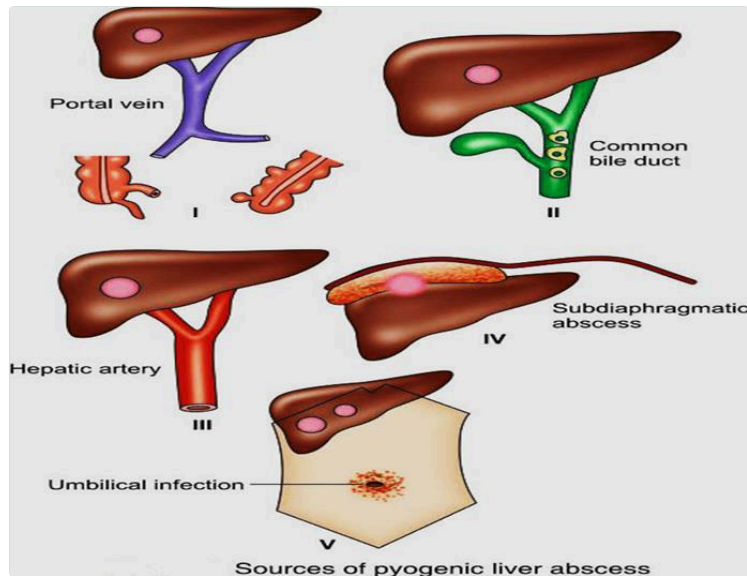


Figure 25.1 — Sources of pyogenic liver abscess

Causative Organisms:

gram -ve bacilli

The microorganisms most commonly isolated from blood and abscess cultures are as follows:

- **E coli – The commonest**
- *Klebsiella pneumoniae*
- *Bacteroides* species
- Streptococci and staph.
- Mixed infection is common

Prognosis

1- When left untreated, pyogenic liver abscess is associated with a mortality of 100 % . 2- With early diagnosis, appropriate drainage, and proper antibiotic therapy, the

Prognosis has improved markedly.

3- Poor prognostic factors are as follows:

- Age older than 70 years
- Multiple abscesses
- Poly-microbial infection
- Presence of associated malignancy or immunosuppressive disease

Clinical picture:

patient is markedly toxic and ill



General symptoms and signs

1- Fever and rigors 2- Toxic look and pallor 3- Weight loss

Local symptoms:

- 1- Right hypochondrial pain
- 2- Repeated vomiting
- 3- History of biliary disease as gall stones or jaundice.
- 4- History of G.I.T. infection as appendicitis.

Local signs:

- 1- Enlarged tender liver
- 2- Tenderness and rigidity in right hypochondrium.

Complications:

- 1- Septic shock → (fever, tachycardia and hypotension) fatal if neglected.
- 2- Rupture into the pleural cavity → empyema.
- 3- Peritoneal rupture → generalized peritonitis (generalized tenderness and rigidity).

Investigations:

Laboratory:

- CBC → Leucocytosis – increased ESR
 - Blood cultures for bacteria to identify the bacteria and determine which antibiotic is needed
 - Renal and liver function tests.

Radiological:

- 1- Abdominal ultrasound to check for an abscess in the liver
- 2- **CT scan** with intravenous contrast, to detect site and size of the abscess cavity.
- 3- Diagnostic Aspirations through US or CT scan for culture and sensitivity



Figure 25.2 — Acute abscess within the right lobe

Management of pyogenic liver abscess

■ Treatment approaches:

1- Initiation of Antibiotics therapy → a combination of two or more antibiotics is given intravenously (e.g. Third generation cephalosporins + Metronidazole). 2-

Percutaneous drainage is the treatment of choice:

- Must be guided by US or CT scan.
- Useful in critical ill patient.
- Used in solitary and superficial abscess.
- Usually followed by placement of a drainage catheter.
- The drain is removed when the abscess cavity collapses, as confirmed by CT.

Contra-indications:

- Ascites
- Coagulopathy (Bleeding tendency)
- Associated malignancy
- Multiple abscesses.

3- Surgical drainage:

- Used in failure of medical and per-cutaneous drainage.
- Used in multiple abscesses.
- In complicated cases with **peritonitis**.
- Open drainage of pyogenic liver abscess may be:

- a) Extra-peritoneal approach.
- b) Trans-peritoneal approach.



Amebic liver abscess

- It is the most frequent extra-intestinal manifestation of *Entamoeba histolytica* infection.
- This infection is caused by the protozoa *E. histolytica*, which enters the portal venous system from the colon.

Pathophysiology

***E. histolytica* exists in 2 forms:**

- The cyst stage is the infective form.
- The trophozoite stage causes invasive disease.

Mode of infection:

- Feco-oral route → cysts are ingested → reach the small intestine → trophozoites are released and settle in the cecum → Amebae then enter the portal circulation and travel to the liver where they form one or more amebic abscesses.
- The abscess cavity has thick wall and contains necrotic tissues and brown, thick fluid (called “anchovy paste”).
- **Right or left lobe ?**
 - The right lobe of the liver is more commonly affected than the left lobe.
 - This has been attributed to the facts that:
 - 1- The right lobe portal blood flow is supplied predominantly by the superior mesenteric vein, whereas the left lobe portal blood flow is supplied mainly by the splenic and inferior mesenteric vein.
 - 2- The right lobe receives more blood than left so the chance of infection is more.

Clinical picture:

A) General symptoms:

- 1- Fever is the most common symptom
- 2- Sometimes rigors (in acute cases)
- 3- Weight loss is more in chronic cases

B) Local symptoms:

- 1- Some patients give a history of dysentery (bloody diarrhea) within the previous few months (key for diagnosis) .



2- Abdominal pain:

- The most common symptom.
- Pain is usually constant and dull aching, frequently located in the right upper quadrant.
- 3- Nausea and vomiting.
- 4- Diarrhea sometimes present

C) General signs:

- 1- Fever present in all patients
- 2- Toxic look

D) Local signs:

- 1- Hepatomegaly
- 2- Right upper abdominal tenderness is a typical finding.
 - When the abscess is located in the left lobe, epigastric tenderness is noted.

3- Signs of peritonitis:

As rebound tenderness, guarding, and absence of bowel sounds, are present when the abscess ruptures into the peritoneal cavity.

Complications:

- 1- Rupture in the pleural cavity → pleural effusion or empyema
- 2- Rupture in peritoneal cavity → peritonitis.
- 3- Rupture in the pericardial sac. → with pericardial effusion in left lobe abscess
- 4- Jaundice when large abscess compressing the biliary tract.

Investigations:

A- Radiological:

- 1- **Abdominal ultrasound** : detect cystic lesion (abscess) in the liver
- 2- **Abdominal CT scan or MRI:**
 - Amebic abscesses usually appear as rounded, well-defined lesions with the presence of complex fluid
 - Thick wall (3–15 mm) and a peripheral zone of edema around the abscess are common and characteristic with sometimes multiple septa are present inside.
 - Usually solitary.



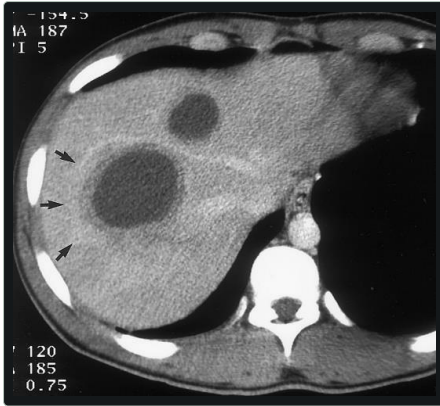


Figure 25.3 — Amebic abscess

B- Liver abscess aspiration with C/S:

- Check for bacterial infection in the amebic liver abscess
- The aspirate is brown, thick fluid (called “anchovy paste”).

C-Lab:

- CBC: Leukocytosis and esinophilia
 - Liver function tests: are elevated in case of marked amebic hepatitis.
 - Serological tests for amebiasis: ELISA
 - Patients with amebic liver abscesses will develop antibodies to *E. histolytica*.
 - Stool testing for amebiasis: Non-specific.

Treatment:

- 1- Amebicidal therapy is generally highly effective. Used in combination with broad spectrum antibiotics
 - 2- Per-cutaneous catheter drainage of amebic abscess is rarely necessary and used only in impending rupture of an abscess. Done under guidance of C.T. or U/S.
 - 3- Surgical drainage: extra-peritoneal or trans-peritoneal approaches
- **Amebicidal therapy:**
 - The treatment of choice is **Metronidazole**, • The cure rate for which is > **90%**.
 - Dose: 500-750 mg IV every 8 hours for 10 days.
 - Resolution of symptoms is fairly rapid and is observed within 3 days in most patients.

Hepato-cellular carcinoma (HCC)



26



Definition

Hepatocellular carcinoma (HCC) is the most common primary malignant tumor of the liver and arises from hepatocytes. It usually develops in a chronically diseased liver, particularly in patients with cirrhosis, but may also occur without cirrhosis, especially in chronic hepatitis B.

Risk Factors

The major risk factors include:

1. **Liver cirrhosis** — the major risk factor regardless of cause.
2. **Chronic viral hepatitis**
 - Chronic hepatitis B — HCC may occur with or without cirrhosis.
 - Chronic hepatitis C — risk is particularly high with advanced fibrosis or cirrhosis.
3. **Alcohol-associated liver disease**
4. **Metabolic dysfunction-associated steatotic liver disease (MASLD)** , particularly with advanced fibrosis or cirrhosis.
5. **Aflatoxin B1** exposure. 6. Other chronic liver diseases associated with advanced fibrosis or cirrhosis.

Clinical Presentation

HCC may be **asymptomatic** , particularly in its early stages, and may be detected during surveillance.

When symptomatic, patients may present with:

- Right upper quadrant or upper abdominal pain
- Weight loss and anorexia
- Fatigue
- Hepatomegaly or a palpable liver mass
- Jaundice
- Ascites
- Deterioration of previously compensated chronic liver disease

New deterioration of liver function or new portal vein thrombosis in a patient with chronic liver disease should raise suspicion of HCC.

HCC Surveillance

The aim of surveillance is to detect HCC at an **early stage when curative treatment may be possible** .

- Patients at sufficiently high risk, particularly those with cirrhosis, should undergo surveillance.



- **Abdominal ultrasound every 6 months** is the standard surveillance approach.
- AFP may be used as an additional biomarker in selected settings.
- Surveillance is not itself diagnostic; a suspicious lesion requires further evaluation.

Diagnosis

1. Ultrasound

Ultrasound is mainly used for surveillance.

A new or enlarging focal liver lesion in an at-risk patient requires further evaluation.

2. Multiphasic CT or MRI

HCC can often be diagnosed **without biopsy** in an appropriate high-risk patient when characteristic imaging findings are present.

The typical pattern is:

- **Arterial phase hyperenhancement (APHE)**
- Followed by **washout** in the portal venous or delayed phase.

3. Alpha-fetoprotein (AFP)

AFP may be elevated in HCC but has limited sensitivity and specificity.

- Normal AFP does **not exclude HCC**.
- Elevated AFP alone does **not establish the diagnosis**.

4. Liver Biopsy

Biopsy is not routinely required when characteristic imaging establishes the diagnosis in an appropriate high-risk patient.

It may be required when imaging is indeterminate or atypical.

Staging

The **Barcelona Clinic Liver Cancer (BCLC)** system is commonly used to stage HCC and guide treatment.

It considers:

- Tumor extent
- Liver function
- General condition/performance status
- Vascular invasion or extrahepatic spread

HCC can be broadly classified into:

- **Early stage** – potentially curable.



- **Intermediate stage** – mainly treated with locoregional therapy in selected patients.
- **Advanced stage** – usually associated with vascular invasion or extrahepatic spread and treated mainly with systemic therapy.
- **Terminal stage** – severe liver dysfunction and/or poor general condition; treatment is mainly supportive.

Treatment

Treatment depends on **tumor stage, liver function, general condition, vascular invasion, and extrahepatic spread**.

1. Curative-Intent Treatment

Liver resection

May provide a potential cure in selected patients with resectable HCC and preserved liver function.

■ Liver transplantation

Treats both the HCC and the underlying diseased liver and is particularly valuable in selected patients with HCC and significant cirrhosis who meet transplant criteria.

■ Thermal ablation

Radiofrequency ablation (RFA) and microwave ablation (MWA) are important potentially curative treatments for selected small HCCs.

2. Locoregional Treatment

Transarterial chemoembolization (TACE) is an important treatment for selected patients with intermediate-stage, liver-confined HCC.

3. Systemic Treatment

Systemic therapy is mainly used for advanced or unresectable HCC, particularly in the presence of vascular invasion or extrahepatic spread.

Modern systemic treatment is based mainly on **immunotherapy and targeted therapies**.

Key Points

- HCC is the most common primary malignant tumor of the liver.
- **Cirrhosis is the major risk factor**, but HCC can occur without cirrhosis, particularly with chronic HBV.
- Surveillance aims to detect HCC at an early, potentially curable stage.
- **Ultrasound every 6 months** is the standard surveillance approach in high-risk patients.



- HCC can often be diagnosed non-invasively by characteristic findings on **multiphasic CT or MRI** .
- **AFP alone cannot diagnose or exclude HCC.**
- Treatment depends on **tumor stage and the condition of the underlying liver** .
- Curative treatments include **resection, transplantation, and thermal ablation** .
- **TACE** is an important locoregional treatment for selected intermediate-stage disease.
- Advanced HCC is mainly treated with **systemic therapy** .

ASCITES



27



Ascites is the pathological accumulation of fluid in the peritoneal cavity , it is either due to a general cause as liver cirrhosis or right ventricular failure or due to a local cause as tuberculosis or malignant invasion of the peritoneum .

Cirrhosis is responsible for 80% of causes of ascites .

Several contributing factors are responsible for ascites formation , they are listed below.

A) Increased hydrostatic pressure:

1. Cirrhosis
2. Hepatic vein occlusion(Budd-Chiari syndrome)
3. Inferior vena cava obstruction
4. Constrictive pericarditis
5. Congestive heart failure I

B) Diminished colloid osmotic pressure:

1. End-stage liver disease with poor protein synthesis
2. Nephrotic syndrome with protein loss
3. Malnutrition
- 4 Protein-losing enteropathy

C) Increased permeability of peritoneal capillaries:

- I . Tuberculous peritonitis
2. Bacterial peritonitis
3. Malignant disease of the peritoneum

D)Leakage of fluid into the peritoneal cavity: .

1. Bile ascites
2. Pancreatic ascites(secondary to a leaking pseudocyst)
- 3.Chylous ascites
4. Urine ascites

Miscellaneous causes:

1. Myxedema
2. Ovarian disease(Meigs' syndrome)
3. Chronic hemodialysis

Mechanism of ascites formation in liver cirrhosis :

- Portal hypertension leads to transudation of fluid in the peritoneal cavity .
- Renal retention of sodium.
- Splanchnic and arterial vasodilatation .



- Increase hepatic lymph production due to congestion of sinusoids .
- Hypoalbuminemia due to decrease hepatic synthesis .

The mechanism involve inappropriate renal sodium and water retention either secondary to vascular changes (underfill and peripheral arterial vasodilatation theory) or as a primary event (overfill theory)

In the underfill theory : stimulation of the rennin- angiotensin – aldosterone system (RAAS) occurs as result of reduced effective circulating volume

Peripheral vasodilatation → diminished effective arterial blood volume → rise in rennin , aldosterone , vasopressin , and sympathetic stimulation → sodium and water retention and renal vasoconstriction → plasma volume expansion + portal hypertension → Ascites

In the overfill theory : The liver send a signal which leads to renal sodium and water retention and in the presence of portal hypertension , fluid will accumulates in the peritoneal cavity

Clinical Picture:

The onset may be sudden due to a precipitating factor or may be gradual

Ascites may be graded as follow:

Grade 1 :Mild ascites: only detectable by ultrasound examination

Grade 2 : Moderate ascites : manifest by moderate symmetrical distention of abdomen

Grade 3 : Large or gross ascites provokes marked abdominal distension

Examination :

- Physical signs suggestive of cirrhosis: Jaundice , palmer erythema, abdominal collaterals.....
- Abdominal distention & thin limbs “spider man’.
- Wide subcostal angle, divarication of recti, umbilicus is shifted downwards, everted ± hernia.
- Scrotal oedema and hydrocele .
- Abdominal wall veins may be apparent due to: caput medusa (portal hypertension) or due to compression of the IVC
- Palpation : Liver and spleen could be palpated by dipping. Fluid thrill could be felt
- Percussion : dullness in the flanks. Shifting dullness.

Secondary effects : Pleural effusion may occur in 5-10% of patients usually right sided this is due to defect in the diaphragm. hydrothorax may be seen in the absence of ascites, due to negative intrathoracic pressure during breathing.

Edema in lower limb: follow ascites, due to hypoalbuminemia.

- Cardiac apex is shifted up & out.

Neck veins could be seen congested: due to increase intrapleural pressure & raised diaphragm.



Investigations :

- Ultrasonography: to diagnose ascites, whether fluid is free or localized, to detect liver cirrhosis

Diagnostic paracentesis and ascitic fluid analysis:

a) Cells: endothelial cells/lymphocytes/PNLs/malignant cells.

b) Culture.

c) Protein: ascitic fluid total protein (AFTP) < 2.5 g/dL.

d) Albumin: is important to calculate Serum-Ascites Albumin Gradient (SAAG)

SAAG = Serum albumin – ascitic fluid albumin.

In cirrhosis, SAAG is > 1.1g/dL.

Differential diagnosis of ascites based on SAAG

Low gradient SAAG < 1.1 g/dl	High gradient ≥ 1.1 g/dl
TB peritonitis	Cirrhosis
Pancreatic Ascites	Fulminant liver failure
Peritoneal carcinomatosis	Alcoholic hepatitis
Biliary ascites	Cardiac ascites
Nephrotic syndrome	Budd – Chiari syndrome

Differential Diagnosis of ascites and its causes:

1-Tuberculous ascites :

- History of TB toxemia.
Matted intestinal loops.
Abdominal lymph nodes may be enlarged.
- Ascitic fluid: Ascitic fluid total protein (AFTP) > 2.5 g/dL, SAAG < 1.1 g/dL , excess lymphocytes. Smear & culture for TB. 4- PCR.
- Laparoscopy and biopsy.

2 -Malignant ascites:

- Symptoms & signs of primary tumor.
- Ascitic fluid: - AFTP > 2.5 g/dL
- SAAG < 1 . 1 g/dL
- Malignant cells. - ”

Increase Lactic acid dehydrogenase.

3-Chylous ascites : It may be caused by malignant lymphoma. The paracentesis sample is characteristic as it is milky white due to ++ fat & clears with ether.



4-Constrictive pericarditis:

- Increase jugular venous pressure. - Chest X-ray : calcification. - ECG & CT of the heart.

5-Budd- Chiari Syndrome:

Increased protein of ascitic fluid. Abdominal ultrasound: congested veins, caudate lobe is prominent. Doppler U/S: shows obstruction of hepatic veins.

6-Ovarian tumor:

- Resonance in the flanks, central distension. - Umbilicus is shifted upwards.

7-Full bladder, pregnancy, flatulence, fatty.

Management of cirrhotic ascites :

- Body weight and urinary volume are measured daily
- Measurement of serum and urinary electrolytes
- Diagnostic paracentesis
- Bed rest is advised

Salt restriction : This is the main stay of therapy in cirrhotic patients with ascites

- Diuretics :

Aldosterone antagonist : Spironolactone (Aldactone) , amiloride : They block

- sodium reabsorption in the distal and collecting tubules , and they conserve potassium

Loop diuretic : Furosemide (Lasix) inhibits sodium , potassium and chloride

- reabsorption from the ascending limb of loop of Henle .
- Two therapeutic approach :

Spironolactone alone : This can be used alone in mild ascites with a dose of 100 –

200 mg daily

- Combination therapy Spironolactone 100 mg and furosemide 40 mg
- Sufficient response means loss of 0.5 Kg/ day or a maximum of 1 Kg /day in the - presence of lower limb edema

If the response is insufficient the dose of spironolactone and furosemide can be - increased gradually on the expense of development of side effects , maximum of dose of furosemid is 160 mg /day and of spironolactone is 400 mg / day .



- Salt-free albumin I.V followed by IV Furosemide increases intravascular volume - and increases diuresis

Small volume paracentesis: for < 5 liters removed, I.V. and in this case volume - expanders are unnecessary .

- **Complications limiting diuretic therapy of ascites**

- Development of hepatic encephalopathy , in the absence of any other precipitating factor (It is better to leave the patient wet and wise than dry and demented !)

Renal impairment: increase of serum creatinine by >100% to a value >2 mg/dl in patients with ascites responding to treatment

Hyponatraemia: a decrease of serum sodium by >10 mmol/L to a serum sodium of <125 mmol/L

Hypo- or hyperkalaemia: a change in serum potassium to <3 mmol/L or >6 mmol/L despite appropriate measures

Incapacitating muscle cramps

Treatment options for refractory ascites

- An ascites is said to be resistant , if there is lack of response after dietary salt restriction& intensive diuretic therapy
- **Large volume paracentesis:** removal of > 5 L of fluid by therapeutic paracentesis with administration of IV albumin (8gm/ L removed) may decrease the post paracentesis hypovolemia. It is safe& effective.
- **TIPS:** bridges a branch of the hepatic vein with a branch of the portal vein . It may increase frequency of hepatic encephalopathy.
- **Liver transplantation.**

Bacterial infection in patients with liver cirrhosis

- Patients with cirrhosis are at risk of developing bacterial infections due to various factors including liver dysfunction , portosystemic dysfunction , bacterial translocation and immune dysfunction
- Bacterial infection in cirrhotics should be treated quickly and effectively with potent antibiotic
- The most common sites for bacterial infection in patients with liver cirrhosis that requires potent empirical antibiotic treatment are : the ascitic fluid (spontaneous bacterial peritonitis) , lung (pneumonia) , Urinary tract (pyelonephritis) and skin (cellulitis)



Spontaneous Bacterial Peritonitis (SBP)

- It represents a serious complication of ascites with cirrhosis , and it affects 10% of cirrhotic patients. The prognosis has improved over time, but mortality rates remain relatively high (about 20% for SBP]

Cause:

Root of infection:

- Hematogenous spread
- Translocation of bacteria through the intestinal wall
 - Organisms: monomicrobial (single organism), E-coli, pneumococci, Klebsiella and gram-positive organisms.
 - Abnormal host defenses: due to .decrease ascitic opsonins and abnormal neutrophil function.
 - SBP is an indication to list the patient for liver transplantation

Clinical Picture & Diagnosis:

- The condition should be suspected in any cirrhotic patient with ascites who clinically deteriorates . The patient becomes drowsy and confused .
- Symptoms and signs of infection : fever, abdominal pain, rebound tenderness, nausea, vomiting& may be diarrhea.
- Diagnostic paracentesis: polymorphs in ascitic fluid > 250 cells/mm³. Ascitic fluid culture.

Treatment:

- Albumin has been recommended in association with antibiotics for the treatment of SBP
- Third-generation cephalosporins are recommended as first-line antibiotic treatment for community-acquired SBP in countries with low rates of antibiotic resistance
- In countries with high rates of antibiotic resistance piperacillin/tazobactam or carbapenem should be considered
- Antibiotic efficacy should be checked with a second diagnostic paracentesis at 48 hours from starting treatment .Suspicion of failure of first-line antibiotic if worsening clinical signs and symptoms and/or increase, or no marked reduction in leucocyte count (at least 25%) in 48 hours
- Example of antibiotics used in SBP : Cefotaxime 2g/8h IV , ceftriaxone 1g/ 12 h ,Ciprofloxacin 500mg/12hours

Prognosis: 25% of patients die. 70% recurrence in the first year.



Prophylaxis:

Primary prophylaxis : Patients with cirrhosis and low ascitic fluid protein concentration (<10 g/L) and/or high serum bilirubin levels are at high risk of developing a first episode of SBP , they can be given primary prophylaxis by norfloxacin 400 mg/day .

Secondary prophylaxis : Norfloxacin 400mg/day for life, after the first attack

Patients doing sclerotherapy should receive norfloxacin 400mg/12 hours days.

Norfloxacin prophylaxis should be stopped in patients with long-lasting improvement of their clinical condition and disappearance of ascites



Renal impairment in liver cirrhosis



28



Hepatorenal syndrome (HRS) : It is a functional liver failure occurring in patients with liver cirrhosis , portal hypertension and ascites , the histology of the kidney is normal , The hallmark of HRS is renal vasoconstriction, although the pathogenesis is not fully understood , however the arterial vasodilatation theory is the most accepted Peripheral and splanchnic vasodilation → Decreased effective blood volume

→ Activation of RAAS, ADH, sympathetic → Increase Renal vasoconstrictors (rennin , ADH, angiotensin II, sympathetic stimulation) this is associated also with decrease renal vasodilators (PGE2, NO)

- **Acute kidney injury (AKI)** : It is a common complication of advanced cirrhosis, and is predictive of a worse outcome.
- Early recognition of AKI and early identification of the phenotype of AKI are essential to initiate appropriate therapy at an early stage and to improve outcome.
- Patients with advanced cirrhosis have reduced muscle mass (especially women) and decreased creatine production. Thus, patients with serum creatinine within the normal range may have markedly impaired renal function.
- The definitions of AKI in cirrhosis have recently been revised to take into account these difficulties. .AKI in cirrhosis is now defined by an increase in serum creatinine ≥ 0.3 mg/dL within 48 hours or a percentage increase of serum creatinine $\geq 50\%$ from baseline which is known or presumed to have occurred within the prior 7 days.

AKI is then categorized into 3 stages of increasing severity :

Differentiating types of AKI

- All types of AKI can occur in patients with cirrhosis (Pre-renal, HRS, acute tubular necrosis (ATN) and post-renal)
- Key point is to differentiate HRS-AKI from ATN
- Classification of HRS was recently revised
 - HRS-AKI (formerly type I HRS)
 - HRS includes renal impairment that fulfills the criteria of HRS but not of AKI (non-AKI-HRS or NAKI) (formerly type II HRS)

Diagnostic criteria for HRS-AKI

Cirrhosis and ascites

No response after 2 consecutive days of diuretic withdrawal and plasma volume expansion with albumin 1 g per kg of body weight

Absence of shock

No current or recent use of nephrotoxic drugs (NSAIDs, aminoglycosides, iodinated contrast media, etc.)



– No macroscopic signs of structural kidney injury, defined as: Absence of proteinuria (>500 mg/day) and absence of microhaematuria (>50 RBCs per high power field) and normal findings on renal ultrasonography

All patients meeting the current definition of **HRS-AKI** should be treated with

vasoconstrictors and albumin

.A distinction must be made between HRS and acute tubular necrosis (ATN) because vasoconstrictors such as terlipressin are only justified in the presence of HRS. Besides criteria for the clinical definition, urinary neutrophil gelatinase (NGAL) associated lipocalin, a biomarker of tubular damage, is significantly higher in patients with ATN compared to those with HRS.

Terlipressin can be administered by IV boluses (1 mg every 4–6 hours) or by continuous IV infusion (2 mg/day)[†]

In case of non-response (decrease in SCr <25% from the peak value) after 2 days, the dose of terlipressin should be increased in a stepwise manner to a maximum of 12 mg/day

Albumin solution (20%) should be used at 20–40 g/day

Serial measures assessing central blood volume can help to titrate the dose of albumin to prevent circulatory overload

Noradrenaline can be an alternative to terlipressin[‡]

Requires a central venous line often in an ICU

Midodrine + octreotide can be an option when terlipressin or noradrenaline are unavailable (but efficacy is much lower)

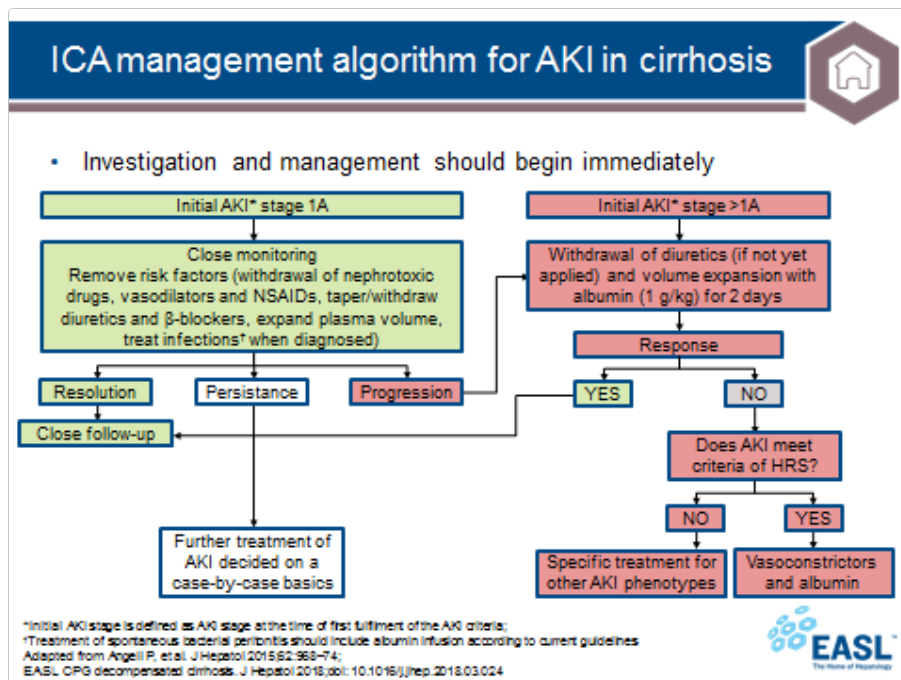


Figure 28.1 — ICA management algorithm for AKI in cirrhosis

JAUNDICE



29



- Jaundice, from the French jaune (yellow), is the yellow-orange discoloration of the skin, conjunctivae, and mucous membranes that results from elevated concentrations of bilirubin in plasma.
- Jaundice becomes evident at plasma bilirubin concentrations of 3 to 4 mg/dL, depending on the patient's normal pigmentation.

Approach to the patient with jaundice

- The bilirubin present in serum represents a balance between input from production of bilirubin and hepatic/biliary removal of the pigment.
- Hyperbilirubinemia may result from (1) overproduction of bilirubin; (2) impaired uptake, conjugation, or excretion of bilirubin; or (3) regurgitation of unconjugated or conjugated bilirubin from damaged hepatocytes or bile ducts.
- In other words hyperbilirubinemia and jaundice may result from A) isolated disorders of bilirubin metabolism, B) liver disease, or C) Obstruction of the biliary tract.
- An increase in unconjugated bilirubin in serum results from either 1) overproduction, 2) impairment of uptake, or conjugation of bilirubin.
- An increase in conjugated bilirubin is due to 1) decreased excretion into the bile ductules or 2) backward leakage of the pigment.
- The initial steps in evaluating the patient with jaundice are to determine (1) whether the hyperbilirubinemia is predominantly conjugated or unconjugated in nature (fractionation of Bilirubin), and (2) whether other biochemical liver tests are abnormal.

Differential Diagnosis of Jaundice and Hyperbilirubinemia

Isolated Disorders Of Bilirubin Metabolism

Unconjugated Hyperbilirubinemia

- Increased bilirubin production

Examples: hemolysis

- Decreased hepatocellular uptake

Examples: drugs (e.g., rifampin)

- Decreased conjugation

Examples: Gilbert's and Crigler-Najjar syndromes

Conjugated Or Mixed Hyperbilirubinemia

Decreased canalicular transport: Dubin-Johnson syndrome



Gilbert's Syndrome

- It is defined as benign, familial, mild, unconjugated hyperbilirubinaemia not due to haemolysis and with normal routine tests of liver function and hepatic histology.
- Jaundice is mild and intermittent.
- Deepening may follow an intercurrent infection or fasting and is associated with malaise, nausea and often discomfort over the liver.
- Patients with Gilbert's syndrome have a deficiency in hepatic bilirubin glucuronidation.
- Hyperbilirubinaemia is life long and not associated with increased morbidity.

Hepatocellular Conditions That May Produce Jaundice

Viral Hepatitis

Hepatitis A, B, C, D, and E

Epstein-Barr virus

Cytomegalovirus

Alcohol

Drug toxicity

Environmental Toxins

Vinyl chloride

Wilson's disease

Autoimmune hepatitis

Cholestatic Conditions That May Produce Jaundice

- Cholestasis is defined as the failure of normal bile to reach the duodenum.
- This may be due to pathology anywhere between the hepatocyte and the ampulla of Vater.
- The term 'obstructive jaundice' is not used, as in many instances no mechanical block can be shown in the biliary tract.

Syndrome Of Cholestasis

- Cholestasis is interference with bile flow or formation.
- This can occur anywhere between the hepatocyte and the ampulla of Vater.
- Functionally, cholestasis is defined as a decrease in canalicular bile flow.
- Morphologically, cholestasis is defined as the accumulation of bile in liver cells and biliary passages.
- Clinically, cholestasis is the retention in the blood of all substances normally excreted in the bile.



- The importance of the distinction between extra and intra-hepatic cholestasis is that symptoms and biochemistry may not separate them.
- There is a need for a diagnostic algorithm to differentiate between the two.

Classification

Extra-hepatic Cholestasis

- Encompasses conditions where there is physical obstruction to the bile ducts.
- Usually this is outside the liver.

Intra-hepatic cholestasis

- Includes those conditions where there is no demonstrable obstruction.

Intrahepatic Conditions Producing Cholestasis

Viral hepatitis

Fibrosing cholestatic hepatitis—hepatitis B and C

Hepatitis A, Epstein-Barr virus

Alcoholic hepatitis

Drug toxicity

Pure cholestasis—anabolic and contraceptive steroids

Cholestatic hepatitis—chlorpromazine

Chronic cholestasis—chlorpromazine

Primary biliary cirrhosis

Primary sclerosing cholangitis

Cholestasis of pregnancy

Benign postoperative cholestasis

Extrahepatic Conditions Producing Cholestasis

Malignant

Cholangiocarcinoma

Pancreatic cancer

Ampullary cancer

Malignant involvement of the porta hepatis lymph nodes

Benign

Choledocholithiasis

Postoperative biliary structures

Chronic pancreatitis





Clinical features of cholestasis

- Prominent features of cholestasis, both acute and chronic, are itching and malabsorption.
- Bone disease (hepatic osteodystrophy) and cholesterol deposition (xanthomas, xanthelasmas) are seen with chronic cholestasis, which is also associated with skin pigmentation.
- In contrast to the patient with hepato-cellular disease where there is malaise and physical deterioration, the cholestatic patient feels well.
- On examination, the liver is usually enlarged.
- Splenomegaly is unusual except in biliary cirrhosis where portal hypertension has developed.
- Steatorrhea is proportional to the degree of jaundice. It is due to the lack of sufficient intestinal bile salts for the absorption of dietary fat and fat-soluble vitamins.
- Hepatic osteodystrophy: Bone disease is a complication of chronic liver disease and in particular chronic cholestasis.
- Bone pain and fractures occur.
- Possible mechanisms are osteomalacia and osteoporosis.
- Bone disease manifests as loss of height, back pain (usually mid-thoracic or lumbar), collapsed vertebrae and fractures particularly of ribs with minimal trauma.

Extra-hepatic Effects Of Cholestasis

- Itching and jaundice are self-evident, but there are numerous other less obvious effects of cholestasis.
- Cardio-vascular responses are abnormal and peripheral vaso-constriction in response to hypotension is impaired.
- The kidneys have an increased susceptibility to hypotension and hypoxic damage.
- The gastric mucosa is more susceptible to ulceration
- Thus although deeply jaundiced patients with cholestasis may appear well apart from itching, there are metabolic and functional changes that under the stress of surgical and non-surgical procedures may result in: acute renal failure, haemorrhage, wound dehiscence and an increased risk of sepsis

Laboratory findings in cholestasis

- All the constituents of the bile show an increased level in the serum. Conjugation of biliary substances is intact but excretion defective
- The serum conjugated bilirubin level is raised
- The serum alkaline phosphatase level is raised, usually to more than three times the upper limit of normal



- Serum γ glutamyl- transpeptidase levels are raised.
- Serum albumin and globulin concentrations are normal.
- The serum aspartate transaminase is usually less than 100iu/l.
- Urine. Conjugated bilirubin is present.

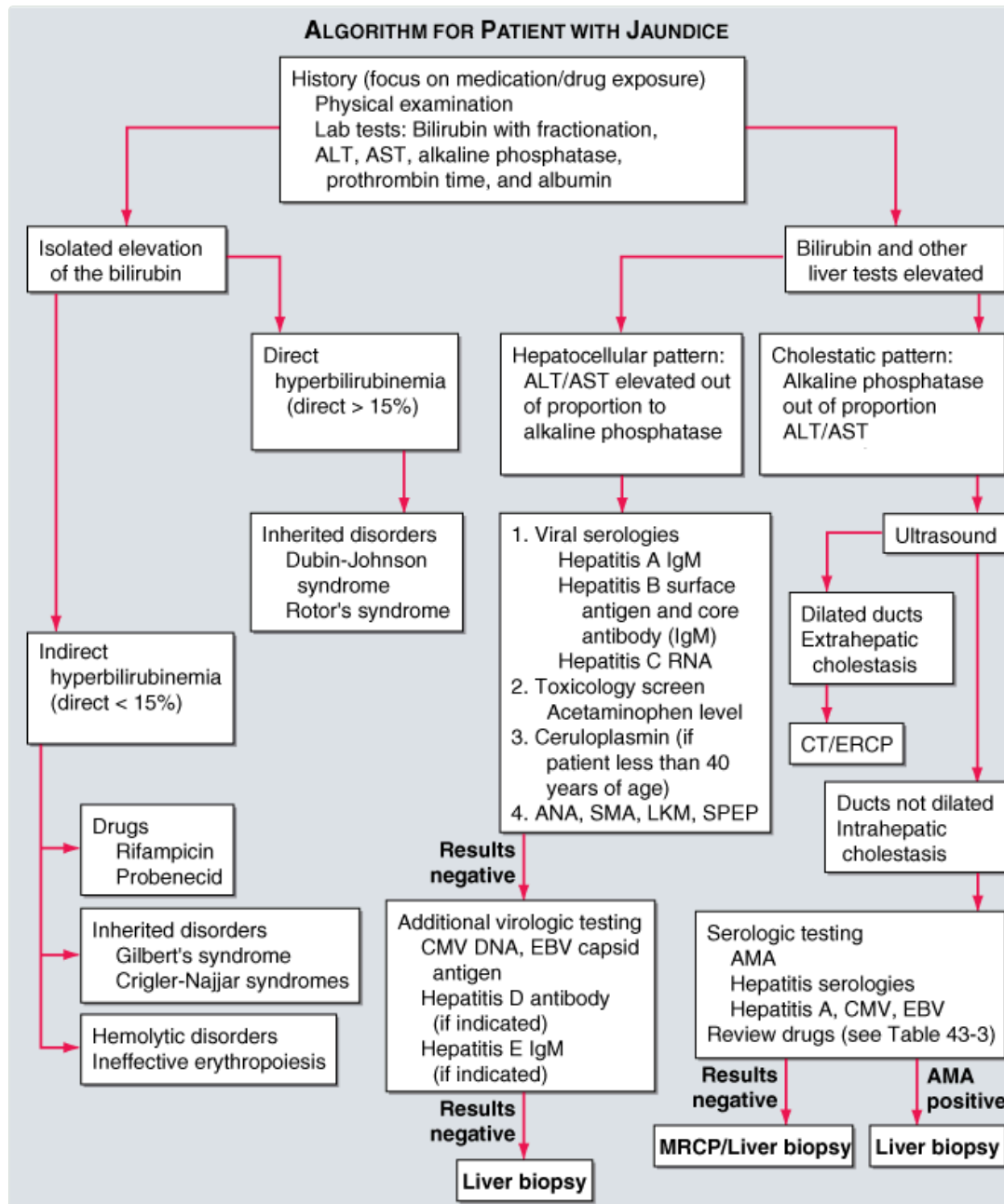


Figure 29.1 — Approach to the patient with jaundice

HEPATIC ENCEPHALOPATHY



30



Objectives

The student will

- 1) Understand the definition and various forms of hepatic encephalopathy
- 2) Know the various causes of hepatic encephalopathy
- 3) Understand the pathogenesis of hepatic encephalopathy
- 4) Know the clinical presentation of hepatic encephalopathy
- 5) Know the diagnostic tools for hepatic encephalopathy
- 6) Understand the basic treatment options for hepatic encephalopathy

- Hepatic encephalopathy is a reversible neuropsychiatric state that complicates liver disease.
- A spectrum of syndromes exists, and it may accompany various forms of liver diseases.
- Types of encephalopathy

Type of encephalopathy	% Survival	Aetiological factors
Acute liver failure	20*	Viral hepatitis Alcoholic hepatitis Drug reactions and overdose
Cirrhosis with precipitant	70–80	Diuresis Haemorrhage Paracentesis Diarrhoea and vomiting Surgery Alcoholic excess Sedatives Infections Constipation
Chronic portal-systemic encephalopathy	100	Portal-systemic shunting Dietary protein intake Intestinal bacteria

Figure 30.1 — Types of encephalopathy

Clinical features

- The picture is complex and affects all parts of the brain.
- There are neurological and psychiatric components.
- For descriptive purposes, features of encephalopathy can be separated into changes in consciousness, personality, intellect and speech.



■ Disturbed consciousness:

Hypersomnia appears early and progresses to reversal of the normal sleep pattern. Further deterioration results in reaction only to intense or noxious stimuli.

■ Personality changes:

These include childishness, irritability and loss of concern for family.

■ Intellectual deterioration

Varies from slight impairment of organic mental function to gross confusion.

■ Speech

Is slow and slurred and the voice is monotonous.

■ The most characteristic neurological abnormality is the 'flapping' tremor (asterixis).

This is due to impaired inflow of joint and other afferent information to the brainstem reticular formation resulting in lapses in posture.

Sometimes arms, neck, jaw, protruded tongue, retracted mouth and tightly closed eyelids are involved, and the gait is ataxic.

Precipitants Of Acute Hepatic Encephalopathy In The Cirrhotic Patient

Electrolyte imbalance

Diuretics

Vomiting

Diarrhoea

Bleeding

Oesophageal and gastric varices

Gastro-duodenal erosions

Mallory-Weiss tear

Drugs

Infection

Spontaneous bacterial peritonitis

Urinary

Chest

Constipation

Large protein meal





Pathogenetic mechanisms

- The basic processes are 1) failure of hepatic clearance of gut derived substances, either through hepato-cellular failure or shunting, and 2) altered amino acid metabolism, both of which result in changes in cerebral neurotransmission. 3) Ammonia is thought to play an important role; other neurotransmitter systems are strongly implicated.

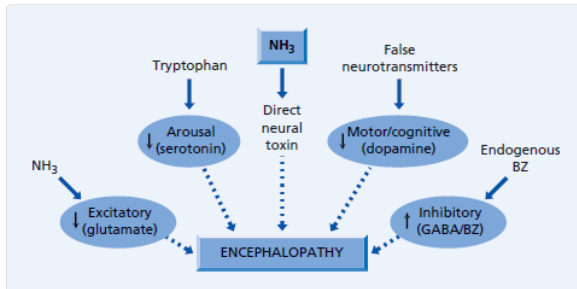


Figure 30.2 — Pathogenetic mechanisms of encephalopathy

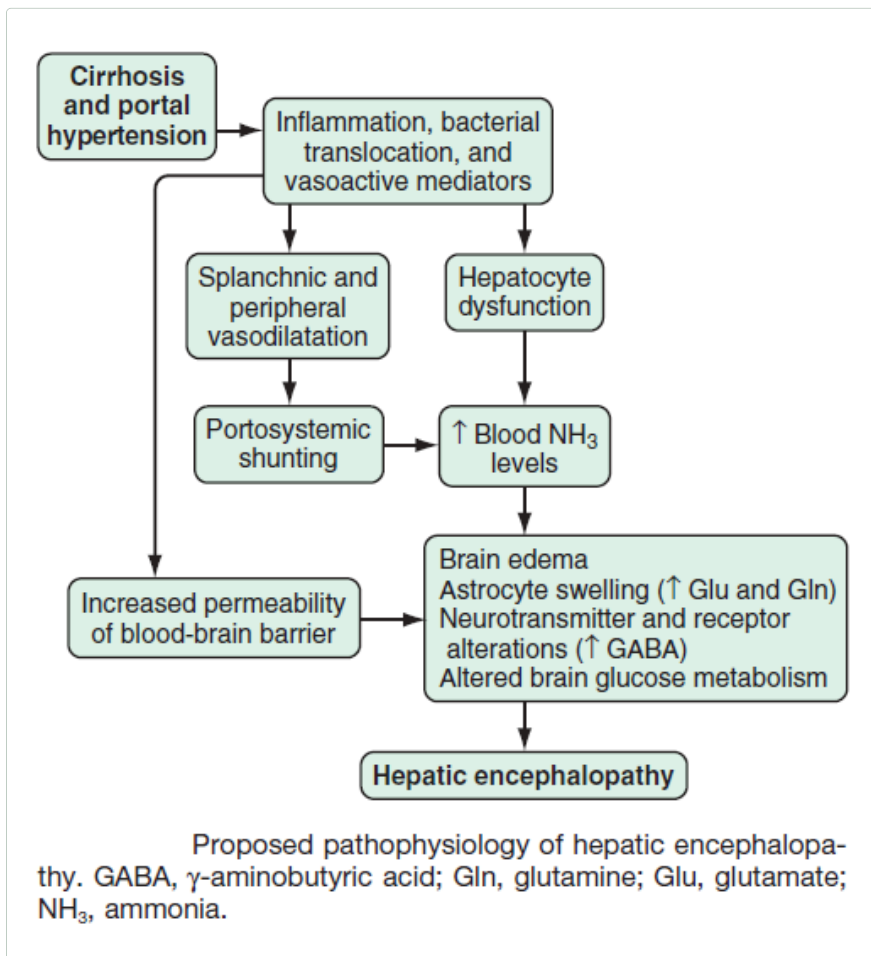


Figure 30.3 — Proposed pathophysiology of hepatic encephalopathy





Treatment of hepatic encephalopathy

Diet

University source discrepancy. Handout p341 gives 20 g/day during an acute attack. The same handout's p279 nutrition update states 1.2–1.5 g/kg/day even with hepatic encephalopathy and permits occasional temporary reduction during severe overt encephalopathy. The 20 g/day statement below is retained as an unresolved source claim, not an editorial endorsement. See [Cirrhosis – Nutrition](#).

- In the acute attack dietary protein is reduced to 20 g/day.
- In patients after an acute episode of coma, a normal protein intake is soon achieved.
- It is important in cirrhotic patients to avoid protein restriction for any longer than is necessary since these patients have a higher than normal protein requirement (1.2 g/kg/day) to remain in positive balance.
- Vegetable protein may be used. The latter is less ammoniagenic and contains small amounts of methionine and aromatic amino acids.

Antibiotics

- Neomycin is a nonabsorbable aminoglycoside given orally in doses of 2 to 6 g per day.
- Metronidazole has become the oral antibiotic of choice for acute management of hepatic encephalopathy when lactulose alone does not achieve adequate results.
 - Rifaximin (Xifaxan) has recently shown to be effective in the treatment of PSE in the dose range of 400 mg 2-3 times daily. It is a nonabsorbable, gut-specific antibiotic that alters the intestinal flora with any systemic side effects.

Lactulose and lactitol

- The human intestinal mucosa does not have an enzyme to split these synthetic disaccharides.
- When given by mouth lactulose reaches the caecum where it is broken down by bacteria predominantly to lactic acid.
- The osmotic volume of the colon is increased and the faecal pH drops.
- The colonic fermentative bacteria prefer lactulose to blood when both are present.
- The growth of lactose-fermenting organisms is favored and organisms such as bacteroides, which are ammonia formers, are suppressed.
- Faecal acidity would reduce the ionization and hence absorption of ammonia (also amines and other toxic nitrogenous compounds).
- Side-effects include flatulence, diarrhea and intestinal pain.

Purgation

Lactulose or lactose enemas may be used and are superior to water

**Sodium benzoate and L-ornithine -L-aspartate**

Sodium benzoate promotes urinary excretion of ammonia and is as effective as lactulose and is less expensive.

L-ornithine-L-aspartate treatment promotes hepatic removal of ammonia by stimulating residual hepatic urea cycle activity and promoting glutamine synthesis, particularly in skeletal muscle.

Levodopa and bromocriptine

If portal-systemic encephalopathy is related to a defect in dopaminergic neurotransmission then replenishment of cerebral dopamines should be beneficial

Dopamine does not pass the blood-brain barrier, but its precursor, levodopa, does and can cause temporary arousal in acute hepatic encephalopathy.

Bromocriptine is a specific dopamine receptor agonist with a prolonged action.

Branched-chain amino acids

In cirrhotic patients the serum branched chain amino acids valine, leucine and isoleucine are low, and aromatic amino acid levels are increased.

Infusions of solutions containing a high concentration of branched-chain amino acids have been used to treat acute and chronic hepatic encephalopathy.

Hepatic transplantation

This may be the ultimate answer to the problem of chronic hepatic encephalopathy



Acute Hepatic Failure



31



ACUTE HEPATIC FAILURE (FULMINANT HEPATITIS) AND JAUNDICE

Objectives

The student will

- 1) Understand the definition of fulminant hepatitis
- 2) Know the various causes of acute hepatic failure
- 3) Know the clinical features of acute hepatic failure
- 4) Know the diagnostic studies in acute hepatic failure
- 5) Know the basic treatment options for acute hepatic failure
- 6) Know the basic mechanisms of jaundice
- 7) Know the different causes of jaundice
- 8) Know the clinical presentation of jaundice
- 9) Know the basic diagnostic studies for jaundice

ACUTE LIVER FAILURE (FULMINANT HEPATITIS)

- Acute liver failure describes the clinical syndrome of severe impairment of liver function (Encephalopathy, Coagulopathy And Jaundice) Within 6 Months Of The Onset Of Symptoms.
- Coagulopathy and Encephalopathy are the twin cardinal features that reflect the severity of liver injury, and both are required to make the clinical diagnosis of ALF.
- The vast majority of patients have no evidence of preexisting liver disease, but 2 exceptions to this requirement have been made: (1) an acute presentation of Wilson disease in a young adult that presents in the setting of previously asymptomatic cirrhosis and (2) Reactivation of HBV infection, either spontaneously or as a consequence of immunosuppression or chemotherapy.

Classification

Classification of acute liver failure				
Liver failure subcategory	Jaundice to encephalopathy	Clinical presentation	Common aetiologies	Prognosis
<u>Hyperacute</u>	0-7 days	Cerebral oedema common	Paracetamol, hepatitis A, ischaemia	Fair
<u>Acute</u>	8-28 days	Cerebral oedema less common	Hepatitis B, drugs	Poor
<u>Subacute</u>	29 days to 12 weeks	Cerebral oedema rare; ascites, peripheral oedema and renal failure more common	Drugs, indeterminate	Very poor

Figure 31.1 — Classification of acute liver failure



Causes:

Causes of acute liver failure	
<u>Infections</u>	
	Hepatitis A, B, C, D, E
	Herpes simplex
	Epstein-Barr virus
	Cytomegalovirus
	Transfusion-transmitted virus (TTV)
	Dengue fever
<u>Drugs and toxins</u>	
	Paracetamol
	Carbon tetrachloride
	Idiosyncratic drug reactions*
	Mushroom poisoning
	Sea anemone sting
<u>Ischaemic</u>	
	Cardiogenic shock
	Hypotension
	Heat stroke
	Cocaine, methamphetamines, ephedrine
<u>Vascular</u>	
	Acute Budd-Chiari syndrome
	Sinusoidal obstruction syndrome
<u>Miscellaneous</u>	
	Wilson's disease
	Acute fatty liver of pregnancy
	Eclampsia/ HELLP syndrome
	Malignancy
	Primary graft non-function after liver transplantation

Figure 31.2 — Causes of acute liver failure

Clinical Features And Diagnosis

The diagnosis of ALF is clinical and based on the presence of encephalopathy in a patient with acute liver injury manifested by coagulopathy.

The encephalopathy is usually overt and ranges from confusion to coma.

1) Encephalopathy

Encephalopathy is mandatory for a diagnosis of ALF. It is classically graded on a scale of 1 to 4

2) Intracranial Hypertension and Cerebral Edema

Cerebral edema has been a hallmark complication of ALF, major cause of death.

The clinical features of cerebral edema include systemic hypertension, decerebrate posturing, hyperventilation, abnormal pupillary reflexes, and ultimately impairment of brainstem reflexes and functions.

3) Hemodynamic Changes and Circulatory Failure

The initial profile is of a hyperdynamic circulation, increased cardiac output, and reduced systemic peripheral vascular resistance.



Progressive disease leads to circulatory failure either as a result of a falling cardiac output or an inability to maintain an adequate mean arterial pressure.

This is a common cause of death in ALF.

4) Infection

Patients with ALF are notoriously prone to infection The predominant bacteria were

Staphylococcus aureus and Streptococci.

5) Renal Failure

In most non-acetaminophen cases, renal failure tends to be a consequence of liver failure and develops later in the course of the disease.

6) Hematologic Abnormalities

In addition to decreased synthesis of coagulation factors by the liver, there is evidence of increased peripheral consumption.

Quantitative and qualitative defects in platelet function are well described in ALF.

Diagnostic Studies

<u>Haematology</u>
Complete blood count: white blood cells, haemoglobin, haematocrit, anti-hepatitis B core
Coagulation panel: prothrombin time/INR, factor V
Blood group
<u>Biochemical</u>
Serum chemistries: sodium, potassium, bicarbonate, chloride, urea, creatinine, calcium, magnesium, phosphate, glucose
Hepatic panel: aspartate aminotransferase (AST), alanine aminotransferase (ALT), alkaline phosphatase, albumin, total protein, total bilirubin
<u>Arterial blood gas</u>
pH, $PaCO_2$, PaO_2 , lactate, ammonia
<u>Virology</u>
Hepatitis B surface antigen and IgM anticore
Hepatitis A (IgM) antibody
Hepatitis C antibody (for underlying chronic infection)
HCV RNA
IgM hepatitis E antibody (in endemic areas)
Hepatitis D antibody if hepatitis B positive
HSV, CMV, EBV PCR (if history of immunosuppression)
Human immunodeficiency virus (if considering transplantation)

Figure 31.3 — Diagnostic studies: laboratory tests



<u>Autoimmune markers</u>
Antinuclear antibody (ANA)
Antismooth-muscle antibody (SMA)
Antiliver/ kidney microsome 1 (ALKM1)
Immunoglobulins
<u>Toxicology</u>
Paracetamol level
Blood alcohol
Urine drug screen
<u>Miscellaneous</u>
Urine copper
Pregnancy test
<u>Microbiology</u>
Blood culture, aerobic and anaerobic
Urine culture and microscopy
Sputum culture and microscopy
<u>Other studies</u>
Chest X-ray, electrocardiogram
Liver ultrasound with Dopplers
Electroencephalogram (EEG), non-contrast head CT (in some cases)
Liver biopsy

Figure 31.4 — Diagnostic studies: additional investigations

Management

1) Neurologic Complications

- Management of the portosystemic encephalopathy of cirrhosis has centered on lactulose and non - absorbable antibiotics. However, lactulose has shown no benefit in acute liver failure.
- There is not enough evidence to recommend non - absorbable antibiotics in acute liver failure. In cirrhosis, L-ornithine and L-aspartate (LOLA) treats encephalopathy by increasing muscle metabolism of ammonia but a recent randomized controlled trial showed no benefit in acute liver failure.
- Therefore, the management of the encephalopathy is essentially that of the underlying liver disease.

2) Treatment of cerebral edema

- Mannitol (0.25 - 1 g/kg i.v. bolus): use for severe elevation of ICP or first clinical signs of herniation, monitor urine output and serum osmolarity.
- Hypertonic saline (30% 5 - 10 mL/h): goal serum sodium 145 - 155 mmol/L, avoid rapid correction.
- Hyperventilation: effects short - lived; may use for impending herniation.

3) Infection

- Aseptic techniques.
- Surveillance for and prompt antimicrobial treatment of infection required.



- Antifungal coverage for patients not responding to broad spectrum antibiotics. • Antibiotic prophylaxis possibly helpful but not proven.

4) Coagulopathy

- Vitamin K (10 mg i.v. or subcutaneous).
- FFP: give only for invasive procedures or active bleeding.
- Platelets: give for platelet counts < 10 000/mm³ or invasive procedures.
- Recombinant activated factor VII (40 µg/kg bolus): give 30 min before procedure if coagulopathy refractory to FFP.
- Cryoprecipitate: for fibrinogen < 100 mg/dL and bleeding.
- Exchange plasmapheresis: allows transfusion of large amount of FFP.
- Prophylaxis for stress ulceration: give H₂ blocker or PPI.

5) Haemodynamics/ renal failure

- Arterial line.
- Pulmonary artery catheterization.
- Volume replacement.
- Check cortisol and cosyntropin stimulation test; hydrocortisone 200 – 300 mg/day for adrenal insufficiency.
- Pressor support (noradrenaline preferred over dopamine or adrenaline) as needed to maintain adequate mean arterial pressure.
- Avoid nephrotoxic agents.
- Continuous modes of haemodialysis if needed, such as venovenous haemodialysis.

6) Metabolic concerns

- Follow closely: glucose, potassium, magnesium, phosphate.
- Consider nutrition: enteral feedings preferred over total parenteral nutrition.

Paracetamol hepatotoxicity

N-acetylcysteine is well established in the management of an acetaminophen overdose and effective in preventing liver injury when administered within 15 hours of acetaminophen ingestion.

NAC is the antidote for paracetamol poisoning, replenishing glutathione that detoxifies the harmful metabolite.

7) Liver transplantation

Hepatic transplantation has to be considered for patients reaching grade 3 and 4 coma due to acute liver failure.

Anatomy of the biliary system



32



Biliary surgery

The gall bladder is second intra-abdominal organ most commonly requiring surgical intervention (after the appendix)

extrahepatic biliary system:

The extra-hepatic biliary tree consists of:

- Right and left hepatic ducts.
- Common hepatic duct. (CHD)
- Gallbladder and cystic ducts.
- Common bile duct (CBD)
- Ampulla of Vater.

parts of gallbladder:

- The gallbladder lies in the gallbladder fossa on the undersurface of the liver.
- This is a pear shaped structure with an average length of 7–12 cm and capacity of 30–50 mL.

The gallbladder consists of:

1. Fundus: Part of the gallbladder projecting beyond the inferior border of the liver.
2. Body: Part of the gallbladder extending from the end of the fundus to the neck of gallbladder.
3. Neck: The neck of the gallbladder forms an S-shaped curve and connects the body of the gallbladder to a narrow infundibulum which is continued as cystic duct.
4. From the neck a small diverticulum may project towards the duodenum. This is known as **Hartman's pouch**.



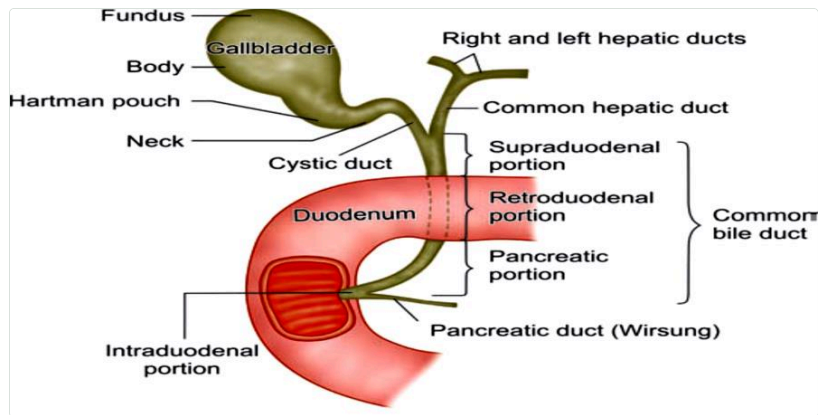


Figure 32.1 — Biliary system anatomy

Extra-hepatic biliary ducts:

- **The right and left hepatic ducts** emerge at the porta hepatis.
- **The left hepatic duct** has a longer course than the right hepatic duct.
- These two ducts join at the confluence to form the common hepatic duct.
- **The common hepatic duct** is 3 cm in length and runs in the free margin of the lesser omentum.
- **The cystic duct** emerges from the neck of the gallbladder is 3 cm long and joins the right margin of the CHD to form the common bile duct.
- **The common bile duct** is 7.5 cm long and is divided into four parts:

1. Supra-duodenal part:

2.5 cm long runs in the free margin of lesser omentum lying to the right of hepatic artery and in front of portal vein.

2. Retro-duodenal part:

Lying behind the first part of the duodenum.

3. **Infra-duodenal part** lies in the groove or in a tunnel in the posterior surface of the head of pancreas.

4. Intra-duodenal part:

Runs obliquely through the wall of the second part of the duodenum, dilates to form the ampulla of Vater and joined by the pancreatic duct opens at the posteromedial wall of the duodenum at the major duodenal papilla.

Sphincters around the bile duct:

1. Sphincter choledochus (Sphincter of Boyden): Sphincter around the terminal part of the common bile duct.



2. Sphincter pancreaticus: This is the sphincter muscle around the terminal part of the pancreatic duct.
3. Sphincter of Oddi: Sphincter around the ampulla of Vater.

Blood supply of gallbladder and biliary tree :

- The gallbladder is supplied by the cystic artery which is usually a branch of right Hepatic artery. The cystic artery arises behind the common hepatic duct, crosses behind and enters the gallbladder.
- An accessory cystic artery arising from the gastro-duodenal artery may also supply the gallbladder.
- The bile duct is supplied by two vertical arteries arising from the hepatic artery running along the bile duct at 3 and 9 o'clock position and giving off circumferential arteries anteriorly and posteriorly.
- The cystic veins draining the gallbladder do not accompany the cystic artery. These veins pierce the fossa for the gallbladder and drains into the intrahepatic part of portal vein.

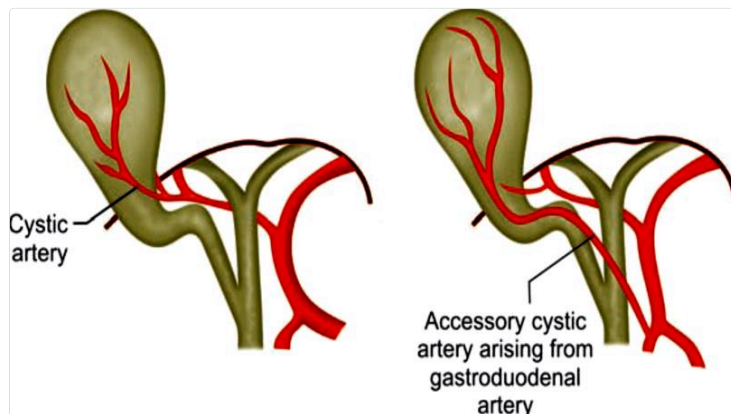


Figure 32.2 — Cystic artery anatomy

N.B. Moynihan's hump:

- This is one of the anomalies in relation to the course of hepatic artery.
- The hepatic artery makes a tortuous course in front of the bile duct near the entry of cystic duct.
- The cystic artery arising from this hump or the hump of right hepatic artery is usually short.
- This is a dangerous anomaly as hepatic artery may be confused with cystic artery and may be clipped during cholecystectomy.



Calot's triangle:

- Superiorly limb: cystic artery
- Right limb: cystic duct
- Left limb: CHD

Important to be identified during cholecystectomy for safety ligation or clipping of the cystic duct and artery and to decrease incidence of bile duct injury

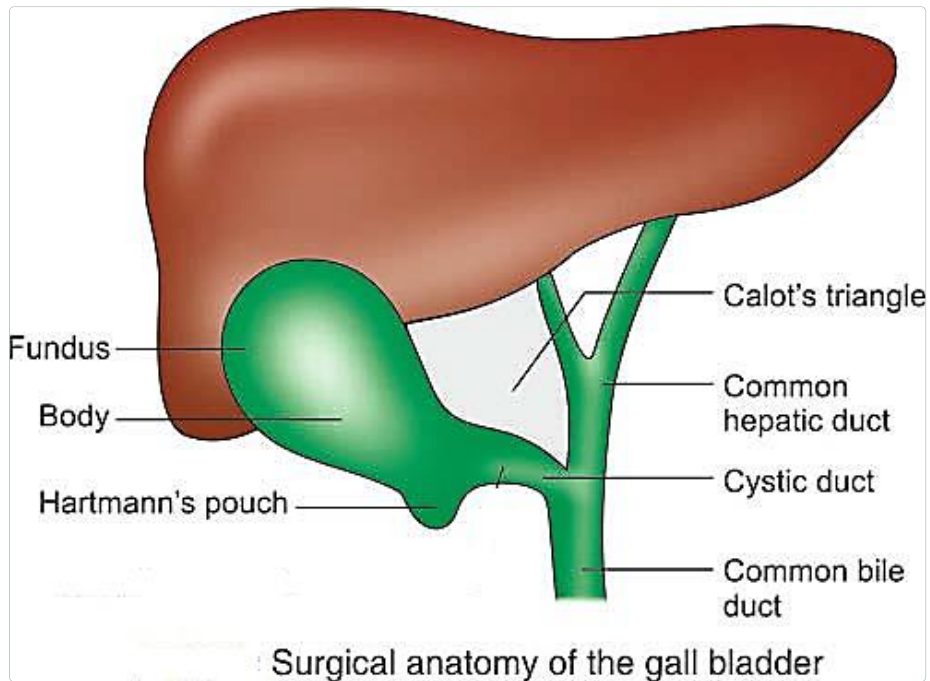


Figure 32.3 — Surgical anatomy of the gallbladder

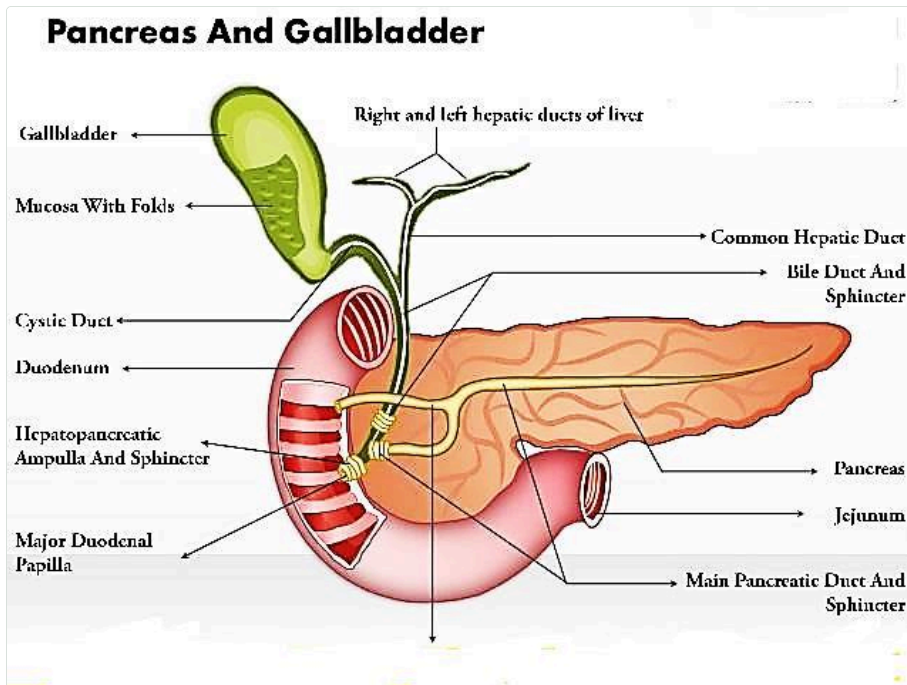


Figure 32.4 — Pancreas and gallbladder

N.B. Arrangement of the structures at free border of lesser omentum =hepato-duodenal ligament:

Formed of 2 layers and contains 3 important structures:

- CBD most anteriorly to the right
- Common hepatic artery in the middle to the left
- Portal vein: posterior
- Epiploic foramen (Winslow): most posteriorly

Clamping of the hepato-duodenal ligament = Pringle's maneuver in severe bleeding from the liver or vascular injury in the liver hilum, hepatic artery or portal vein.

Diseases of the gall bladder



33



- Congenital diseases
- Gall bladder stones
- Inflammations (acute and chronic cholecystitis)
- Complications (e.g. obstructive jaundice)
- Cancer G.B.

Congenital anomalies of the gall bladder

1. Agenesis: absent gall bladder
2. Double gall bladders
3. Ectopic (intra-hepatic , left side,..)

4. Phrygian cap:

- It is the most common congenital anomaly related to the shape of gall bladder
- There is a septum or fold between the fundus and body

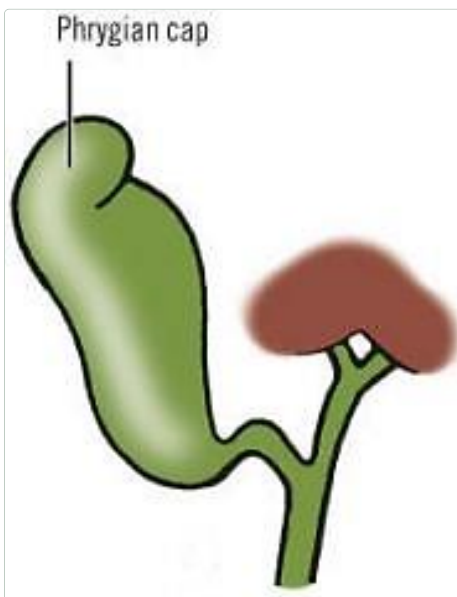


Figure 33.1 — Phrygian cap

5. Anomalies related to the position of the cystic duct and artery

6. Choledochal cyst:

- It is congenital cystic dilatation of the extra-hepatic bile duct mostly the CBD
- Clinically presented usually before 10 years.

Complications:

- Rupture with biliary peritonitis
- Stasis and cholangitis



- Liver abscess
- Stone formation
- Malignant transformation with cholangiocarcinoma
- Portal hypertension and portal vein thrombosis

Types of choledochal cysts:

Type	Description
Type I	Fusiform dilation of the common bile duct (80-90% of the cases).
Type II	Bile duct diverticulum (0-2% of the cases).
Type III	Saccular dilation of the common bile duct inside the duodenum wall, or choledochocele.
Type IV	Multiple intra- and extrahepatic bile dilations (10-15%). Variant A: intra- and extrahepatic affectation. Variant B: involves only the extrahepatic bile duct.
Type V	Fusiform or saccular cystic dilation of the intrahepatic bile ducts, whether or not associated to hepatic fibrosis (Caroli disease).

Figure 33.2 — Types of choledochal cysts

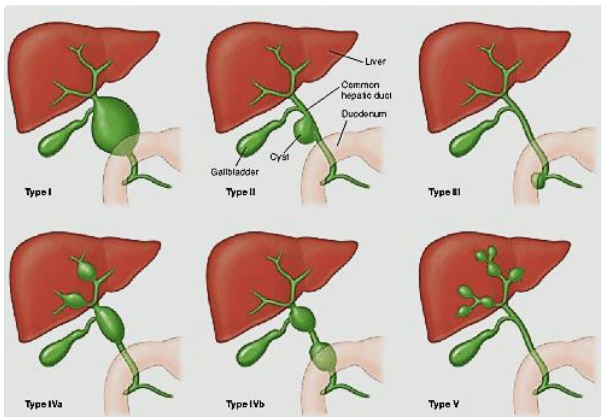


Figure 33.3 — Choledochal cyst classification diagrams

Treatment of choledochal cyst:

- Excision of the cyst with biliary reconstruction according to the type of the cyst
e.g. excision with bilio-enteric anastomosis.

Gall stones and chronic cholecystitis



34



Stone means solid substance formed from the constituents of bile within the biliary system.

Pathology

Incidence:

Gall stones are common; they are more common in fatty fertile females around 40 years, with the incidence rising with age.

Types of stones:

Gallstones are composed of two basic components, **cholesterol and bile pigment**.

Cholesterol stones:

1- Pure cholesterol stones

2- Mixed stones

1 - Pure Cholesterol stones (20%)

They are whitish, usually solitary and big (2.5 cm).

■ The process of cholesterol stone formation:

1- Cholesterol elimination occurs as a result of its secretion into bile as:

a- Cholesterol itself b- bile salt (25:1)

2- In bile, the cholesterol is kept in solution by secreted **phospholipids**.

■ Causes of Cholesterol stones:

- Gall bladder stasis → precipitation of cholesterol
- Prolonged fasting → decreases water and increases concentration
- **Any factor that increases plasma cholesterol** will increase bile cholesterol as:
 - a) Obesity: (common in females and older ages) b) High fat in the diet c) Metabolic conditions (diabetes, familial hyper-lipidaemias)
 - **Pregnancy** → stasis → Precipitation
 - Stones may also occur as a result of **bile salt depletion** which occurs in mal-absorption of bile salts in the **distal ileum**, and cirrhosis.

■ Cholesterol is prevented from precipitation by:

- Bile salts (25 : 1)
- Non-steroidal anti-inflammatory drugs
- Caffeine





Figure 34.1 — Cholesterol stone

Cholesterol stone

II - Mixed stones

- Formed of cholesterol + bilirubin + debris + bacteria
- Main causes are stasis and super-added infection and hypercholesterolemia.
- Most common type of stones → **75 % or more.**
- **Mixed stones** are **usually multiple dark or light brown and faceted.**

III - Pigment stones (5 – 10%)

Stones may occur as a result of increase in bilirubin load due to haemolytic anaemias .

- Stones are multiple small and black or brown.
 - a) **Brown pigment stones** formed of bilirubin + calcium soaps+ bacteria

b) Black pigment:

high bilirubin + low calcium.



Figure 34.2 — Pigment and mixed stones



Clinical picture of gall stones

- 1- Non complicated gall stones are **Asymptomatic**
- 2- Fatty dyspepsia (common in fatty, fertile females around 40 years).

3 - Biliary pain (colic):

Biliary pain is caused by: **Obstruction** or **inflammation**. • Pain occurs when the neck of the gall bladder is obstructed by a stone and gall bladder contraction continues under the influence of cholecystokinin.

Character:

- The resultant visceral pain is therefore dull, poorly localised, • usually extends across the right hypochondrium, epigastrium and may radiate to the back, and can be felt at lower end of right scapula.
 - This pain is of rapid onset and usually lasts a number of hours
 - Relief occurs only when the stone either dislodges or, passes through the cystic duct into the bile duct. The patient remains unwell for 1-2 days.
- 4- Patient may be presented with complication as jaundice, pancreatitis, acute cholecystitis,

Abdominal examination:

- During the attack → tender epigastrium and right hypochondrium

Investigations

1 - Ultrasound:

- a) Ultrasound has a major role in the diagnosis of gallstones.
- b) The typical appearance is acoustic shadowing (dark area) radiating beyond the stone
- c) Determine the state of gall bladder (inflamed or not) and wall thickness
- d) Determine the diameter of the CBD and IHBR
- e) Determine the state of liver (cirrhotic or fibrotic)

2- Plain X-Ray abdomen → stones are mostly radiolucent (not used nowadays)

3 - Lab.

- a) CBC → leukocytosis in associated acute cholecystitis
- b) Liver enzymes
- c) Total and direct bilirubin (for biliary obstruction)
- d) ALP & GGT (for biliary obstruction)
- e) PT & PTT (for bleeding disorders)





Treatment:

Gallstones: clinical situations and recommended management:

Incidental (asymptomatic):

elective laparoscopic cholecystectomy (in diabetics or multiple small stones).

Incidental (at laparotomy):

remove the gall bladder if technically suitable.

Biliary pain:

Elective laparoscopic cholecystectomy.

Acute cholecystitis:

Urgent cholecystectomy or delayed 6 weeks.

Post-endoscopic sphincterotomy for common bile duct stone: elective laparoscopic cholecystectomy.

Gallstone pancreatitis:

laparoscopic cholecystectomy

Types of operations:

- 1- Open cholecystectomy
- 2- Laparoscopic cholecystectomy (best)

Why do you prefer laparoscopic cholecystectomy?

Laparoscopic cholecystectomy has been established as gold standard for the treatment of Gall stone diseases

- Surgery is safe in the hands of a trained surgeon
- Less pain, less hospital stays
- Better Cosmetic
- Early return to work





Complications of Gall Stones

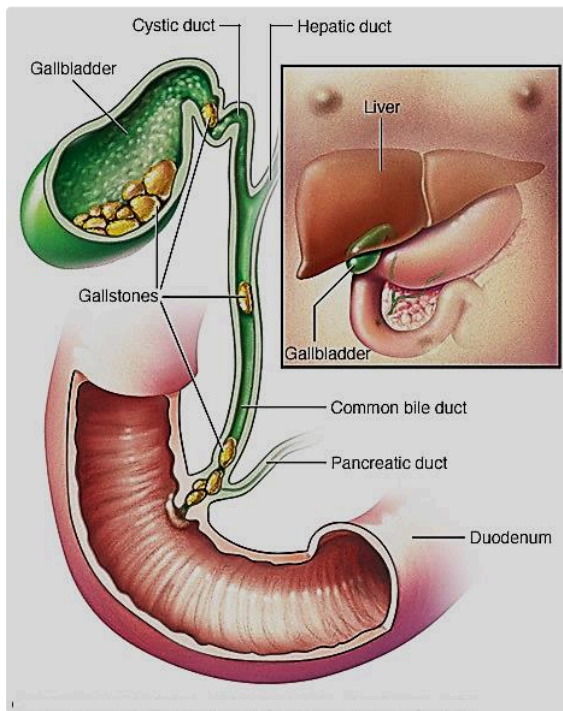


Figure 34.3 — Gallstone complications: biliary anatomy

A. Stones remaining in the gall bladder

1. Biliary colic
2. Acute cholecystitis and its sequelae—gangrene/perforation/local abscess/biliary peritonitis
3. Chronic cholecystitis
4. Mucocele of gallbladder
5. Empyema of gallbladder
6. Carcinoma of gallbladder.

B. Stones migrated into the bile duct

- 1- Obstructive jaundice
2. Recurrent cholangitis
3. Acute pancreatitis.

C. Stones migrated into the intestine Gallstone ileus.

→

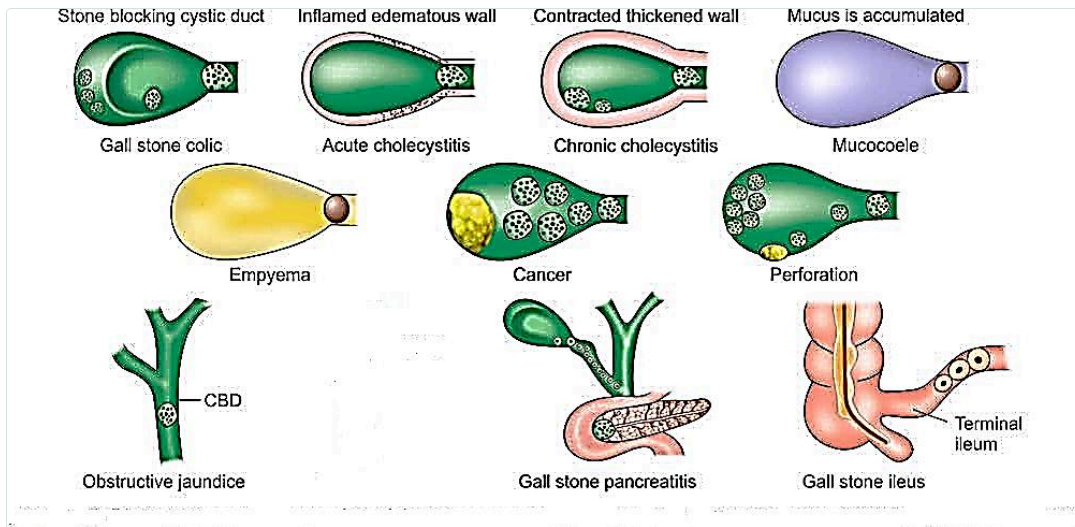


Figure 34.4 — Complications of gall stones

Acute cholecystitis

It means acute inflammations of the gall bladder

Types:

I- **Acute calcular cholecystitis** → 90 %

II- **Acute non- calcular cholecystitis** → 10 %

Etiology:

Commonest organism causing acute cholecystitis is *E. coli*

Routes of infection in acute cholecystitis:

1- Hematogenous— infection reaches the gallbladder via the cystic artery. **2- Through bile**—enteric organisms gains entry into the portal circulation and filtered by the liver and excreted in bile and reaches the gallbladder.

Risk factors for calcular cholecystitis

- Female sex
- Obesity
- Drugs (especially hormonal therapy in women)
- Pregnancy
- Increasing age (> 40 years)



Risk factors for non-calicular cholecystitis: (low resistance)

Critical illness

Major surgery or severe trauma or burns

Generalized Sepsis

Long-term total parenteral nutrition (TPN)

Prolonged fasting

Diabetes mellitus

Pathology of acute calcular cholecystitis

a) Stone obstructing the cystic duct the most important initiating factor b)

Distension of the gall bladder with mucus retention → **Mucocele** c) More distension of the G.B. → ischemia of the wall → infection → **acute empyema** of the G.B.

d) More ischemia of the wall → **gangrene of the G.B. and perforation** → peritonitis

e) The perforation may occur at the fundus of the gallbladder or at the site of impaction of stone

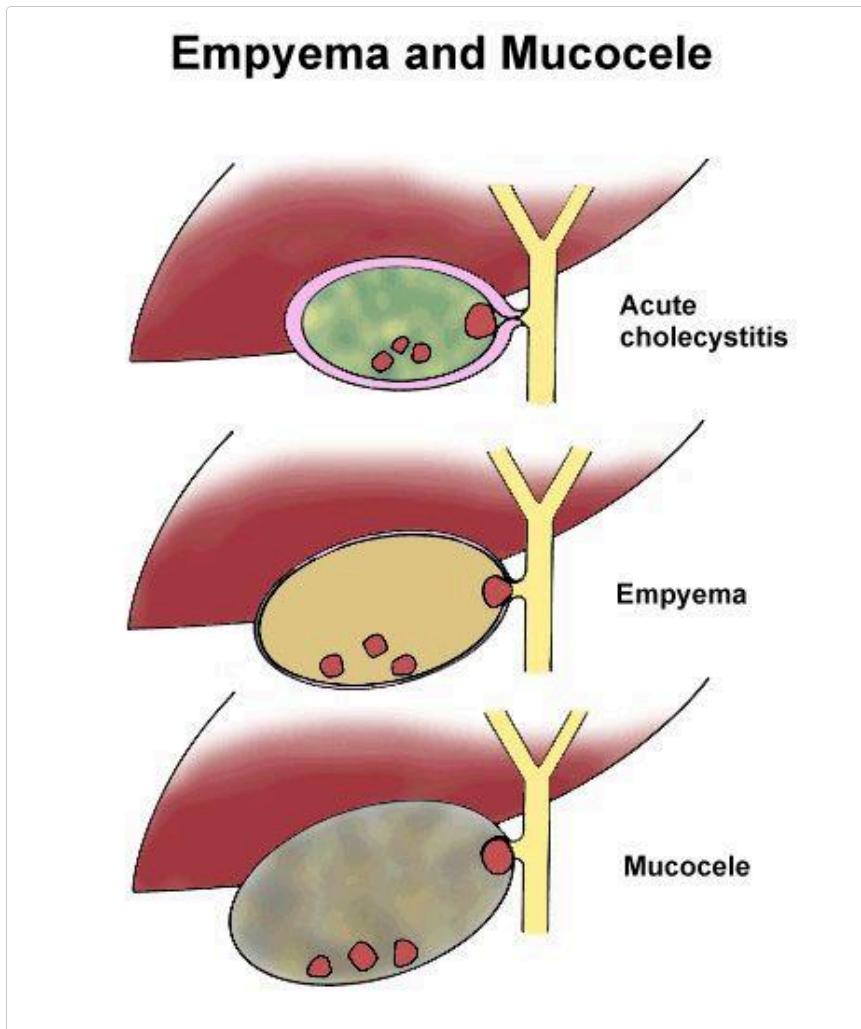


Figure 34.5 — Empyema and mucocele

Symptoms of acute cholecystitis

1- Right Upper abdominal pain is the most common presenting symptom

- The pain may radiate to the right shoulder or scapula
- Pain frequently begins in the epigastric region and then localizes to the right upper quadrant.
- Pain firstly is colicky but almost always becomes constant dull aching

2- Nausea and vomiting.

3- Fever

Signs:

Fever, tachycardia, toxic look

Tenderness in the Right hypochondrium or epigastric region, with guarding • and rebound tenderness



Palpable and tender gallbladder.

+ve Murphy's sign:

patient gets severe pain and holds his breath on dipping • the thumb of the examiner deep to the right costal margin at the region of the gall bladder.

D.D. of acute upper abdominal pain:

1. Acute cholecystitis
2. Acute cholangitis
3. Perforated peptic ulcer
4. Acute pancreatitis
5. Acute liver abscess
6. Acute sub-hepatic appendicitis
7. Intestinal obstruction
8. Acute pyelonephritis and renal colic

The sequelae of acute cholecystitis

Resolution:

Inflammation subsides and patient recovers.

Gangrene:

Infection may lead to gangrenous change in gallbladder manifested by • increasing pain, toxemia and appearance of rebound tenderness.

Perforation of gallbladder. Perforation may be:

- **i. Localized**—localized abscess formation manifested by severe pain, fever with chill and rigors and extreme tenderness in right upper quadrant of abdomen. **ii. Generalized perforation**—leading to generalized biliary peritonitis manifested by generalized pain abdomen, muscle guard or rigidity and extreme tenderness all over abdomen.
- iii. Perforation into a neighboring viscus** most commonly duodenum

Investigations

CBC → Leukocytosis.

SGPT and SGOT levels may be elevated in cholecystitis or with CBD • obstruction.

High Bilirubin and alkaline phosphatase may indicate CBD obstruction

Amylase/lipase assays are used to assess for pancreatitis; amylase may also be • mildly elevated in cholecystitis. Also differentiate the acute pancreatitis. (as D.D.)

Urinalysis is used to rule out pyelonephritis and renal calculi (as D.D.)



Married females of childbearing age should undergo pregnancy test. (as D.D.) •

Radiological investigations

1- Ultrasonography:

Is the preferred **initial imaging test** for the diagnosis of acute cholecystitis.

- Gallbladder wall thickening and edema (>2 -4 mm)
- Gallbladder distention (diameter > 4 cm, length > 10 cm)
- Peri-cholecystic fluid

2-Hepatobiliary scintigraphy (HIDA SCAN)

- An imaging test used to view the liver, gallbladder, bile ducts, and duodenum.
- If there is no sign of the radioactivity detected in the gallbladder → Diagnostic for acute cholecystitis (edema and cystic duct obstruction)
- Not done routinely

3- CT scan can identify complications of acute cholecystitis (Perforation)

4- Plain x-ray abdomen erect. (To exclude perforated P.U.)

Management

- Treatment of acute cholecystitis depends on the severity of the condition and the presence or absence of complications.
- **The initial treatment** includes:
 - a) Bowel rest, NPO (nothing per-oral) and nasogastric suction.
 - b) IV fluids, Analgesia, IV antibiotics: (3rd generation cephalosporins and Metronidazole)
 - c) Vomiting can be treated with antiemetics
- **Surgical procedure:**

1- Laparoscopic cholecystectomy:

(standard for surgical treatment of cholecystitis)

2- Open cholecystectomy.

Indications:

- Diagnosis is established within 40 hours of the attack
- Failure of response to medical treatment
- Presence of complications as perforation
- after 6 weeks in postponed cases

N.B. Surgery could be safely done within 48 hours of diagnosis. After that time the edematous tissues at the gall bladder area makes surgery to be difficult with high incidence of bile duct injury, so some prefer surgery after 6 weeks



Ascending cholangitis

Definition : spreading **infection** of the **bile ducts** (cholangitis)

Pathogenesis :

- 1 - **Bacteria** ascend through the CBD from its junction with the duodenum.
- 2- It tends to occur if the bile duct is already obstructed by gallstones, strictures or tumors.
- 3- High pressure → reflux of bacteria to blood → bacteremia → septicemia

Signs and symptoms

• I **Charcot's triad** :

- 1- Right upper abdominal pain,
- 2- Jaundice,
- 3- Fever and rigors

II- **Reynolds' pentad.**

Charcot's triad + Hypotension + Confusion

Treatment:

- 1- Combined antibiotics against gram – ve bacilli (3rd generation cephalosporins + Metronidazole)
- 2- Intravenous fluids
- 3- Treatment of the cause of bile obstruction (ERCP)

Chronic calcular cholecystitis

Clinical picture:

- May be asymptomatic and discovered accidentally
- Recurrent attacks of epigastric or right hypochondrial pain (persistent pain) • Attacks of severe biliary colic
- Nausea & vomiting
- Flatulent dyspepsia with intolerance to fatty meals
- Tenderness in right hypochondrium (Murphy's sign) with deep inspiration.

Investigations → as before



D.D.

Chronic duodenal ulcer
Chronic gastric ulcer
Chronic pancreatitis
Recurrent appendicitis
Right-sided renal calculus
Chronic pyelonephritis.

Treatment :

Cholecystectomy either open or laparoscopic is the ideal treatment for symptomatic patients. **Or** in those who are diabetics or having small G.B. stones that are liable for migration and producing complications



Obstructive Jaundice



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CAUSES OF OBSTRUCTIVE JAUNDICE:

According to the pathology:

1. Congenital:

Biliary atresia - choledochal cyst.

2. Inflammatory:

Ascending cholangitis- chronic pancreatitis.

3. Obstructive:

CBD stones- biliary stricture- parasitic infestation (Ascaris).

4. Neoplastic: • Carcinoma of head of pancreas

- Cholangiocarcinomas (CBD)
- Klatskin tumour (CHD).

5. Extrinsic compression of CBD or CHD :

- by lymph nodes or tumours.
- MIRIZZI Syndrome

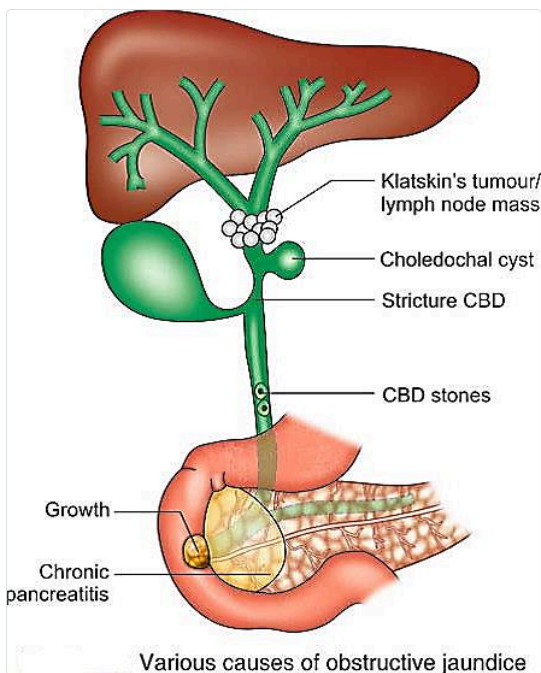


Figure 35.1 — Causes of obstructive jaundice



Another classification: According to site of obstruction:

I-Luminal causes :

CBD stones - parasitic infestation

II-Mural causes:

- Biliary atresia - Choledochal cyst
- Biliary stricture (Inflammatory - iatrogenic - malignant)
- Ascending cholangitis - sclerosing cholangitis
- Cholangio-carcinomas (CBD) - Klatskin tumour

III- Extral-mural: • Compression by lymph nodes

- Cancer head of pancreas
- **MIRIZZI Syndrome** = Hartmann's Pouch stone

■ Mirizzi syndrome:

- The presence of a stone impacted in the cystic duct or the gall bladder neck (Hartmann's pouch), causing inflammation and external compression of the common hepatic duct and thus biliary obstruction and jaundice.

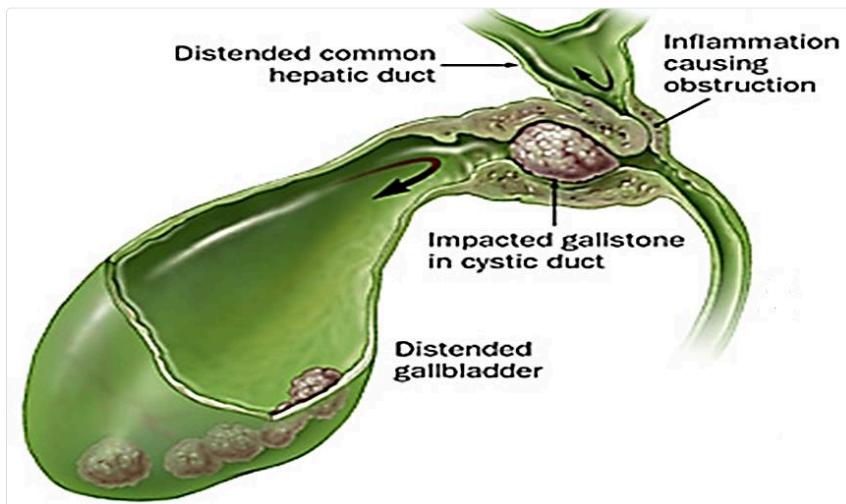


Figure 35.2 — MIRIZZI Syndrome

N.B. The most common causes of biliary obstruction are biliary stones followed by malignancy (cancer head of pancreas and cholangio- carcinoma)

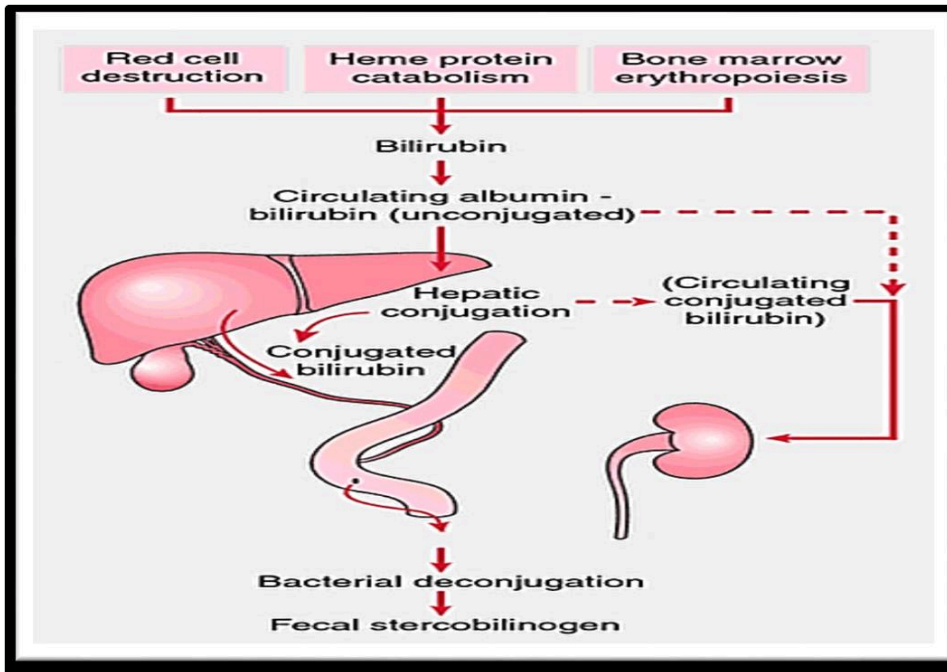


Figure 35.3 — Bilirubin metabolism

Complications of obstructive jaundice

The complications are proportional to the **duration and intensity** of the jaundice.

A. Secondary biliary cirrhosis (liver cirrhosis due to prolonged cholestasis).

B. **Acute cholangitis** is the most common complication, can cause septic shock and death.

C. **Acute renal failure**. may be due to nephrotoxic effects of bile salts, direct bilirubin, endotoxins, or inflammatory mediators.

D. **Failure of bile salts** to reach the intestine results in:

1- **Fat mal-absorption** with steatorrhea.

2- The fat-soluble vitamins **A, D, E, and K** are not absorbed, resulting in **vitamin deficiencies**. Vitamin K deficiency leads to **decreased prothrombin synthesis** → disordered haemostasis with an abnormally prolonged **PT** and bleeding tendency. 3- **Endo-toxemia** → **nephrotoxic** → **Acute renal failure**

Calcular obstructive jaundice

Aetiology:

- GB stone (commonest) → migrate to CBD.



- Primary stones of CBD usually (pigment Brown)
- Meritzzi syndrome

Commonest organisms:

E-coli, klebsilla usually complicate the biliary stasis

Clinical picture:

Symptoms:

- History of fatty dyspepsia
- Yellowish discoloration of skin and sclera (intermittent jaundice)
- Clay colored stool (decreased stercobilin in stool)
- Dark urine (tea- colored) → increased direct bilirubin in urine
- Itching (bile salts)
- Sometimes biliary colic
- **Charcot's triad in case of associated infection (cholangitis):**
 - 1- Jaundice
 - 2- Pain
 - 3-Fever

Raynaud's pentade in associated cholangitis:

- a) Charcot's tirade b) Hypotension (septicemia) c) Altered mental status (septicemia)
- The jaundice is not deep usually light yellow (D.D. of olive green in malignant obstruction).
 - On examination: G.B not palpable (chronic fibrotic wall) except rarely if its neck is obstructed too (double stones) the back pressure causes bladder distension

Investigations:

Laboratory:

- **Raised :**
Direct and total bilirubin (up to 10 – 12 mg%) - Alkaline phosphatase – Gamma glutamyl transferase.

Mild elevation :

SGOT & SGPT

Prolonged:

- Prothrombin time (due to avitaminosis K)



Decreased urobilinogen in urine.

Radiological:

1- Ultrasound this is most useful initial investigation:

- Most important to show intrahepatic bile ducts dilatation
- Measure the diameter of CBD (normal up to 9 mm) (dilated in OJ)
- Comment on IHBR dilatation (dilated in OJ)
- Comment on the status of the GB and its stones
- Visualize CBD stones or areas of narrowing
- Comment on liver state (cirrhotic or not)

2- CT scan :

Done on suspected cases of malignant O J to detect pancreatic tumors

3- ERCP: the investigation of choice in obstructive jaundice

- ERCP is an endoscopic and radiologic investigation to visualize both the biliary and pancreatic duct systems.
- Endoscopically, the ampulla of Vater is identified and cannulated.
- A contrast agent is injected into these ducts, and x-ray images are taken • It can be also therapeutic for stone extraction by Dormia basket or insertion of a stent in CBD.

Risks of ERCP :

Ascending infections, - perforations, - pancreatitis

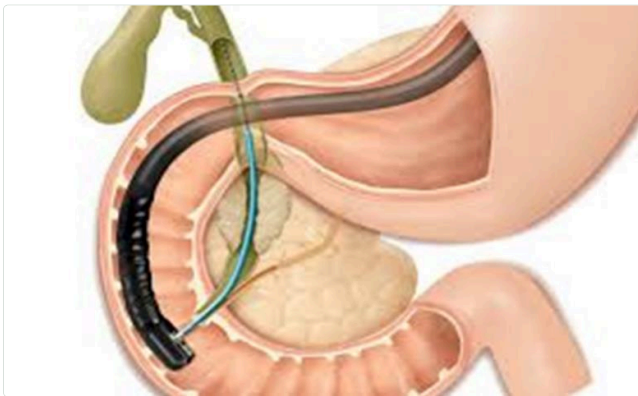


Figure 35.4 — ERCP: procedure diagram

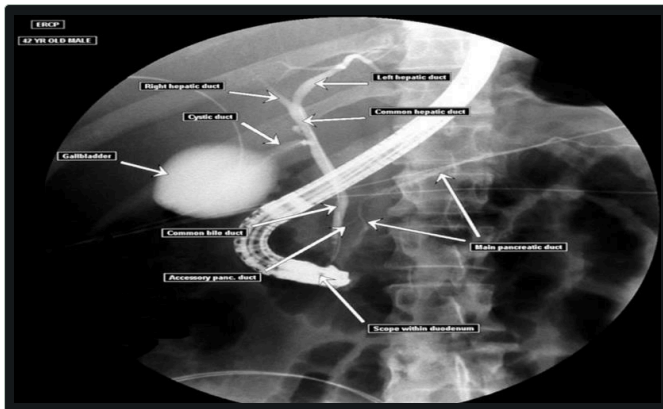


Figure 35.5 — Normal ERCP

4- MRCP

- A sensitive non-invasive method of detecting biliary and pancreatic duct stones, strictures, or dilatations within the biliary system.
- It is also sensitive TO detect cancer.

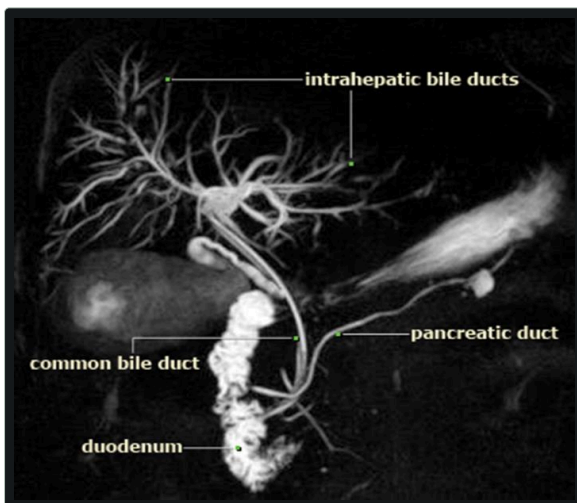


Figure 35.6 — Normal MRCP

5- PTC: percutaneous trans- hepatic cholangiography

- The liver is punctured to enter the peripheral intrahepatic bile duct system. • It is especially useful for lesions proximal to the common hepatic duct.

Management of stone CBD

- 1) Support the liver by correction of the general condition by I.V. fluids for hydration (glucose 10%, Ringer's lactate)
- 2) Support the kidney by Mannitol → increases renal blood flow (hypotension and hyperbilirubinemia together causes renal shut down)



- 3) Prevent infection by antibiotics, using 3rd generation cephalosporins
- 4) Prevent bleeding by correction of the avitaminosis (K) by I.V. vitamin K administration.
- 5) **a) ERCP** extraction of calculi followed by laparoscopic **cholecystectomy**

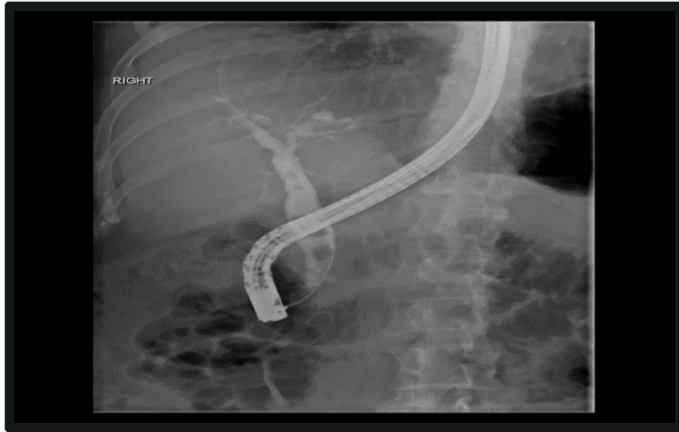


Figure 35.7 — ERCP: Stones in CBD

OR

b) Open cholecystectomy and common bile duct exploration to remove the stone.

Done when ERCP was failed or not available

T- Tube insertion which is made of Latex.

Horizontal limb is suited to lie in the supra-duodenal part of CBD.

Vertical limb comes out straight from the abdomen.

T- Tube cholangiography is done on the 5th day and tube is extracted at • the **10th** day after 24-hour of occlusion without problems (fever, leakage, or pain).

Early slippage of the tube (within 3-5 days) should be re-introduced

Excess bile comes out through the tube indicate distal obstruction, and

ERCP is indicated

OR

C) Cholecystectomy and Choledcho- duodenostomy:

- In dilated CBD (more than 1.5 cm)
- Stricture of the lower end of CBD
- Anastomosis is done between the proximal dilated CBD and first part of the duodenum

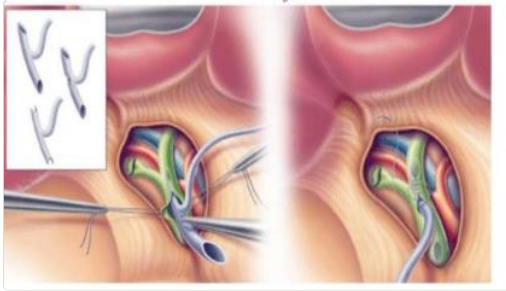


Figure 35.8 — Bile duct exploration

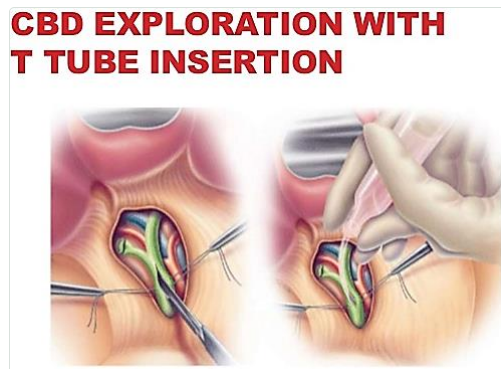


Figure 35.9 — CBD exploration with T-tube insertion

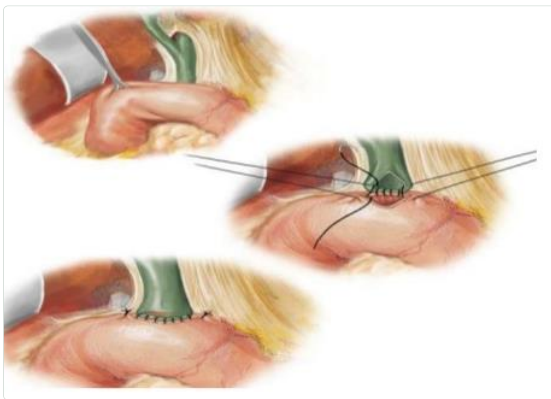


Figure 35.10 — Choledocho-duodenostomy

Differences between calcular and malignant obstructive jaundice:

Calcular obstructive jaundice

- Usually a fatty fertile female / middle age
 - Intermittent jaundice (can be progressive if stone is impacted)
 - Jaundice Usually not reaching high levels (deep yellow)
 - Pain is colicky in nature, and typical for biliary colic
 - G.B not palpable except rarely if its neck is obstructed too (double stone)



(Courvoisier law)

- No weight loss.
- Ultrasound is conclusive (stones, thick wall G.B., stone in CBD, Dilated CBD)
- ERCP is diagnostic and therapeutic
- Prognosis is good

Malignant obstructive jaundice:

- Usually, old age male
 - Progressive jaundice except very rarely in ampullary tumors where sloughing can give temporary decrease
 - Jaundice usually reaching high levels (deep or olive-green jaundice).
 - Pain is constant and referred to the back in pancreatic tumor, while it is absent in CBD tumors
 - G.B is palpable (healthy G.B. wall) except in Klatskin tumors (tumor of the confluence of CHD)
 - Associated weight loss is severe (Cachexia)
 - Abdominal mass may be felt in the epigastrium, in advanced cases
 - Ultrasound may not conclusive
 - C.T is conclusive and ERCP is diagnostic and can be palliative by inserting a stent in CBD
 - Bad prognosis

D.D. of acute pain in right upper quadrant of abdomen includes:

1. Acute cholecystitis
2. Acute cholangitis
3. Acute pancreatitis
4. Acute peptic ulcer
5. Peptic ulcer perforation
6. Liver abscess
7. Acute retro-cecal sub-hepatic appendicitis
8. Acute intestinal obstruction
9. Right sided renal colic
10. Acute right sided pyelonephritis
11. Acute myocardial infarction
12. Basal pneumonia

Acute pancreatitis



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It is an acute inflammation with auto-digestion of the pancreas.

Causes:

- A- Gall stones: 40 %
- B- Alcohol: 35%
- C- Idiopathic: 10 – 30 %
- D- Post- ERCP: 4%
- E- Trauma: external or surgical (during splenectomy)
- F- Infections: as mumps, CMV
- G- Hyper-calcemia (hyper-parathyroidism)
- H- Auto-immune – drug induced

Gall stones:

40% of cases

- Gall stones passing into the bile duct and temporarily lodging at the sphincter of Oddi.
- Transient obstruction of the pancreatic duct by a stone impacted in the ampulla of Vater.
- Acinar cell injury occurs secondary to increasing pancreatic duct pressures caused by obstructive biliary stones at the ampulla of Vater, • Reflux of infected bile into the pancreatic duct causes activation of the pancreatic enzyme leading to pancreatitis.
- The risk of a stone causing pancreatitis is inversely proportional to its size

Alcohol:

35 % of cases

- Leads to premature enzyme activation and release.
- Alcohol can cause both acute and chronic pancreatitis.
- Alcohol causes spasm of sphincter of Oddi leading to obstruction of flow of pancreatic juice.
- Alcohol is a cellular metabolic poison and causes damage of parenchymal cells.
- Alcohol causes precipitation of proteins and calcium in the pancreatic duct resulting in ductal obstruction.
- Alcohol increases the permeability of the pancreatic duct to the enzymes which leaks out into the surrounding tissues resulting in pancreatic damage.
- Alcohol causes reduction of blood supply of the pancreas resulting in ischemic damage to the pancreas.



Pathogenesis of acute pancreatitis

- A. Alcohol use or gall stones can cause Acute pancreatitis by injury of the acinar cells.
- B. Once cellular injury occurs, trypsinogen activated to trypsin
- C. Premature activation of trypsin → Leads to activation of elastase, chymotrypsinogen, phospholipase A2, complement, and kinins → **Auto-**digestion of the pancreas
- D. Elastase will digest wall of capillaries on the surface of pancreas and surrounding tissues → **hemorrhages**
- E. Lipases will break down fats into FFA that combine with calcium from the blood → **calcium soaps** (white patches over the pancreas and omentum)
- F. Macrophages release cytokines that lead to local and systemic inflammatory responses. (Tumor necrosis factor-alpha (TNF-α), interleukin (IL)-6, and IL-8). G. These mediators of inflammation cause an increased pancreatic vascular permeability, leading to hemorrhage, edema, and pancreatic necrosis.
- H. As the mediators are excreted into the circulation, systemic complications (**SIRS**) such as :
 - Acute respiratory distress syndrome (ARDS), • pleural effusions, • Renal failure.
 - Hypotension and shock state

Pathological types of acute pancreatitis,

- **Acute edematous pancreatitis (80 %)** : parenchymal edema and peripancreatic fat necrosis occur first.

Necrotizing pancreatitis (20 %):

When necrosis involves the parenchyma, accompanied by hemorrhage..

- **Pancreatic abscess** : result from infection on top of necrotizing pancreatitis.

Clinical picture of acute pancreatitis:

History of Previous biliary colic OR alcohol consumption (major causes of acute pancreatitis)

1- Abdominal pain:

- Severe, starts 12-24h after eating a large meal or consuming alcohol, • Radiate to the back
- Pain is worse on laying supine & better after sitting or leaning forward.

2- **Bad general condition**, hypotension, shock in severe cases, and Toxic look.

3- Nausea and vomiting

4- Jaundice



5- Low grade fever

6- Tachycardia

Abdominal examination:

- Abdominal distension → mainly epigastric
 - Tenderness, guarding rigidity and rebound tenderness more in the epigastrium
 - Hyper resonance (in presence of ileus)
 - Shifting dullness in marked abdominal fluid collection.
 - Silent abdomen by auscultation due to ileus.
 - **Special signs: might be absent**
 - **Grey turners sign.:** (reddish- discoloration along the flanks resulting from retroperitoneal blood dissecting along tissue planes)

Cullen's sign.:

(bluish discoloration around the umbilicus resulting from hemo-peritoneum)



Figure 36.1 — Grey Turner's sign



Figure 36.2 — Cullen's sign

Cullen's sign Grey turner's sign

D.D.

- a) Perforated peptic ulcer.
- b) Acute cholecystitis & biliary colic.
- c) Acute Intestinal obstruction.
- d) Renal colic.





Investigations:

Laboratory:

1- Serum amylase:

it remains elevated in the first 48 hours and returns normal (half-life 10 hours)

2- Lipase:

(diagnostic test): It is elevated for 7-14 days.

3- Others:

- ↑ WBC (15000 - 30000). - ↑ LDH > 500 U/dl
- ↑ Glucose . - ↓ Albumin.
- ↓ Ca in serum. - ↑ AST. Bilurbin, Alkaline Ph.
- **ABG: show Hypoxia.**

Radiological:

1- Plain x- ray abdomen erect and supine:

- Exclude perforated viscus (air under diaphragm)

Sentinel loop:

air filled loop in the left upper quadrant.(duodenum)

Colon-cut off sign:

gas-filled segment of transverse colon abruptly ending at the area of pancreatic inflammation.

2- U/S:

- Gall stones.
- Biliary obstruction.(dilated CBD)
- Presence free peritoneal collection
- Pseudo-pancreatic cyst.

3- C.T:

- diagnostic even with normal amylase
- Enlarged, oedematous pancreas.
- Abscess & hemorrhage around the pancreas
- Presence of gas bubbles in CT scan indicates pancreatic abscess.
- Pseudo-cyst

N.B. Severe acute pancreatitis

According to the **Atlanta classification**, severe acute pancreatitis is evidenced by the following:

- **Evidence of organ failure**



- Systolic blood pressure below 90 mm Hg
- [P a O₂] 60 mm Hg or lower, • Serum creatinine level 2 mg % or higher, • Local complications (eg, necrosis, abscess, pseudo-cyst)
- Ranson score = 3 or higher.

Ranson's criteria

At admission or diagnosis

Age over 55 years

White blood count over 16,000/cu mm

Blood glucose over 200 mg/dl

Serum Lactic dehydrogenase (LDH) over 350 U/l

Serum glutamic oxaloacetic transaminase (AST) over 250 U/l

During initial 48 h

Hematocrit fall greater than 10% points

Blood Urea nitrogen rise more than 5 mg/dl

Arterial Po₂ below 60 mm Hg

Serum calcium below 8 mg/dl

Base deficit >4 meq/l

Estimated fluid sequestration more than 6000 ml

Figure 36.3 — Ranson's criteria

- < 3 criteria: mortality = 1 %
- 3 - 4 : mortality = 15 %
- 5 - 6 : mortality = 40 %
- > 7 : mortality = 100%

Complications of acute pancreatitis:

I- General

- Shock & renal failure:(pancreatic necrosis is associated with leakage of fluid in the pancreatic bed & peritoneum
 - Hypocalcemia: sequestration of Ca in fat necrosis.(saponification). • Hypo-albuminemia: Capillary permeability→ascites and edema.. • Hyperglycemia: due to disruption of pancreatic islets (↓ insulin) . • Hypoxia: Respiratory distress → end in respiratory failure.

II- Pancreatic complications

1- Necrosis. - Abscess

2- Pseudo- pancreatic cyst:

- A circumscribed collection of fluid rich in pancreatic enzymes, blood, and necrotic tissue, appear several weeks after pancreatitis.



- Typically located in the lesser sac behind the stomach
- Pancreatic pseudo-cysts are usually complications of pncreatitis.
- It differs from true cysts, which are lined by epithelium; in that they are lined with granulation and fibrous tissues.

Clinically :

- Patient complain of epigastric pain
- History of pancreatitis weeks ago
- Epigastric swelling
- Recurrent vomiting (pressing the duodenum or stomach)
- Fever in case of infection
- C.T. scan is diagnostic

Treatment of acute pancreatitis:

1. Bed rest & Oxygen.
2. NPO & NG tube.
3. IV saline to support the circulation.
4. Pethidine (75-100 mg) to control pain (avoid morphine because it causes spasm of sphincter of oddi)
5. Dopamine (if shock not respond to the fluid replacement)
6. Ca gluconate IV only if there is hypo-calcemia.
7. Fresh frozen plasma for coagulopathy.
8. Serum albumin to correct hypo-albumenemia .
9. Insulin in complicated hyperglycemia.
10. Parenteral nutrition if there are severe pancreatitis & ileus.
11. Antibiotics: prophylaxis to prevent infection
12. ERCP: if sever pancreatitis results from stone in biliary duct.
13. **Treatment of**

pseudocyst:

- Many are resolved spontaneously.
- If it is not resolved after 4 - 6 weeks, surgery is done.
- Cysto-gastrostomy or cysto-jejunostomy is used through open surgery.
- Endoscopic: endoscopic cystogastrostomy
- Percutaneous drainage (C.T guided)

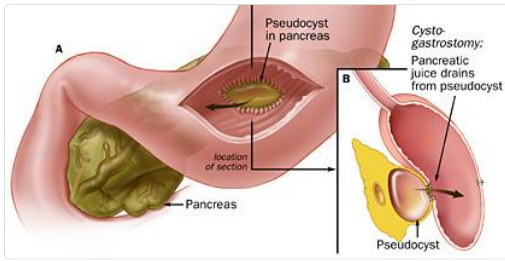


Figure 36.4 — Cysto-gastrostomy.

14- Gallstone pancreatitis

- It is optimal for patients admitted with gallstone pancreatitis to undergo cholecystectomy before discharge.
- Patients discharged with gallstone pancreatitis without a cholecystectomy are at high risk for recurrent bouts of pancreatitis.
- **(ERCP) with sphincterotomy** and stone extraction is indicated in cases with impacted stones.

15- Percutaneous placement of a drainage tube into the fluid collection or abcess under the guidance of ultrasonography or CT scanning.

Cancer pancreas



37



Etiology:

- Pancreatic cancers can arise from the exocrine (95 %) and endocrine portions (5 %) of the pancreas.
- The exocrine portion, include the ductal epithelium and acinar cells.
- Approximately 75 % of all pancreatic carcinomas occur within the head or neck of the pancreas.
- 15-20% occurs in the body of the pancreas.
- 5-10 % occurs in the tail.

Pathology:

- Age: old age
- Sex: more in males

Predisposing and risk factors:

- **Smoking** is the most common environmental risk factor for pancreatic carcinoma.
 - Obesity
 - dietary factors: Consumption of red meat increases the risk
 - Diabetes mellitus
 - Long-standing, chronic pancreatitis increases the risk for the development of pancreatic cancer.
 - Genetic predisposition
 - Multiple endocrine neoplasia.

Microscopic picture:

- 80% are adenocarcinomas of the ductal epithelium
- Endocrine pancreatic tumors include: insulinoma – gastrinoma

Macroscopic:

mass in the head, body or tail of pancreas.

Spread:

- 1- Local→ nearby organs as stomach, duodenum, sup. mesentric vessels, IVC
- 2- Lymphatics: peri-pancreatic lymph nodes, celiac and porta-hepatis lymph nodes
- 3- blood: liver, lungs, bones and brain



Clinical picture: cancer head of pancreas

- 1- Epigastric pain → radiating to back
- 2- Obstruction of the bile duct → **picture of malignant obstructive**
 - **jaundice:**
 - Old aged male, marked weight loss
 - Deep jaundice (olive green)
 - Tea-coloured urine, clay stool
 - Itching
 - Malabsorption symptoms
 - Bleeding tendencies
 - Superficial thrombophlebitis
 - Duodenal or gastric obstruction → gastric outlet obstruction → Vomiting.
 - Distended gall bladder (Courvoisier law)
 - Hard fixed mass in the epigastrium

Investigations:

I- Lab.

- CBC : anemia
 - Liver function: bilirubin, alkaline phosphatase, Albumin • Renal functions
 - PT- PTT: elevated
 - Glucose : in endocrine tumors
 - CA19-9

II- ERCP →

- diagnostic – biopsy
- put a stent in CBD for drainage (palliative or pre-operative preparation)

III- Ultrasound → primary investigation in obstructive jaundice (state of gall bladder, CBD, Liver, ascites)

IV- Endo-luminal ultra-sound:

to detect local spread

V- PTC (Percutaneous trans-hepatic cholangiography) → Put a stent for drainage.





Figure 37.1 — ERCP: source duct image

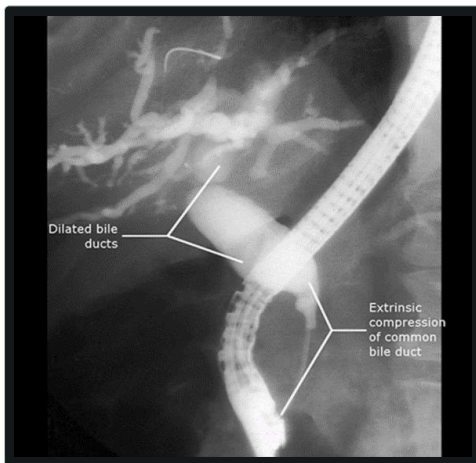


Figure 37.2 — ERCP: malignant stricture ERCP: cancer head

V- CT scan - MRI

- Site and size of mass
 - Detect metastases
 - Detect ascites.





Figure 37.3 — C.T. : cancer head of pancreas

Treatment of cancer head

Operable and resectable cases:

A- Whipple operation

B- Total pancreatectomy

■ Preoperative preparations:

1- Preoperative stenting of the CBD through the ERCP

To drain the bile → decrease bilirubin → improves the general condition 2- Correct hypo-prothrombinemia → by **intravenous vitamin K** administration. 3-

Intravenous fluid administration → for proper hydration of the kidneys and improves kidney function

4- Glucose 25% → central line for liver support

5- Broad spectrum antibiotics to guard against infection.

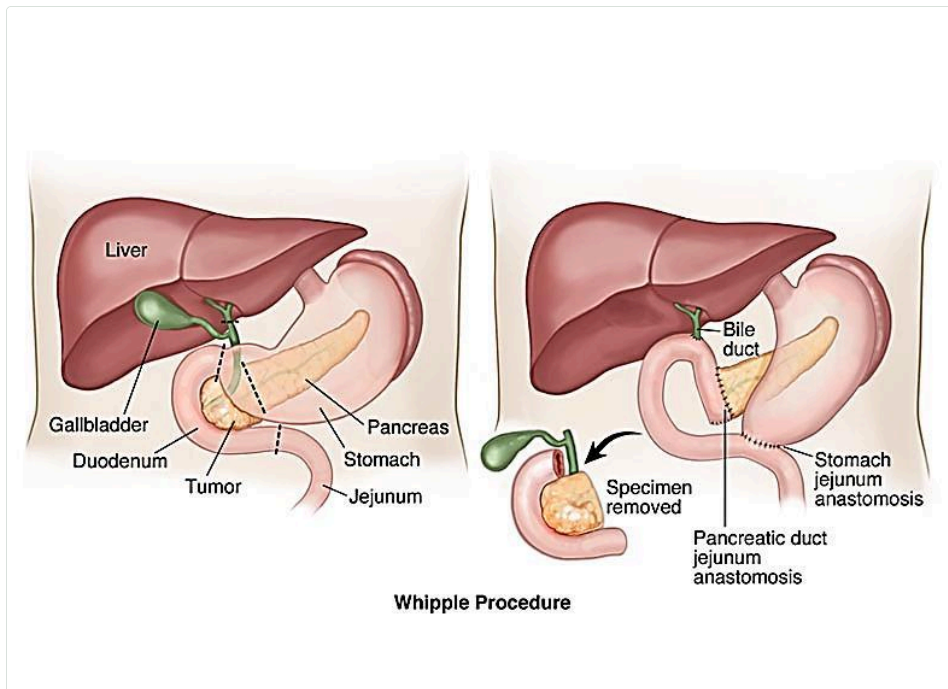


Figure 37.4 — Whipple procedure

■ Whipple's operation:

- 1- Resection of head and neck of pancreas including the tumour
- 2- Resection of duodenum and distal stomach
- 3- Resection of G.B. and lower CBD
- 4- Gastro-jejunostomy
- 5- Pancreatico-jejunostomy
- 6- Hepatico-jejunostomy
- 7- Postoperative chemotherapy

Inoperable cases:

Very bad prognosis

1- Palliative bypass:

- Gastro-jejunostomy
 - Choledocho-jejunostomy
 - Jejunio-jejunostomy
- 2- Palliative stent in the CBD through ERCP
 - 3- PTC stenting
 - 4- Radio-Therapy and chemotherapy

Acute Abdomen



38



Learning Objectives

The student will

- 1) Understand the definition of acute abdomen
- 2) Know the various causes of acute abdomen, both medical and surgical causes
- 3) Know the clinical presentation of acute abdomen
- 4) Know the diagnostic studies needed to be done in acute abdomen

Definition

The term acute abdomen describes a syndrome of Sudden Abdominal Pain with accompanying symptoms and signs that Focus Attention On The Abdominal Region and in which the pain has been Present For Less Than 24 H.

The ultimate task for evaluating a case of an acute abdomen is to eventually conclude would the case be managed conservatively (medical acute abdomen) or would it be referred to a surgeon (surgical acute abdomen).

Causes

Abdominal Causes Of Acute Abdomen (surgical)	
Gastrointestinal ❖ Appendicitis ❖ Perforated peptic ulcer ❖ Intestinal obstruction ❖ Intestinal perforation ❖ Intestinal ischemia ❖ Colonic diverticulitis ❖ Meckel diverticulitis ❖ Inflammatory bowel disease	Gynecological ❖ Ruptured ovarian cyst ❖ Ovarian torsion ❖ Ectopic pregnancy ❖ Acute salpingitis ❖ Pyosalpinx ❖ Endometritis ❖ Uterine rupture
Pancreatic, biliary, hepatic, and splenic ❖ Acute pancreatitis ❖ Acute cholecystitis ❖ Hepatic abscess ❖ Ruptured or hemorrhagic hepatic tumor ❖ Acute hepatitis ❖ Acute cholangitis ❖ Splenic rupture	Urological ❖ Ureteral stone ❖ Pyelonephritis
	Retroperitoneal ❖ Aortic aneurysm ❖ Retroperitoneal hemorrhage

Figure 38.1 — Abdominal causes of acute abdomen: surgical



Extra Abdominal Causes Of Acute Abdomen (medical)	
Thoracic ❖ Myocardial infarction ❖ Acute pericarditis ❖ Lower lobe pneumonia ❖ Pneumothorax ❖ Pulmonary infarction	Metabolic ❖ Diabetic ketoacidosis ❖ Addisonian crisis ❖ Acute porphyria ❖ Hyperlipoproteinemia ❖ FMF
Hematological ❖ <u>Sickle cell crisis</u> ❖ <u>Acute leukemia</u>	Neurological ❖ Herpes zoster ❖ <u>Tabes dorsalis</u> ❖ Nerve root compression
	Drug-related ❖ Lead toxicity ❖ Narcotic withdrawal

Figure 38.2 — Extra-abdominal causes of acute abdomen: medical

Clinical features:

- Sudden severe pain characterizes such events as a perforated ulcer, a ruptured viscus, a ruptured ectopic pregnancy, a spontaneous pneumothorax, or a dissecting aortic aneurysm.
- More gradual development of pain is typical of acute pancreatitis, acute cholecystitis, intestinal obstruction, bowel perforation, diverticulitis, and intraabdominal abscess.
- Vomiting Usually accompanies an acute abdomen to a variable degree.
- A clinical rule is that pain precedes vomiting in disorders requiring surgical treatment, whereas vomiting precedes pain in medically treated disorders.

Vomiting can be persistent, as with intestinal obstruction.

- Long-standing obstruction may result in feculent vomiting due to the proximal growth of colonic bacterial flora.
- Vomiting of bloody material suggests that the bleeding lesion is above the ligament of Treitz.

Clinical examination:

- Temperature elevation suggests sepsis from intraabdominal inflammation and infection.
 - Tachycardia typically accompanies an acute abdomen.
 - Blood pressure may also be elevated, although hypotension and shock may develop in patients with a perforated viscus or severe sepsis.
 - Local abdominal examination
 - The abdomen may be distended due to ascites or intestinal obstruction.



- In thin patients with complete obstruction, hyperperistalsis may be visible on the abdominal wall.
- Ecchymosis in the flanks (Grey Turner's sign) or around the umbilicus (Cullen's sign) may indicate hemorrhagic pancreatitis or a ruptured ectopic pregnancy. • Peristalsis is increased in partial or complete bowel obstruction; typically there are rushes and high-pitched sounds.
- Hypo- or aperistalsis develops in intestinal ileus from a variety of causes, including peritonitis, electrolyte imbalance, severe inflammation (e.g., toxic megacolon, pancreatitis), and long-standing intestinal obstruction.
- The examiner should listen for vascular bruits and friction rubs.
- Tympany typically is present in the distended abdomen of intestinal obstruction or toxic megacolon.
- Patients with an acute abdomen typically have abdominal tenderness.
- If peritonitis is present, either localized or generalized, there may be guarding and rigidity.
- Rebound tenderness should be elicited by gently depressing the abdominal wall with one or two fingers, then quickly releasing the pressure.
- Cough tenderness also indicates peritonitis.
- One should remember that the signs of peritonitis may be blunted in elderly or severely ill patients.
- Thus, presence of little or no tenderness does not exclude peritonitis in these patients.



Physical signs in case of acute abdomen

Murphy's sign	Cessation of inspiration during right upper quadrant examination	Acute cholecystitis
Cullen's sign	Periumbilical bluish discoloration	Retroperitoneal hemorrhage Pancreatic hemorrhage AAA rupture
Grey Turner's sign	Bluish discoloration of the flanks	Retroperitoneal hemorrhage Pancreatic hemorrhage AAA rupture
Kehr's sign	Severe left shoulder pain	Splenic rupture Ectopic pregnancy rupture
Obturator sign	Pain with flexed right hip rotation	Appendicitis
Psoas sign	Pain when raising a straight leg against resistance	Appendicitis (right side)

Evaluating a case with acute abdomen

- Laboratory tests: Complete blood count, Serum electrolytes, Arterial blood gases, Serum amylase level, Liver profile, and Urinalysis
- Imaging studies
- Ultrasonography
- CT
- IVP

Treatment

- The General Approach:

This includes intravenous fluids, nothing by mouth, and in most instances nasogastric suction to decompress the stomach and prevent additional air from entering the bowel. In some patients, it may be necessary to pass an additional long tube to decompress the bowel.

Careful attention to volume of fluid intake and urine output is important.

Serum electrolytes and arterial blood gases should be monitored

- Specific treatment depends on the cause of the acute abdomen.



Abdominal trauma



39



Liver trauma

Only a major violence usually leads to liver injury as it is well protected
80% of cases of liver injury associated other intra-abdominal injuries.

Grades of liver injury

. Grade-I:

- Hematoma: Subcapsular, nonexpanding hematoma <10% surface area.
- Laceration: Capsular tear, nonbleeding < 1 cm of parenchymal depth.

. Grade-II:

- Hematoma: subcapsular, nonexpanding haematoma, 10–50% surface area.
- Laceration: Capsular tear, active bleeding, 1–3 cm parenchymal depth.

. Grade-III:

- Hematoma: Subcapsular hematoma >50% surface area or expanding, ruptured hematoma with active bleeding.
- Laceration: Parenchymal laceration >3 cm depth.

. Grade- IV:

- Hematoma: Ruptured intraparenchymal hematoma with active bleeding.
- Laceration: Parenchymal disruption involving 25–50% of hepatic lobe.

Grade- V:

- Hematoma/laceration: Parenchymal disruption involving >50% of hepatic lobes.
- Injuries of major hepatic veins or juxtahepatic vena cava.

. Grade- VI:

- Vascular injury: Hepatic avulsion.

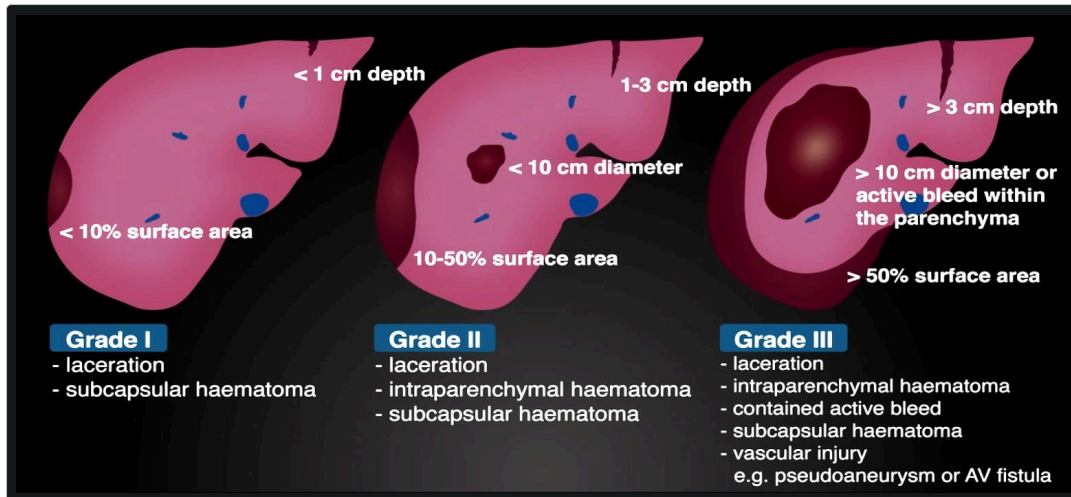


Figure 39.1 — Liver injury grades I-III

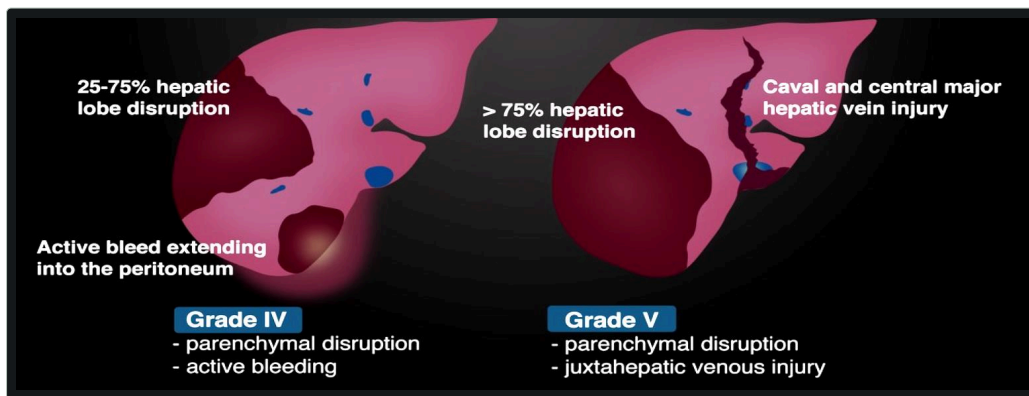


Figure 39.2 — Liver injury grades IV-V

Diagnosis of liver injury

■ Type of trauma:

1. Severe blunt trauma in the right lower chest with fracture ribs or to the right upper quadrant of abdomen.
2. Penetrating or missile injury in the right upper quadrant.

Clinical picture:

History of abdominal trauma mainly in the right side of the abdomen and • lower rib cage

Asymptomatic cases (sometimes in small sub-capsular hematoma or

- small tear)

Pain in right hypochondrium

Picture of hemo-peritoneum

- Marked abdominal pain



- Abdominal distension (due to ileus and or hemoperitoneum)
- Vomiting
- Generalized abdominal tenderness, rebound tenderness and rigidity.

Picture of hypovolemic shock.

- Drowsiness, loss of consciousness
 - Rapid weak pulse
 - Hypotension
 - Tachypnea

Investigations for diagnosis of liver trauma

- Baseline CBC: for Hemoglobin level (determine severity of bleeding).
- **Diagnostic peritoneal lavage**—positive in patients with intraperitoneal hemorrhage.
- **Ultrasonography** of abdomen—diagnostic in majority of patients.
- **CT scan of abdomen**, accurately diagnose the grades of injury.
Both detects the presence of hemoperitoneum, liver tears and hematomas and detect associated abdominal injuries as splenic trauma



Figure 39.3 — Lacerated liver and splenic hematoma

Indications of conservative treatment of hepatic injuries

1. In majority of hepatic injuries bleeding stops spontaneously.
2. About 80% of hepatic injuries are now treated by non-operative means.
3. Patient hemodynamically stable on admission.
4. Achievement of hemodynamic stability with intravenous fluid.
5. Not requiring excessive blood transfusion for hepatic injury.
6. No other injury which requires laparotomy



Conservative treatment of hepatic injury includes:

- Intravenous fluids infusion
- Blood transfusion if required
- Analgesics
- Antibiotics
- Intensive monitoring every hourly:
 - Pulse, blood pressure, respiration and abdominal signs
 - **Serial contrast CT scan or ultrasonography** to assess the progress of hematoma or to assess continuing bleed. Free contrast in and around the liver indicates continuing bleed.

Indications of surgery for hepatic trauma

- Laparotomy is indicated for persistent or recurrent bleeding.
- Shocked patient with a distended or distending abdomen.
- Temporary response to intravenous fluid followed by hemodynamic instability.
- Failure of conservative treatment due to ongoing or recurrent bleeding. • Associated injury requiring laparotomy as bowel injuries.
- Penetrating trauma

Lines of surgical management:

Anti-shock measures

- Local compression
- Pringle's maneuver
- Suturing of the tear
- Peri-hepatic packing
- Definitive liver surgery as resection
- On exploratory laparotomy and evacuation of hemoperitoneum a minor liver injury is found which is not bleeding. What to do?

The strategy in such situation is not to explore the hematoma.

- On exploration there is minor liver injury, which is actively bleeding, what to do?

The strategy in such situation is to **compress** the tear and maintain the compression for 15 - 20 minutes. If the bleeding stops the tear sites should not be opened.

- What should be the operative strategy in patients with major liver injury and active bleeding, which could not be controlled by compression?

1. When compression fails to stop the bleeding then inflow occlusion is done by applying a non-crushing clamp in the hepato-duodenal ligament to occlude the hepatic artery and portal vein. **(Pringle's maneuver) = Inflow occlusion**

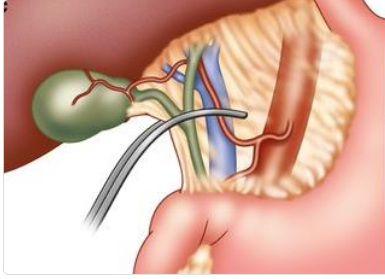


Figure 39.4 — Pringle's maneuver: inflow occlusion

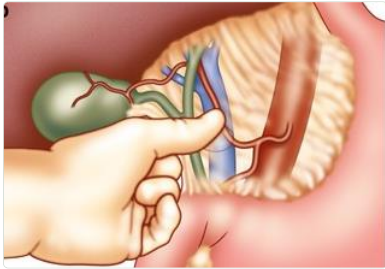


Figure 39.5 — Pringle's maneuver

2. If bleeding is controlled with inflow occlusion, then the possible source of bleeding is either hepatic artery or portal vein branches. Intermittent release of the clamp will allow to identify the site of bleeding, which may then be **suture ligated**. Direct suturing requires a good view into the depths of the injury.
3. If bleeding could not be controlled by suturing, **peri-hepatic packing** would be the approach as a damage control measure. Patient is returned to theater and re-explored after 24–36 hours when he is hemodynamically stable.
4. On re-exploration the pack is removed and bleeding reassessed.
 - There may be no bleeding and no evidence of necrosis or bile leaks. No resection is required in such situations.
 - The bleeding may stop but there may be non-viable or macerated liver tissue, which require debridement or resection.
 - If bleeding recurs on pack removal, decision is to be taken whether to **repack** or to do definitive procedure. If the patient's condition is stable segmental **liver resection** may be done by an experienced surgeon.
 - If bleeding is not controlled by inflow occlusion then the likely source of bleeding is either hepatic vein or intrahepatic inferior vena cava injury (high mortality rate).

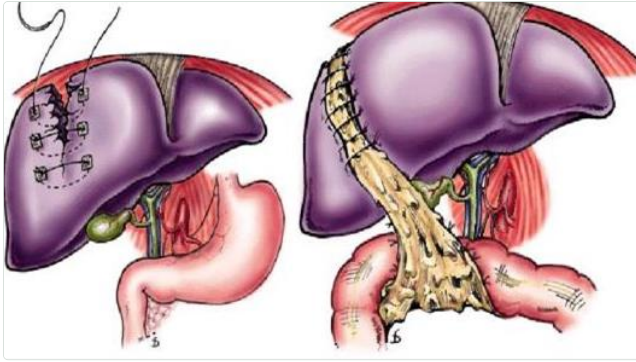


Figure 39.6 — Suturing the tear with omental patch

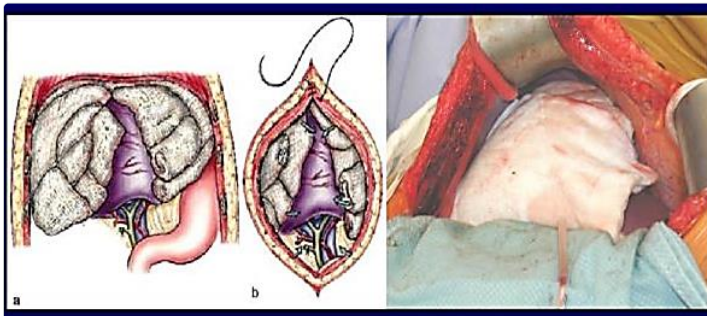


Figure 39.7 — Peri-hepatic packing

Complications following hepatic trauma:

1. Deranged liver function test—bilirubin, transaminase, alkaline phosphatase and GGT level may rise. Serum albumin level may fall.
2. External biliary fistula may occur due to minor or major ductal injury. ERCP and stenting may allow healing. Development of delayed biliary stricture may require appropriate surgery at a later date.
3. Hemobilia is rare complication. Hepatic artery angiography and selective embolization may help.
4. Sub-phrenic abscess and septicemia.
5. Pulmonary complications (ARDS).
6. Multisystem organ failure (irreversible shock).

Splenic trauma



40



Most common indication for laparotomy after blunt trauma

Spleen is the most commonly injured abdominal organ in blunt trauma

Mechanism of trauma:

- **Blunt trauma** as in MCA, falling from height and sports.
- **Sharp penetrating trauma** as in stab wound.
- **Injuries:**
 - 1- External direct trauma transfer from chest wall.
 - 2- Direct punctures from rib fracture (9th, 10th, 11th ribs).

Clinical Types of splenic injuries:

1- Delayed rupture spleen

a) Patient presented with splenic hematoma and managed conservatively. b) Presented days later signs and symptoms of intra-peritoneal hemorrhage. **2- Classical picture of rupture spleen** → discussed below

3- Fatal rupture spleen:

- Those with severe trauma with main splenic vessels injury and macerated spleen
- Death may occur in delayed management.

Diagnosis of splenic trauma:

I- Clinical II- Radiological III- DPL

I- Clinical: Classic picture of rupture spleen

- A. Picture of hypovolemic shock
- B. Picture of hemoperitoneum

General Symptoms:

- a. History of trauma (**blunt** as MCA or **sharp** as in stab wound) b. Symptoms of hypovolemic shock (cold sensation, thirst, drowsy, loss of consciousness)
- c. Symptoms of associated injuries (e.g. fractures).

Local symptoms:

a) Abdominal pains:

1. **Firstly**, left hypochondrial pain
2. Later, generalized abdominal pain (peritoneal irritation due to hemo-peritoneum)
3. Left upper quadrant pain referred to left shoulder.



b) Left sided abdominal swelling (hematoma) c) External wound in case of stab or sharp trauma.

General signs:

a) Signs of hypovolemic shock in

actively bleeding spleen:

1- Rapid weak pulse 2- Hypotension 3- tachypnea

3- Unconsciousness 4- dry mouth b) Signs of associated injuries (fractures, head injury,.....)

Local signs (abdominal examinations):

■ Inspection:

- 1- External wound or hematoma
- 2- Bruising over left upper quadrant.

3-Kehr sign:

left upper quadrant pain, with referred left shoulder Pain when the patient lying down and leg elevated.

4- Abdominal distension (due to ileus)

■ Palpation:

Signs of peritoneal irritation due to hemo- peritoneum:

1- Abdominal tenderness (firstly left hypochondrial then generalized 2- Rebound tenderness and guarding rigidity

■ Percussion:

- 1- Hyper resonance due to ileus
- 2- Shifting dullness from blood in the peritoneal cavity

■ Auscultation:

Silent abdomen due to ileus.

Investigations:

I-Radiological:

■ Ultrasound and - CT scan:

Noninvasive, rapid, low cost

Diagnose liver, spleen, renal traumas.

Presence of intra-peritoneal fluid (blood), Replace DPL

Put grading system for splenic trauma



Figure 40.1 — Splenic trauma: CT view



Figure 40.2 — CT: splenic trauma

■ **Splenic injury grading system:**

1- Grades I and II are considered minor injuries.

2- Grade III:
moderate injuries

3- Grades IV and V:
severe injuries (fatal)

II. DPL: diagnostic peritoneal lavage

- Blood aspirate from the lower quadrants of the abdomen indicates hemo-peritoneum
 - Sensitive
 - Not frequently done nowadays



Grade	Type	Description
I	Tear	Capsular tear, < 1cm
	Hematoma	- Sub-capsular, < 10% of spleen surface
II	Tear	1-3 cm parenchymal
	Hematoma	- Sub-capsular, 10-50%, of spleen surface
III	Tear	3 -10 cm parenchymal
	Hematoma	- Sub-capsular, > 50% of spleen surface
IV	Tear	> 10 cm Involving parenchymal segmental vessels injury
V	Tear	Shattered spleen Hilar injury

III - Laboratory investigations:

1- CBC and hematocrit value →

- Assess blood loss
- Guide for resuscitation and blood volume replacement.
- 2- ABG, pH and bicarb. assessment
- 3- Renal functions tests.
- 4- Routine preoperative investigations.
- 5- PT & PTT.

Management of splenic injuries

1 - Management of hypovolemic shock

- Insert I.V. line – central venous line and measure C.V.P.
 - Blood samples for cross matching and lab. investigations.
 - insert urinary catheter to assess renal blood flow and degree of shock and response to resuscitation
 - insert naso-gastric tube
 - starts crystalloid fluid resuscitation
 - starts blood transfusion once prepared
 - stop external bleeding wounds
 - ABC system for assessment of this poly-trauma patient (describe)

2 - Indications for Surgery : Urgent laparotomy

In **Hemodynamic instability** and continuous hemorrhage:

- SBP < 90 mmHg



- HR > 120 beats/min
- No response to resuscitation.
- Radiological evidence of major lacerations
- **Steps:**

Midline incision preferred (exploratory)

Rapid evacuation of blood and clots to assess other sources of injury: liver, mesentery,

Control of the vascular pedicle of spleen by clamping

Splenic mobilization:

1. Define lienorenal ligament- divide under direct vision
2. Incision of phrenocolic ligament with the spleen retracted medially.
3. Ligation of short gastric vessels

A- Indications of splenectomy intra-operatively:

1. Unstable patient
2. Extensive injury with continued bleeding
3. Bleeding from hilar injury.

Divide splenic artery + vein after triple ligations individually: Avoid Pancreatic tail injury

Complications after Splenectomy:

- 1- Pancreatic tail injury → pancreatic fistula
 - 2- Post-splenectomy thrombocytosis
 - 3- Increase in risk for fatal pneumococcal infections and sepsis
- More in children than adults, so vaccination is important post-operatively.
- 4- Other post-operative complications as bleeding, infections, hernias,.....

B- Splenorrhaphy: mainly indicated in children

- 1- Superficial hemostatic agents as gelatin sponge or topical thrombin for grade I and II
 - 2- Suture repair
 - 3- Mesh Wrapping
- N.B. 1/3 of splenic mass needed to maintain immune-competence.

3- Non-operative management

Mainly used in pediatric age group for fear of opportunistic Infection as Pneumococcal

- **Rules:**
- a- Hemodynamic stability, b- Adequate monitoring available c- Injury grade: I or II



Penetrating trauma of the abdomen

Causes:

1. Stab wound
 2. Gun shot
- Most commonly associated with vascular and/or visceral injuries
 - Multiple injuries are common
 - The associated injuries depend on the site of penetrating trauma

Lists of tissues injuries in penetrating trauma:

1. Skin- subcutaneous tissues
 2. Abdominal wall muscles (tears- hematoma)
 3. Parietal Peritoneum (determines the trauma either penetrating or non-penetrating)
 4. Liver injuries (epigastrium- right hypochondrium)
 5. Duodenum (epigastrium- right hypochondrium)
 6. Stomach (epigastrium- left hypochondrium)
 7. Colon (anywhere according to the site of trauma)
 8. Small intestine (anywhere according to the site of trauma)
 9. Spleen (left hypochondrium)
 10. Kidneys (loins)
 11. Vascular injuries: anywhere
- Retroperitoneal hematoma
 - Major vascular injuries (aorta, iliac, IVC,.....)

Clinical presentations:

Symptoms:

- History of trauma
- External bleeding
- Symptoms of hypovolemic shock (.....)
- Abdominal pain
- Abdominal distension



General Signs:

- Signs of hypovolemic shock (.....)



- Signs of neurogenic shock
- Signs of septic shock (neglected cases)

Local signs:

i- In perforated viscus:

- Abdominal distension
- Tenderness, rebound tenderness & guarding rigidity of the abdomen
- Hyperresonance

■ Silent abdomen ii- In solid organ injuries (liver – spleen) as before iii- In vascular injuries: picture of hemoperitoneum

- Abdominal distension
- Tenderness, rebound tenderness & guarding rigidity of the abdomen
- Hyperresonance + shifting dullness
- Silent abdomen iv- Teat for peritoneal penetration via introduction of sterile catheter through the external wound.

Investigations:

1. CBC
2. Radiological:
 - a) Plain x-ray abdomen erect: for detection of air under diaphragm in perforated viscus
 - b) Pelvi-Abdominal ultrasound:
 - Fluid and /or blood collections
 - Solid organ injuries and grade of injury c) CT scan in stable cases

Treatment

1. Antishock measures in shocked patient (mention in details)
2. Conservative in-patient management with vital stable patients and negative radiological findings
3. Urgent Exploration in:
 - Vital unstable irrespective resuscitation
 - Evidence of hemoperitoneum or peritoneal irritation clinically
 - Radiological evidences of abdominal collections
 - Radiological evidence of perforated viscus
 - Evisceration of omentum or intestine through the external wound
 - Any doubt go for exploration
 - Exploration can be done either through laparotomy or via laparoscopy



Procedures:

Deal with the associated injuries accordingly after proper abdominal exploration:

- a) Gastric injuries: suturing in 2 layers
- b) Small intestine: resection anastomosis – suturing in 2 layers in small injuries
- c) Colon according to the site of injury ranging from right hemi- colectomy to proximal diverting colostomy or ileostomy
- d) Major vascular injuries: repair accordingly
- e) Retroperitoneal hematoma: drains are left inside
- f) Finally peritoneal toilet with normal saline and drains are left inside.

رَبِّ زِدْنِي عِلْمًا

[سورة طه : 114]



محاضرات فيديو مراجعة شاملة

COMPLETE MEDICAL VIDEO LECTURES



ENT



Forensic
Medicine



Special Sense
Eye



Ophthalmology



Rheumatology



Orthopedic



Clinical
Toxicology



Cardiovascular
System



Cardiology



Respiratory
System



Chest



Gastrointestinal
Tract



General
Surgery



Special
Surgery



Urinary
System



Pediatric 1



Pediatrics 2



Surgery



Neurology



Endocrinology



Obstetrics



Nephrology



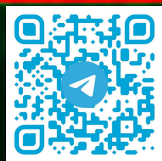
GIT



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شرح مبسط
ومنظم



محتوى شامل
ومحدث



مناسب لجميع
مراحل الدراسة



راجع في أي وقت
ومن أي مكان